# LizardNet: Lizard Connectome-Inspired Network for Fast Few-Shot Hyperspectral Classification

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***Abstract*—Hyperspectral image (HSI) classification requires spectral selection, boundary-sensitive spatial processing, and low model complexity. We present LizardNet, a lightweight neuro-dynamic framework whose architecture draws on three organizational features of the reptilian visual pathway: retinal oil-droplet filtering, horizontal-cell lateral inhibition, and shallow tectal integration. LizardNet implements these features with three modules. The connectomics-inspired sparse spectral gating (SSG) module uses an ODE-inspired steady-state score and Top-$K$ hard gating. The lateral inhibition and anisotropic diffusion (LIAD) module applies a partial differential equation (PDE)-based, boundary-aware spatial update. The tectal low-rank multipath integration (TMI) module uses nuclear-norm regularization to constrain multipath fusion. On Indian Pines (IP) with 5% training samples, LizardNet obtains 94.63% overall accuracy (OA); on Trento, it obtains 98.75% OA. CPU-only FP32 measurements on a Jetson Orin Nano 8 GB give 25.14 samples/s, 39.78-ms mean latency, and 0.191-J incremental energy per sample. Under the same protocol, DGPF-RENet gives 3.18 samples/s, 314.22-ms mean latency, and 1.854-J incremental energy per sample. These results quantify the accuracy and computational cost of using explicit spectral, spatial, and low-rank constraints for HSI classification. The code and trained models are available at: https://github.com/yuhua666/LizardNet**

***Index Terms*—Bioinspired neural architecture, connectomics, hyperspectral image classification, lateral inhibition, lightweight model, spectral gating.**

## I. Introduction

HYPERSPECTRAL imaging records narrow-band spectral signatures over a broad wavelength range and supports applications including precision agriculture, environmental monitoring, and urban planning [1]. Hyperspectral image (HSI) classification presents three main difficulties. First, hyperspectral cubes contain hundreds of correlated bands, and the discriminative subset depends on the task. Wavelength-selective filtering in biological photoreceptors provides one reference for spectral reduction [2], [3]. Second, materials with similar spectra can produce ambiguous class boundaries, which requires spatial processing that responds to local transitions. Retinal lateral inhibition provides a biological reference for this operation [4]. Third, per-class annotations often number only in the tens or hundreds, which increases overfitting in large models. Recent few-shot methods address data scarcity through cross-domain feature disentanglement [5] or semantic-anchor-enhanced prototype learning (SAEPL) [6]; these methods use auxiliary losses and multitask optimization. Together, these issues motivate spectral selection, boundary-aware spatial processing, and parameter control within the architecture.

To tackle these issues, contemporary deep learning approaches have explored two main directions. The first focuses on increasing model capacity through Transformers [7], [8], [9], [10], state-space models such as Spectral-Mamba [11] and S2Mamba [12], and graph convolutional networks (GCNs) [13], [14], [15]. While these architectures improve long-range modeling, their practical computational cost can remain substantial because of dense token interactions, repeated state-space scans, or graph construction and message passing, depending on the architecture. The second direction targets efficiency through lightweight design [16], tensor decomposition [17], network pruning [18], or knowledge distillation (KD) [19]. These methods usually reduce parameters through post hoc structural simplification, which removes connections from an initially dense model and may discard fine-grained spectral information. In both directions, data-driven optimization determines most of the network topology. Biological neural systems provide a different design reference: genetic and developmental programs constrain their connectivity patterns [20]. Fig. 1 shows spectral noise and blurred semantic boundaries in several conventional CNN and Transformer classification maps. The proposed model uses predefined spectral and spatial constraints to preserve finer spatial–spectral structure.

Computational neuroscience provides a useful design perspective. Barlow's *efficient coding hypothesis* [21] describes sensory systems as transmitting relevant information under metabolic constraints. Sparse retinal ganglion-cell activity [22] and energy-constrained coding in ectothermic vertebrates [23], [24] are consistent with this view. Connectomic studies further

Received 26 May 2026; revised 7 August 2026 and 22 August 2026; accepted 11 September 2026. Date of publication 22 September 2026; date of current version 12 October 2026. This work was supported in part by the National Natural Science Foundation of China through the General Program under Grant 72571281 and in part by the Excellent Young Scientist Fund of Hunan Province under Grant 2025JJ40066. *(Corresponding authors: Yingmei Wei; Yanming Guo.)*

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Digital Object Identifier 10.1109/TGRS.2026.3736683

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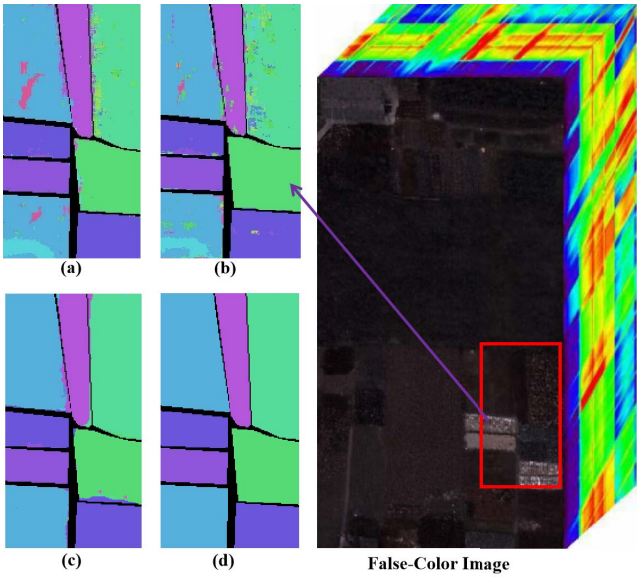



Fig. 1. Classification maps on WHU-Hi-HongHu. (Right) Hyperspectral cube. (a) CNN-based prediction with blurred adjacent regions. (b) Transformer-based prediction with fragmented errors. (c) LizardNet prediction. (d) Ground-truth labels.

show that efficient coding depends on both the computations performed and the topology through which signals propagate [25], [26]. Lizards provide a relevant example because their basal metabolic rate is approximately 1/7–1/10 that of a mammal of similar body mass.

Three vertebrate visual-processing motifs are relevant to our design. Lizard cone oil droplets provide the species-specific reference for wavelength-dependent filtering [2], [27]. Published gecko recordings provide a lizard-specific reference for separating steady and transient photoreceptor responses [28]. Center–surround interaction in turtle horizontal cells supplies a broader reptilian reference for spatial antagonism [29]. The optic tectum supplies a comparative vertebrate reference for compact parallel integration [30], [31], [32]. These observations motivate spectral selection, boundary-aware spatial processing, and compact feature integration. The scope of this mapping is a mechanism-level design abstraction limited to these selected visual-processing properties.

LizardNet translates this mapping into explicit constraints on spectral routing, spatial diffusion, and feature integration. The framework treats HSI classification as a routing problem with a fixed module-level topology. Each of its three modules formalizes one biological processing stage and addresses one HSI classification problem.

The three modules address the three HSI challenges in sequence. First, the connectomics-inspired sparse spectral gating (SSG) module suppresses spectral redundancy through an ODE-inspired steady-state score followed by Top-$K$ hard gating. The resulting binary mask sets unselected channels to zero and imposes an $L_0$ sparsity constraint. Second, the lateral inhibition and anisotropic diffusion (LIAD) module addresses spatial boundary ambiguity. Its partial differential equation (PDE)-inspired update limits diffusion across strong boundaries and promotes consistency within homogeneous regions. Third, the tectal low-rank multipath integration (TMI) module promotes a compact effective-rank representation and small-sample generalization. Nuclear-norm regularization encourages low effective rank in its multipath fusion weights. The shallow constrained topology and feature routing provide overall parameter and FLOP efficiency.

Top-$K$ selection, Perona–Malik diffusion, and nuclear-norm regularization are the established operations. The conceptual contribution of LizardNet is a constraint-allocation principle that assigns them to three different control variables in one ordered feature route: SSG constrains spectral support, LIAD constrains spatial propagation on that selected support, and TMI constrains the effective rank of the subsequent fusion map. The framework-level contribution lies in the placement and coupling of the support, propagation, and fusion-capacity constraints; the three elementary operations retain their established definitions.

We evaluate LizardNet on Indian Pines (IP), DFC2013 Houston, WHU-Hi-HongHu, and Trento. With 5% of the IP samples used for training, it obtains 94.63% overall accuracy (OA). On Trento, it obtains 98.75% OA. Across these datasets, LizardNet uses 0.75M–0.94M parameters and 2.52M–2.90M FLOPs, with the lowest FLOPs among the evaluated lightweight models under the common $15 \times 15$ input protocol. The experiments therefore assess both classification accuracy under limited supervision and the computational cost of the proposed constraints.

Our contributions are threefold.

1) We map the three-stage saurian visual pathway (oil-droplet filtering → lateral inhibition → tectal integration) to three HSI operations (spectral selection → boundary-aware spatial processing → compact feature integration). This mapping links each architectural constraint to a specific biological mechanism.
2) We propose LizardNet, a lightweight neuro-dynamic framework that allocates explicit constraints to three successive control variables: SSG constrains spectral support through steady-state hard routing, LIAD constrains spatial propagation through edge-conditioned diffusion, and TMI constrains fusion capacity through a nuclear-norm penalty. This ordered allocation and coupling constitute the framework-level contribution; the individual Top-$K$, Perona–Malik, and nuclear-norm operations remain established techniques.
3) Experiments on four benchmarks report 98.75% OA on Trento and 94.63% OA with 5% training samples on IP. LizardNet uses 0.75M–0.94M parameters and 2.52M–2.90M FLOPs across the four datasets and records the lowest FLOPs among the evaluated lightweight models under the common $15 \times 15$ input protocol.

## II. Related Works

### A. Deep Spectral–Spatial Representation Learning

Contemporary HSI classification increasingly uses sequence models to capture long-range spectral dependencies. Representative examples include SpectralFormer [7], SSFTT [8],

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the morphological Transformer [9], HSI-BERT [33], and the group-aware hierarchical Transformer [10]. State-space alternatives such as SpectralMamba [11] and S$^2$Mamba [12] extend sequence modeling to spectral–spatial data. These methods encode global context through dense token interactions or repeated state-space scans, which can create substantial computational overhead on high-resolution scenes. Recent alternatives focus on multiscale fusion and classifier geometry. MMD-MLP uses LiDAR guidance, local–global directional MLP blocks, and multiresolution multiscale representations for HSI classification [34]. Fu et al. [35] combined an attention-enhanced ResBlock-in-ResBlock feature extractor with an equiangular tight-frame classifier. These designs emphasize feature aggregation and the output classifier. LizardNet emphasizes sparse routing, edge-conditioned diffusion, and low-rank integration within a compact topology.

Parallel to sequence models, GCNs attempt to explicitly model geometric and relational structures within hyperspectral data [13], [14], [15]. By constructing graph representations over pixels or superpixels, these methods aim to capture spatial dependencies and non-Euclidean relationships that are difficult to encode in grid-based architectures.

These methods predominantly rely on heuristic adjacency matrices, such as $k$-nearest-neighbor graphs or learned similarity graphs. Their topology can therefore be sensitive to hyperparameter choices and data distribution. Connections derived from statistical similarity may retain redundant edges and inefficient propagation paths.

Recent few-shot HSI methods directly address labeled-data scarcity. Zhang et al. [5] proposed an orthogonal feature disentanglement network (OFD-Net) that separates domain-shared and domain-specific features to reduce negative transfer. Yan et al. [6] introduced SAEPL, which uses trainable anchors and contrastive objectives to align class prototypes across domains. SelfMTL combines self-supervised contrastive representation with meta-transfer learning for hyperspectral target detection [36]. TSCCD uses temporal self-construction and cross-domain learning for unsupervised hyperspectral change detection [37]. The latter two methods address scarce annotations and distribution shifts in target-detection and change-detection settings. LizardNet addresses small-sample HSI classification through sparse routing and low-rank integration constraints that regularize the network architecture and loss.

### B. Connectomic Neuro-Dynamics

Continuous-time neural models and sparse connectivity provide two approaches to computational efficiency. Continuous-time formulations describe state evolution within the network, and connectomic studies describe sparse, modular, and selectively routed biological pathways [25], [38]. Connectome-driven models further associate specific wiring motifs, including recurrent microcircuits and hierarchical pathways, with the resulting representation. Standard deep networks commonly obtain sparsity through pruning or dense optimization, which adds training and design steps [39], [40]. These observations motivate the use of predefined sparse pathways in LizardNet.

Architectures that encode anatomical wiring priors can reduce parameter redundancy and make information flow explicit [41]. In such models, the topology determines the available propagation paths and part of the inductive bias. LizardNet follows this approach by specifying sparse spectral routing, boundary-conditioned spatial interaction, and low-rank integration within a neuro-dynamic framework.

### C. Lightweight Model Design Strategies

Model pruning has become a fundamental technique for reducing redundancy and improving computational efficiency in deep neural networks. Recent advances have shifted from heuristic criteria toward structured and optimization-driven strategies. For instance, Guan et al. [42] proposed a differentiable pruning framework for end-to-end sparsity learning, while Lin et al. [43] introduced cross-layer dependency modeling to enhance pruning consistency. Furthermore, Fang et al. [44] developed DepGraph to generalize structured pruning via dependency graphs, enabling hardware-friendly compression across diverse architectures. Pruning remains less studied in HSI classification. Existing lightweight HSI methods primarily use architectural simplification [45], [46], leaving explicit sparsification of spectral redundancy as an open direction.

Low-rank decomposition provides a complementary paradigm by leveraging the intrinsic low-dimensional structure of data and model parameters. Cai et al. [47] applied low-rank matrix factorization for network compression, while Zhuang et al. [48] and Wang et al. [49] demonstrated that low-rank and tensor-based models can effectively capture spectral–spatial correlations in HSIs. More recently, deep low-rank models [50] integrate neural networks with low-rank priors for enhanced representation learning. Generative priors provide another form of spectral–spatial modeling. HDiff-HIR uses a hierarchically conditional diffusion model for hyperspectral image reconstruction [51]. Its reconstruction objective focuses on recovering spectral–spatial detail. LizardNet uses a discriminative low-complexity routing objective for classification. However, most existing methods focus on reconstruction or restoration tasks, with limited exploration in lightweight classification. Therefore, how to jointly exploit spectral redundancy and spectral–spatial dependencies under structured compression remains an open challenge.

Model compression techniques such as quantization and KD have been widely explored to improve the efficiency of deep neural networks. Model quantization reduces memory footprint and accelerates inference by representing weights and activations with low-bit precision. Early studies [52] established the foundation for efficient integer-based inference, while subsequent works have focused on improving the accuracy of posttraining quantization (PTQ). For example, Li et al. [53] proposed BRECQ, which minimizes PTQ error through block reconstruction, and Liu et al. [54] extended PTQ to Vision Transformers by addressing the sensitivity of attention mechanisms to quantization noise.

In parallel, KD transfers knowledge from a large teacher model to a compact student model to maintain performance under resource constraints. Recent advances move

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beyond conventional logit-based distillation toward more adaptive strategies. For instance, Lan et al. [55] introduced label revision and data selection mechanisms to improve knowledge quality, while comprehensive studies [56] summarize multilevel distillation paradigms, including feature- and relation-based knowledge transfer. Despite these advances, the integration of quantization and KD in HSI classification remains limited, where high spectral dimensionality and strong interband correlations pose additional challenges for efficient compression and knowledge transfer.

### D. Biological Inspiration for Efficient HSI Modeling

As ectothermic organisms, lizards operate under strong metabolic constraints, with their sensory and neural processing capabilities tightly coupled to environmental temperature through a nonlinear thermal performance curve [57]. Metabolic rates follow an exponential temperature dependence governed by the temperature coefficient $Q_{10}$ (typically 2–3.9 [58], [59]), so even moderate thermal fluctuations propagate through ventilation and oxygen transport [60] into reduced synaptic transmission, firing dynamics, and visual perception [61]. These constraints establish an intrinsic *resource budget* that has driven the saurian visual system toward maximally information-efficient processing.

These biological observations motivate three requirements for HSI classification: robustness to environmental and sensor variation, selective processing of high-dimensional spectra, and separation of classes with similar signatures. LizardNet addresses these requirements with structural sparsity, selective spectral routing, and boundary-aware feature propagation.

*1) Secondary Quantitative Validation and Design Relevance:* No new animal experiments were conducted in this study. All physiological values below come from published primary studies and are used for secondary mechanism-level analysis. Loew et al. [27] measured visual pigments and oil droplets microspectrophotometrically in 17 Caribbean anoline species and reported four vitamin-$A_1$-based pigment maxima at approximately 365, 455, 495, and 564 nm. They also identified five recurring oil-droplet classes: yellow and green droplets in long-wavelength-sensitive cones, green droplets in medium-wavelength-sensitive cones, and colorless droplets in short-wavelength- and ultraviolet-sensitive cones. The colorless short-wavelength-sensitive droplet absorbed strongly between 350 and 400 nm; the ultraviolet-sensitive droplet showed a different absorption profile in this interval. We use these observations as evidence for multiple, unequally filtered pathways before downstream integration. The comparison operates at the pathway-organization level. SSG implements selective routing by globally summarizing the HSI channels, ranking their input-dependent scores, retaining a fixed number $K$ of pathways, and setting the remaining pathways to zero. Under an identical backbone and scoring input, Top-$K$ binary gating raises OA from 91.34% to 94.42% and reduces FLOPs from 2.90M to 2.77M relative to continuous soft weighting (see the ablation study in Section V-G). The published measurements motivate selective pathway organization, and the HSI ablation measures the classification effect of the corresponding routing operation.

Kleinschmidt and Dowling's intracellular recordings [28] from *Gekko gekko* provide a second lizard-specific constraint. Steady backgrounds produced graded, sustained hyperpolarization, and single-receptor increment thresholds followed Weber's law for approximately 3.5 log units. Sodium aspartate abolished the initial response transient, which the authors associated with possible removal of horizontal-cell feedback, and preserved the steady adaptation characteristics. This separation of transient and steady responses defines the scope of our abstraction. SSG evaluates the analytical equilibrium $s^*$ for a static HSI patch, and LIAD represents spatial feedback and boundary interaction. Temporal photoreceptor dynamics lie outside the model scope. The deterministic solution $s(t) = s^* + (s(0) - s^*)e^{-t/\tau}$ establishes $s^*$ as the analytical equilibrium and avoids a stepwise numerical approximation. In the spectral gating ablation (Section V-G), this steady-state score adds 0.21 percentage points of OA over the same Top-$K$ gate with a conventional GAP score. The physiological result supports the separation of steady and transient responses, and the ablation measures the classification effect of the equilibrium score.

Our quantitative spatial reference is the turtle-retina study by Piccolino et al. [29], which is used as broader reptilian evidence. The two studies above provide the lizard-specific evidence in this analysis. Piccolino et al. [29] stimulated turtle small-field luminosity horizontal cells with circular spots and annuli and found that dim-spot responses decreased when spot diameter exceeded approximately 1000 $\mu$m. Bright spots larger than approximately 1000 $\mu$m produced a delayed antagonistic component, and peripheral annular stimulation under a central background produced depolarizing responses [29]. These results show that retinal interaction depends on spatial extent and center–surround configuration. LIAD represents this spatial selectivity through gradient-conditioned conductance: diffusion remains high within homogeneous regions and decreases across strong boundaries. In the spatial operator ablation (Section V-G), the Base setting obtains 91.83% OA, isotropic diffusion obtains 93.18%, spatial attention obtains 93.86%, and LIAD obtains 94.63%. The diffusion step analysis in Section V-G shows that a second update adds 0.18 percentage points at 7% higher cost; five updates lower OA to 94.28% at 26% higher cost. The results support bounded, edge-conditioned interaction, and the five-step result quantifies the effect of repeated diffusion.

*2) Cross-Domain Relevance and Validation Boundary:* The published values constrain three design decisions in LizardNet: multiple differentially filtered pathways motivate competitive spectral selection; the measured separation of transient and steady responses motivates direct equilibrium scoring for static inputs; and spatially contingent antagonism motivates boundary-conditioned, finite-step diffusion. These observations provide mechanism-level constraints. We did not perform a direct physiological-to-feature Pearson or Spearman correlation. The physiological and computational variables do not have commensurate axes: the network channels represent dataset-specific HSI measurements or their preprocessing outputs, while a $15 \times 15$ remote sensing patch has no biologically defined conversion to retinal micrometers. Pairing these variables by index would impose an arbitrary correspon-

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dence and yield a scientifically uninterpretable correlation. We therefore evaluate each borrowed mechanism through a matched computational comparison: soft and hard spectral gating, conventional and steady-state scoring, and isotropic and boundary-conditioned diffusion with an iteration sweep. The resulting evidence establishes mechanism-level consistency with selected published reptilian retinal properties.

## III. Theoretical Foundations of Connectomic Priors

This section presents the mathematical definitions used by the three LizardNet modules. Section IV gives the architectural details and biological motivation.

### A. Optimal Spectral Transport via Neural Stochastic Differential Equations

SSG represents selective synaptic transmission with a dynamic variable for each spectral pathway. Let $x \in \mathbb{R}^{H\times W\times C}$ be the input hyperspectral tensor. The channelwise significance vector $s(t) \in \mathbb{R}^C$ obeys a neural SDE extending the leaky integrate-and-fire model

$$ds_c(t) = \underbrace{-\frac{1}{\tau}\left(s_c(t) - \mathcal{F}_{\text{syn}}(x_c)\right)dt}_{\text{Drift (Deterministic LIF)}} + \underbrace{\beta\sqrt{s_c(t)}dW_t}_{\text{Diffusion (Stochastic Noise)}} \quad (1)$$

where $\tau$ is the membrane time constant, $\mathcal{F}_{\text{syn}}(\cdot)$ is the global spatial aggregation, $\beta$ controls the synaptic-noise volatility, and $W_t$ is a Wiener process.

*Proposition 1 (Asymptotic Stability of the Deterministic Drift):* For bounded inputs, the deterministic component of (1) admits a unique stable equilibrium $s^*$. Under bounded diffusion, the full SDE possesses a stationary distribution concentrated near $s^*$. The Lyapunov candidate $V(s) = \frac{1}{2}\sum_c (s_c - s_c^*)^2$ satisfies $\dot{V} = -\frac{1}{\tau}\sum_c (s_c - s_c^*)^2 \le 0$, giving global convergence to $s^* = \mathcal{F}_{\text{syn}}(x)$; the diffusion term yields a positive-recurrent Markov process whose invariant measure concentrates near $s^*$ as $\beta \to 0$.

*1) Scope of the Steady-State Solution:* HSI patches are static inputs, so SSG uses the equilibrium response of the deterministic drift. The closed-form value $s^* = \mathcal{F}_{\text{syn}}(x)$ is computed directly during inference and avoids iterative temporal simulation. The model therefore represents steady spectral selection; transient photoreceptor timing remains outside its scope.

Under the sparsity budget $\|M\|_0 = K$, the most informative channel subset maximizes the trace of the Fisher-information matrix $\mathcal{I}(\theta)$

$$\Omega_K^* = *argmax_{\Omega:|\Omega|=K} \text{Tr}\left(\mathcal{I}\left(\theta; x_\Omega\right)\right). \quad (2)$$

The Fisher-information criterion motivates the score-ranking heuristic used by SSG. SSG ranks the channel scores as a tractable approximation to (2). Table XIII evaluates the implemented heuristic: Top-$K$ gating increases OA from 91.34% to 94.42% relative to soft gating under the same backbone and scoring input. This ablation measures the empirical effect of hard selection; it does not constitute a direct experimental verification of the Fisher-information objective.

### B. Variational Formulation of Retinal Lateral Inhibition

The LIAD module solves a variational problem on the image manifold. Let $u : \Omega \to \mathbb{R}$ be the feature map; edge sharpening minimizes the total variation energy

$$E(u) = \int_\Omega \left(\tfrac{1}{2}(u - u_0)^2 + \lambda\phi\left(\|\nabla u\|\right)\right) d\mathbf{x} \quad (3)$$

where $\phi(\cdot)$ is a robust potential (e.g., Perona–Malik) penalizing smoothing across edges. The Euler–Lagrange gradient descent yields the anisotropic diffusion PDE

$$\frac{\partial u}{\partial t} = \nabla\cdot\left(\frac{\phi'\left(\|\nabla u\|\right)}{\|\nabla u\|}\nabla u\right) - (u - u_0). \quad (4)$$

With $\phi(s) = 1 - e^{-s^2/K^2}$, the conductance recovers $g(\|\nabla u\|) = \exp(-\|\nabla u\|^2/K^2)$, producing a piecewise-constant scene approximation that transforms fuzzy class boundaries into sharp transitions.

### C. Low-Rank Tectal Integration and Generalization Bounds

TMI represents compact parallel integration by constraining the routing matrix $W_c$ to low rank. By the Eckart–Young–Mirsky theorem, the best rank-$r$ approximation satisfies $\min_{\text{rank}(\hat{W})\le r}\|W^* - \hat{W}\|_2 = \sigma_{r+1}(W^*)$. When the singular values of the HSI feature correlation matrix decay rapidly, a small $\sigma_{r+1}$ gives a correspondingly small rank-$r$ approximation error.

For a hypothesis class $\mathcal{H}$ with nuclear-norm-bounded weights $\|W\|_* \le B$, the Rademacher complexity scales as

$$\mathfrak{R}_S(\mathcal{H}) \le \frac{B}{\sqrt{N}}\cdot O\left(\log\left(\min(m,n)\right)\right) \quad (5)$$

where $N$ is the sample size. This Rademacher complexity bound motivates the nuclear-norm term used in TMI. Penalizing the nuclear norm controls the quantity $B$ appearing in the bound. The parameter sensitivity analysis in Section V-G shows, at $\alpha = 10^{-3}$, the training–validation gap decreases on all four datasets, and OA increases relative to $\alpha = 0$. These observations are consistent with the intended regularization effect and do not constitute a direct empirical proof of the bound.

## IV. LizardNet

HSI classification requires joint spectral and spatial information to separate materials with similar signatures. LizardNet maps the input to a structure-constrained representation that combines spectral selection with boundary-aware spatial processing. The spectral gate removes redundant channels, and the spatial operator limits feature mixing across detected boundaries.

LizardNet is a biology-inspired HSI framework built from three modules. Its design uses selected organizational motifs reported in reptilian visual systems: competitive spectral routing, boundary-conditioned lateral interaction, and compact parallel integration. The forward pass composes the three modules as follows:

$$h = \mathcal{I}_\psi\left(\mathcal{B}_\phi\left(\mathcal{S}_\theta(x)\right)\right) \quad (6)$$

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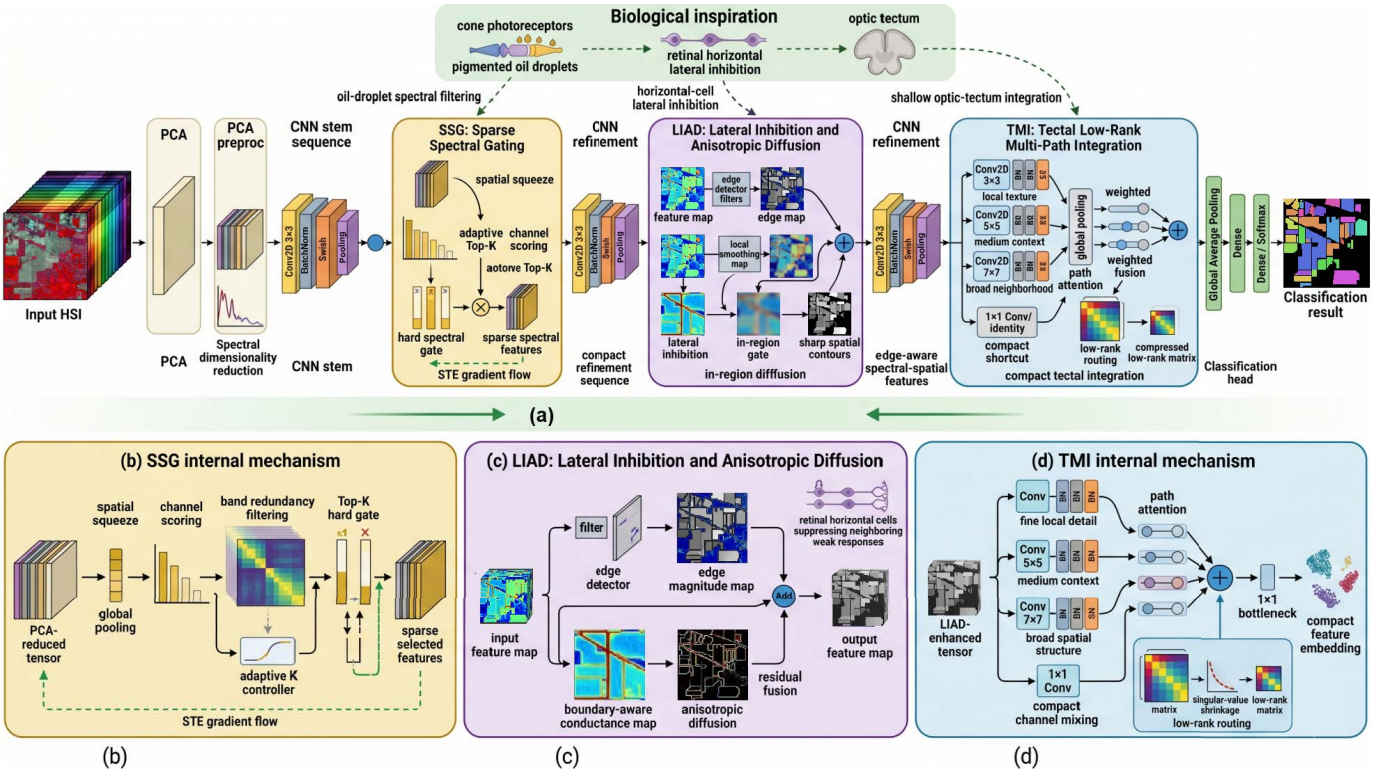



Fig. 2. Schematic of LizardNet. (a) Overall LizardNet architecture, processing HSI through a biologically inspired cascade: SSG, LIAD, and TMI. (b) Internal mechanism of the SSG module. It performs spatial squeezing, band redundancy filtering, and adaptive Top-$K$ hard gating for spectral selection, maintained by STE gradient flow. (c) Internal mechanism of the LIAD module. This module fuses an edge magnitude map with anisotropic diffusion, guided by a conductance map, to suppress neighboring weak responses. (d) Internal mechanism of the TMI module. It integrates multiscale spatial context ($3 \times 3$, $5 \times 5$, and $7 \times 7$) and channel mixing via path attention, regulated by low-rank matrix shrinkage for compact feature embedding.

where $x \in \mathbb{R}^{C \times H \times W}$ represents the input hyperspectral volume. Here, $\mathcal{S}_\theta$ denotes a connectomics-inspired SSG module (modeling early visual selection), $\mathcal{B}_\phi$ is a LIAD module (modeling retinal horizontal cells), and $\mathcal{I}_\psi$ is a TMI module. The parameters $\theta, \phi$, and $\psi$ are the learned end-to-end, subject to the architectural constraints imposed by the reptilian wiring diagram.

LizardNet consists of three core modules: SSG, LIAD, and TMI. SSG computes the closed-form equilibrium score motivated by the ODE and then applies Top-$K$ gating to select spectral pathways. Given the spatial noise and boundary ambiguity prevalent in HSIs, the LIAD module incorporates a PDE-based boundary-aware mechanism to emulate biological LIAD. LIAD sharpens object contours and suppresses background variation through gradient-conditioned conductance. TMI uses nuclear-norm regularization to promote low-effective-rank feature integration and small-sample generalization. The shallow constrained topology and feature routing account for the overall parameter and computational efficiency. Fig. 2 depicts the overall LizardNet architecture.

The three constraints act at noninterchangeable locations in (6). SSG changes which channels enter the spatial stage, LIAD changes how those routed features propagate across local boundaries, and TMI limits the effective rank after multipath features have been formed. Consequently, LIAD receives the hard-routed tensor $x_s$. The nuclear-norm penalty applies only to the downstream fusion matrix $W_c$ and does not act on the gating scores or diffusion kernel. This ordered coupling establishes a structured dependency among spectral selection, boundary-aware propagation, and downstream rank regularization, with each constraint assigned to a distinct stage.

### A. Biological Design Mapping: From the Saurian Visual Pathway to Network Modules

LizardNet draws on three processing stages reported for the saurian visual pathway [3], [32]: 1) colored oil droplets in cone photoreceptors provide wavelength-dependent filtering at the retinal outer segment; 2) horizontal and bipolar cells support lateral interaction and selective signaling in the outer and inner plexiform layers; and 3) the optic tectum integrates parallel inputs in a compact laminar structure. The three LizardNet modules, $\mathcal{S}_\theta$ (SSG), $\mathcal{B}_\phi$ (LIAD), and $\mathcal{I}_\psi$ (TMI), translate these stages into spectral selection, boundary-aware spatial processing, and compact multipath integration.

### B. Connectomics-Inspired SSG

In saurians, cone oil droplets modify spectral sensitivity before downstream neural processing [2], [27]. This wavelength-dependent filtering motivates an initial spectral organization stage. LizardNet then applies SSG as a computational routing abstraction: it scores the HSI channels and passes a fixed number of the highest-scoring channels to the spatial backbone. The operation captures selective pathway

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use at the mechanism level and does not reproduce species-specific cone sensitivities or photoreceptor–bipolar synaptic physiology.

We formulate spectral gating with an ODE-inspired dynamic system. Its equilibrium has a closed-form expression, so SSG computes the steady-state value directly during inference

$$s_c^{\star} = \sigma\left(\frac{1}{HW}\sum_{i,j} x_{c,i,j}\right) \tag{7}$$

where $\sigma(\cdot)$ is the sigmoid transfer function. Direct evaluation gives the deterministic equilibrium for a static input; the module does not model the time course of physiological adaptation.

We therefore describe SSG as an ODE-inspired steady-state module. Its implemented operations are the closed-form equilibrium score and the subsequent Top-$K$ hard gate, which imposes an $L_0$ sparsity constraint.

To impose structural sparsity, we derive a hard-gating mask $M \in \{0,1\}^{C\times 1\times 1}$ through Top-$K$ competition. Competitive inhibition motivates the ranking operation at a functional level; the binary mask is the computational implementation used in SSG. For scores $s^{\star}$, the selection is

$$M_{c,1,1} = \mathbb{I}\left(c \in \mathrm{TopK}\left(s^{\star}\right)\right) \tag{8}$$

where $\mathbb{I}(\cdot)$ is the indicator function. We use the fixed ratio $K = \lfloor 0.8C \rfloor$ in the implementation; the selected indices remain input dependent. This defines a sparse spectral subspace $\mathcal{S}_K$

$$\mathcal{S}_K = \left\{x' \in \mathbb{R}^{C\times H\times W} \,\middle|\, x'_{c,:,:} = 0 \quad \forall c \notin \mathrm{TopK}\left(s^{\star}\right)\right\}. \tag{9}$$

The SSG module $\mathcal{S}_\theta$ applies the binary mask to form a sparsely routed feature tensor

$$\mathcal{S}_\theta(x) = x_s = x \odot M. \tag{10}$$

*1) Biological Motivation for $L_0$ Sparsity:* SE-Net [62] uses soft attention and assigns a continuous weight to every channel. SSG uses a binary Top-$K$ mask: the $K$ selected channels are retained, and the remaining channels satisfy $M_c = 0$. This operation imposes the spectral constraint $\|x_s\|_{2,0} = K < C$ and restricts downstream computation to a fixed subset of $K$ selected channels. From the information bottleneck perspective [63], the operation approximates $\min I(X; X_s)$ subject to $I(X_s; Y) \geq I_0$. PCA and Top-$K$ gating form a two-stage spectral reduction procedure. PCA supplies a linear subspace projection, and Top-$K$ gating supplies task-dependent channel selection. This organization corresponds to the two biological motifs used in the model design: presynaptic spectral filtering and selective routing.

### C. LIAD Module

Retinal center–surround organization and lateral interaction provide the biological motivation for LIAD. Turtle horizontal-cell recordings supply the broader reptilian spatial reference used in this work [29]; the Perona–Malik equation supplies the computational diffusion model [64]. LIAD represents edge-conditioned interaction at the mechanism level, with its scope defined by the gradient-conditioned computation below.

We first compute an edge potential $E$. A center–surround receptive field can be represented by the isotropic DoG kernel $\mathrm{DoG}(x,y) = G_{\sigma_1}(x,y) - G_{\sigma_2}(x,y)$. The Sobel operators $S_x$ and $S_y$ approximate first derivatives along the horizontal and vertical axes and provide two directional edge components

$$E = \sqrt{(x_s * S_x)^2 + \left(x_s * S_y\right)^2 + \varepsilon} \tag{11}$$

where $S_x$ and $S_y$ compute the fixed horizontal and vertical image gradients. These directional components provide the edge signal used by the spatial conductance function.

We model boundary-dependent spatial propagation with an anisotropic diffusion PDE. The equation follows the Perona–Malik form [64]; in LIAD, its conductance term represents reduced interaction across strong edges and stronger interaction within homogeneous regions:

$$\frac{\partial U(t)}{\partial t} = \nabla \cdot \left(g\left(\|\nabla E\|_2\right) \nabla U(t)\right) \tag{12}$$

with initial condition $U(0) = x_s$. At strong edges, $\|\nabla E\|$ is large and $g$ approaches 0, which limits feature mixing across the boundary. Within homogeneous regions, $\|\nabla E\|$ is small and $g$ approaches 1, which promotes local consistency. This conductance pattern implements contour-sensitive processing for HSI regions with similar spectral signatures.

Discretizing (12) via a forward Euler scheme yields the updated feature map $\tilde{x}_s$. For the implemented single update with step size $\Delta t$, this operation is explicitly

$$U^{(1)} = U^{(0)} + \Delta t\, \nabla \cdot \left(g\left(\|\nabla E\|_2\right) \nabla U^{(0)}\right), \quad U^{(0)} = x_s. \tag{13}$$

Because $0 < g \leq 1$, the standard explicit-diffusion condition $0 < \Delta t \leq 1/4$ (unit 2-D grid) provides a conservative stability range. The local truncation error is $O(\Delta t^2)$. Repeated steps accumulate first-order global error and increase diffusion cost linearly. LIAD implements one edge-aware residual update for local boundary refinement. Long-range diffusion requires additional steps, and excessive iteration may over-smooth small objects. Table XV reports the accuracy and cost for $T = 1, 2, 3$, and 5. We then route $\tilde{x}_s$ through multiscale receptive fields

$$f_k = \phi_k\left(\tilde{x}_s\right), \quad k \in \{3, 5\}. \tag{14}$$

Here, $\phi_3$ and $\phi_5$ are $3 \times 3$ and $5 \times 5$ convolutions that capture fine and medium spatial context, respectively. The two kernel sizes are computational design choices and are not assigned to specific lizard retinal cell types.

*The "Innate Prior + Learned Correction" Hybrid Inductive Bias Paradigm:* The final module $\mathcal{B}_\phi$ fuses these pathways with an explicit edge prior

$$x_e = \mathcal{B}_\phi(x_s) = W_f\left[f_3, f_5\right] + \lambda E. \tag{15}$$

The Sobel-derived edge potential $E$ is a fixed filter that supplies an edge prior. The learnable convolutions $\phi_3$ and $\phi_5$ provide task-specific multiscale features. The coefficient $\lambda$ controls the amplitude of the additive edge residual. At $\lambda = 0$, the fused representation contains the learned term $W_f[f_3, f_5]$ alone; increasing $\lambda$ strengthens the edge residual relative to this unit-coefficient branch. Feature norms determine the realized contribution of each term. Table XVI identifies $\lambda = 0.5$ as

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the empirical optimum for OA, average accuracy (AA), and $\kappa$, and we use this value in all experiments. The downstream projection further calibrates the combined representation.

For SSG, the deterministic drift in (1) has the closed-form solution $s(t) = s^* + (s(0) - s^*)e^{-t/\tau}$, with $s^* = \mathcal{F}_{\text{syn}}(x)$. Direct evaluation of $s^*$ equals the deterministic equilibrium. The implemented formulation omits transient timing and stochastic variance and is intended for static HSI patches. Tasks involving temporal retinal dynamics require an explicit time-dependent formulation.

### D. Tectal Low-Rank Multipath Integration

The optic tectum motivates compact parallel integration at a comparative vertebrate level [30], [31], [32]. TMI transfers this motif to HSI classification through two spatial branches followed by a shared fusion projection. Nuclear-norm regularization encourages a small set of dominant response modes. This is a biology-inspired architectural constraint, not a claim that the two branches reproduce identified lizard tectal pathways.

The compact laminar organization of the optic tectum motivates integration through a limited set of response modes. TMI represents this design motif with nuclear-norm regularization of its routing matrix, which encourages a low effective rank.

Within each TMI block, we define two spatial processing branches operating on the boundary-enhanced features $x_e$

$$p_1 = \gamma_1\left(x_e * K_{3\times 3}\right), \quad p_2 = \gamma_2\left(x_e * K_{5\times 5}\right). \tag{16}$$

Here, $\gamma_i$ denotes the branch nonlinearity. Each branch is followed by batch normalization [65] to stabilize its activation scale before fusion.

These streams are integrated via a fusion matrix $W_c$

$$Z = \left[p_1, p_2\right], \quad z = W_c Z. \tag{17}$$

The dual-stream architecture applies TMI to the original features, $G_1 = \text{TMI}(x_e)$, and to the attention-enhanced features, $L_2 = \text{TMI}(\hat{x}_e)$. The first stream preserves the broader spectral–spatial context, and the second emphasizes selected features. Concatenation retains the two streams as separate channel groups and allows the downstream layer to learn their relative contribution.

During training, we regularize $W_c$ with the nuclear norm. The continuous-time expression below represents gradient descent on the regularized objective

$$\frac{dW_c}{dt} = -\eta\nabla_{W_c}\left(\mathcal{L}_{\text{task}} + \alpha\|W_c\|_*\right) \tag{18}$$

where $\|\cdot\|_*$ penalizes the sum of singular values and encourages a small number of dominant response modes. In the implementation, $W_c$ is the weight matrix of the $1 \times 1$ fusion convolution, and the nuclear-norm term is applied to this matrix during training. We use $\alpha = 1 \times 10^{-3}$ for all four datasets, selected by the sensitivity analysis in Table XVIII. This regularizer promotes a low effective rank but leaves the inference architecture, parameter count, and FLOPs unchanged; no posttraining factorization or pruning is applied. The final population readout is given by $h = \rho(z)$.

TABLE I
HARDWARE AND SOFTWARE CONFIGURATIONS

| Hardware | | Software | |
|---|---|---|---|
| CPU | Intel(R) Xeon(R) 8255C | OS | Linux Ubuntu |
| RAM | 50 GB | CUDA | 11.2 |
| GPU | NVIDIA RTX 2080 Ti (11GB) | Framework | TensorFlow 2.9.0 / Keras 2.1.6 |
| Platform | Autodl Server | Language | Python 3.8 |

## V. EXPERIMENT

### A. Experimental Environment and Settings

To ensure the reproducibility of the results and maintain experimental consistency, all evaluations in this study were conducted on a standardized hardware and software platform. This unified computational environment minimizes performance fluctuations caused by external variables, providing a stable baseline for comparative analysis. The detailed environment version and parameters are summarized in Table I.

We utilized the Adam optimizer with a batch size of 32, an initial learning rate of $1e^{-3}$, and a decay rate of $1e^{-3}$. Models were trained for 64 epochs, taking into account the training time and the processing capability of the device. To ensure statistical significance, all reported results are the mean of ten independent runs.

### B. Data Description

To ensure reproducibility, we provide detailed descriptions of the four hyperspectral datasets used in this study. The specific class partitions and sample sizes are reported in Table II.

*1) Indian Pines:* Acquired by the AVIRIS sensor over northwestern Indiana, this dataset comprises 145 × 145 pixels with a spatial resolution of 20 m. It includes 220 spectral bands (0.4–2.5 $\mu$m) and covers 16 land-cover classes. The scene is predominantly agricultural, featuring crops with similar spectral signatures, making it a standard benchmark for fine-grained classification.

*2) DFC2013 Houston:* Released for the 2013 IEEE GRSS Data Fusion Contest, this dataset was captured by the ITRES CASI sensor over the University of Houston campus. It consists of 144 spectral bands with a high spatial resolution of 2.5 m. The 15 land-cover classes include complex urban structures, presenting challenges related to shadow effects and intraclass variability of urban materials.

*3) WHU-Hi-HongHu:* Provided by the RSIDEA Research Group at Wuhan University, this UAV-borne dataset focuses on precision agriculture in Hubei Province, China. It contains 270 spectral bands and a massive 940 475 pixels across 22 classes. The dataset's high spatial resolution and diverse crop types make it an ideal testbed for evaluating model scalability and generalizability.

*4) Trento:* Acquired by the AISA Eagle sensor over a suburban area in Trento, Italy, this dataset features 63 spectral bands (0.402–0.989 $\mu$m) and 1-m spatial resolution. Although it has fewer bands, the spectral similarity between vegetation

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TABLE II
SAMPLE DISTRIBUTION AND CLASS PARTITION FOR HSI DATASETS

| INDIAN PINES | | DFC2013 HOUSTON | | TRENTO | | WHU-HI-HONGHU | |
|---|---|---|---|---|---|---|---|
| CLASS | SAMPLES | CLASS | SAMPLES | CLASS | SAMPLES | CLASS | SAMPLES |
| ALFALFA | 46 | HEALTHY-GR | 1251 | APPLE-TREES | 4030 | RED-ROOF | 14041 |
| CORN-NOTILL | 1428 | STRESSED-GR | 1254 | BUILDINGS | 2903 | ROAD | 3512 |
| CORN-MINTILL | 830 | SYNTHETIC-GR | 697 | GROUND | 552 | BARE-SOIL | 21821 |
| CORN | 237 | TREES | 1244 | WOODS | 9199 | COTTON | 163258 |
| GR-PASTURE | 483 | SOIL | 1242 | VINEYARDS | 10501 | CO-FI | 6218 |
| GR-TREES | 730 | WATER | 1225 | ROADS | 3205 | RAPE | 44557 |
| GR-PA-MO | 28 | RESIDENTIAL | 1268 | | | CH-CA | 24103 |
| HAY-WI | 478 | COMMERCIAL | 1244 | | | PAKCHOI | 4054 |
| OATS | 20 | ROAD | 1252 | | | CABBAGE | 10819 |
| SO-NO | 974 | HIGHWAY | 1227 | | | TU-MU | 12394 |
| SO-MI | 2455 | RAILWAY | 1235 | | | BR-PA | 11015 |
| SO-CL | 593 | PA-LOT 1 | 1233 | | | BR-CH | 8954 |
| WHEAT | 205 | PA-LOT 2 | 469 | | | SM-BR-CH | 22507 |
| WOODS | 1265 | TE-CO | 428 | | | LA-SATIVA | 7356 |
| BL-GR-TR-DR | 386 | RU-TR | 660 | | | CELTUCE | 1002 |
| ST-ST-TO | 93 | | | | | FI-CO-LE | 7262 |
| | | | | | | RO-LE | 3010 |
| | | | | | | CARROT | 3217 |
| | | | | | | WH-RA | 8712 |
| | | | | | | GA-SP | 3486 |
| | | | | | | BR-BE | 1328 |
| | | | | | | TREE | 4040 |

types (e.g., vineyards versus apple trees) poses a significant challenge for classification models.

### C. Evaluation Metrics

Performance is evaluated using three complementary metrics: OA, AA, and the Cohen Kappa coefficient ($\kappa$).

The average classification accuracy for each category is expressed in terms of the AA. It is calculated as follows:

$$\mathrm{AA} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{Recall}_i \tag{19}$$

where $Recall_i$ denotes the classification accuracy of category $i$ and $n$ denotes the number of categories.

OA is the proportion of correctly classified samples in the multiclass confusion matrix

$$\mathrm{OA} = \frac{\sum_{i=1}^{r} x_{ii}}{N}. \tag{20}$$

Here, $x_{ii}$ is the number of samples from class $i$ assigned to class $i$, $r$ is the number of classes, and $N$ is the total number of test samples.

Cohen's kappa is computed from the multiclass confusion matrix as

$$\kappa = \frac{N\sum_{i=1}^{r} x_{ii} - \sum_{i=1}^{r} x_{i+}x_{+i}}{N^2 - \sum_{i=1}^{r} x_{i+}x_{+i}}. \tag{21}$$

In this expression, $x_{i+}$ and $x_{+i}$ are the row and column marginals for class $i$, respectively.

### D. Comparison With State-of-the-Art Methods

To further evaluate the effectiveness of LizardNet, it was compared with 14 representative hyperspectral image classification methods, including traditional approaches, CNN-based models, and Transformer-based techniques: DGPF-RENet [66], Cnn2D [67], DBDA [68], DwT [69], FDSSC [70], Hamida [71], He [33], Lee [72], RvT [73], CS2DT [74], DSFormer [75], HyperKAN [76], Swin-HSI [77], and MambaHSI [78]. The classification performance was assessed using four standard HSI datasets: IP, DFC2013 Houston, WHU-Hi-HongHu, and Trento. All reported results represent the mean of ten experimental trials, rounded to two decimal places.

*1) Results on the IP:* The IP dataset contains a limited and imbalanced set of labeled samples across 16 land-cover categories. We evaluate LizardNet at training proportions of 5%, 10%, 15%, and 20% to measure its behavior as the amount of supervision increases. Tables III and IV report the results.

The 5% IP split contains only one training sample for class 7 (grass-pasture-mowed) and two for class 9 (oats). Table III shows that DBDA and Hamida obtain 0.00% accuracy for some classes under this split, and Cnn2D and He also give lower classwise results. LizardNet maintains higher aggregate OA, AA, and $\kappa$ under the same sampling protocol.

Based on the quantitative results across all training ratios, we observe that the proposed LizardNet model consistently achieves the highest classification performance among the compared deep-learning-based methods.

1) *Few-Shot Regimes (5% and 10%):* At 5% training, LizardNet obtains 94.63% OA, 89.26% AA, and 93.87% $\kappa$, the highest values among the compared methods in this setting. At 10%, its OA and AA increase to 98.11% and 95.19%, respectively.
2) *Higher Data Regimes (15% and 20%):* LizardNet obtains 99.29% OA at 15% training and 99.48% OA at 20%. Under the 20% setting, eight of the 16 classes reach 100.00% classwise accuracy (classes 1, 4, 7, 8, 9, 13, 14, and 16).

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TABLE III

CLASSIFICATION RESULTS ON THE IP DATASET (PART 1: 5% AND 10%)

| TRAIN SAMPLE | CLASS | MODEL | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D | DBDA | DWT | FDSSC | HAMIDA | HE | LEE | RVT | CS2DT | DSFORMER | HYPERKAN | SWIN-HSI | MAMBAHSI | DGPF-RENET | OURS |
| 5% | 1 | 7.75 | 66.67 | 50.00 | **98.85** | 10.53 | 50.85 | 30.30 | 73.24 | 50.00 | 100.00 | 74.42 | 14.12 | 52.38 | 62.95 | 81.82 |
| | 2 | 81.38 | 75.81 | 54.20 | 85.55 | 76.91 | 80.13 | 58.32 | 83.58 | 75.99 | 55.11 | 97.69 | 76.25 | 94.15 | 91.06 | **92.72** |
| | 3 | 87.26 | 74.12 | 63.73 | 74.06 | 69.58 | 72.62 | 59.29 | 81.40 | 66.87 | 72.44 | 77.65 | 80.41 | 88.35 | **92.22** | 92.18 |
| | 4 | 71.29 | 85.89 | 55.12 | 76.80 | 65.46 | 76.46 | 60.14 | 84.43 | 94.19 | 96.62 | 82.49 | 40.17 | 88.95 | 91.68 | 91.15 |
| | 5 | 93.05 | 92.45 | 83.55 | 95.97 | 90.13 | 93.48 | 82.30 | 93.70 | **98.45** | 88.52 | 81.82 | 87.16 | 86.76 | 94.55 | 89.61 |
| | 6 | 96.83 | 97.52 | 93.53 | 96.11 | 96.31 | 97.03 | 91.19 | 93.84 | 93.04 | **99.07** | 98.65 | 95.39 | 98.91 | 95.89 | 97.41 |
| | 7 | 68.75 | 77.14 | 77.14 | 93.10 | 7.14 | 61.54 | 71.19 | 77.42 | 92.86 | **100.00** | 81.48 | 7.42 | 43.75 | 60.37 | 75.56 |
| | 8 | 62.78 | 97.74 | 97.19 | 99.23 | 92.37 | 96.58 | 95.56 | 96.78 | 89.85 | **100.00** | 95.18 | 95.81 | 99.28 | 99.63 | 99.78 |
| | 9 | 37.33 | 0.00 | 18.99 | 10.00 | 0.00 | 64.29 | 64.29 | 90.00 | 43.75 | **100.00** | 73.68 | 21.36 | 15.62 | 68.42 | 43.16 |
| | 10 | 78.38 | 75.58 | 64.06 | 87.47 | 76.45 | 74.86 | 74.71 | 86.55 | 66.34 | 70.75 | 91.86 | 67.89 | 87.49 | 92.63 | **93.10** |
| | 11 | 80.39 | 78.72 | 72.35 | 87.55 | 82.34 | 81.32 | 56.80 | 89.94 | 72.80 | 65.28 | 93.15 | 87.14 | 97.66 | 96.37 | **96.50** |
| | 12 | 77.45 | 70.00 | 49.41 | 91.40 | 73.35 | 70.28 | 47.70 | 77.40 | 66.55 | 74.87 | 92.59 | 45.28 | 84.93 | 91.56 | **91.67** |
| | 13 | 93.27 | 97.01 | 94.29 | 83.33 | 99.74 | 100.00 | 97.24 | 97.99 | 100.00 | 100.00 | 98.40 | 96.52 | 93.91 | 96.46 | 98.26 |
| | 14 | 62.08 | 94.61 | 90.71 | 94.23 | 96.29 | 94.07 | 94.17 | 89.63 | 91.74 | 92.27 | **99.74** | 96.35 | 99.14 | 98.94 | 97.74 |
| | 15 | 87.25 | 85.34 | 76.89 | 83.24 | 85.30 | 72.94 | 72.83 | 69.13 | 88.88 | 95.37 | 60.23 | 57.19 | 86.54 | **97.38** | 93.95 |
| | 16 | 68.22 | 91.19 | 76.52 | 89.34 | 85.16 | 80.77 | 96.17 | 96.55 | 97.67 | **100.00** | 94.12 | 83.47 | 93.75 | 74.38 | 93.60 |
| | OA | 75.89 | 82.69 | 72.15 | 88.20 | 82.93 | 82.97 | 70.51 | 87.11 | 77.54 | 76.69 | 91.61 | 80.44 | 93.35 | 94.44 | **94.63** |
| | AA | 72.09 | 78.74 | 69.86 | 84.14 | 69.19 | 79.20 | 72.01 | 86.35 | 80.56 | 88.14 | 87.07 | 65.75 | 81.97 | 87.78 | **89.26** |
| | κ | 73.18 | 80.48 | 68.68 | 86.65 | 80.43 | 80.51 | 66.63 | 85.34 | 73.90 | 73.65 | 90.50 | 77.41 | 92.40 | 93.66 | **93.87** |
| | TRAIN (S) | 4.53 | 18.46 | 38.95 | 68.85 | 41.87 | 261.16 | 57.46 | 27.95 | 4.98 | 15.03 | 13.45 | 18.83 | 8.60 | 53.09 | 11.42 |
| | TEST (S) | 1.13 | 33.27 | 31.64 | 51.20 | 29.42 | 26.89 | 43.33 | 6.58 | 8.88 | 5.96 | 1.05 | 1.00 | 22.88 | 4.53 | 1.78 |
| 10% | 1 | 72.57 | 97.50 | 39.64 | **98.77** | 91.36 | 63.33 | 75.00 | 79.07 | 96.15 | 98.39 | 85.37 | 39.52 | 21.95 | 85.71 | 96.10 |
| | 2 | 87.93 | 88.03 | 71.29 | 93.91 | 93.77 | 82.44 | 82.35 | 88.17 | 97.97 | 73.46 | **98.54** | 78.41 | 91.69 | 96.40 | 97.57 |
| | 3 | 88.56 | 79.84 | 65.16 | 93.06 | 89.61 | 82.94 | 77.92 | 95.68 | 95.71 | 91.30 | 76.22 | 81.33 | 95.33 | 96.47 | **97.66** |
| | 4 | 95.13 | 94.88 | 52.29 | 87.69 | 92.53 | 83.25 | 70.46 | 90.21 | 97.75 | 99.44 | 86.89 | 69.45 | 86.44 | 95.70 | **99.30** |
| | 5 | 94.93 | 94.57 | 81.69 | 93.79 | 95.14 | 96.72 | 91.30 | 72.57 | **98.53** | 90.66 | 79.71 | 83.61 | 92.40 | 96.28 | 95.66 |
| | 6 | 96.44 | 98.57 | 98.41 | 98.50 | **99.62** | 98.20 | 94.75 | 96.48 | 98.27 | 99.25 | 99.21 | 99.14 | 98.96 | 97.63 | 99.16 |
| | 7 | 19.16 | 64.94 | 86.27 | 53.19 | 80.00 | 92.00 | 81.82 | 89.29 | 95.83 | 100.00 | 100.00 | 76.47 | 1.04 | 87.69 | 85.60 |
| | 8 | 95.38 | 100.00 | 97.95 | 98.51 | 99.31 | 97.49 | 97.39 | 97.56 | 97.28 | 100.00 | 95.42 | 100.00 | 99.94 | 99.77 | **100.00** |
| | 9 | 82.93 | 65.45 | 27.91 | 87.80 | 61.54 | 94.44 | 94.44 | 75.00 | 100.00 | 100.00 | 88.89 | 0.00 | 0.00 | 90.00 | 66.67 |
| | 10 | 80.62 | 87.29 | 65.08 | 88.82 | 90.48 | 86.53 | 83.56 | 90.64 | 90.76 | 80.37 | 92.52 | 85.42 | 90.79 | 97.03 | **97.21** |
| | 11 | 63.66 | 88.38 | 76.39 | 88.99 | 93.70 | 88.19 | 85.78 | 94.36 | 98.14 | 76.28 | 94.02 | 91.56 | 96.69 | 98.10 | **98.75** |
| | 12 | 63.28 | 91.89 | 54.80 | 79.52 | 93.01 | 88.91 | 76.41 | 86.32 | 94.86 | 84.24 | 91.83 | 63.84 | 92.44 | 95.66 | **96.35** |
| | 13 | 96.35 | 92.50 | 98.11 | 98.14 | 99.46 | 100.00 | 98.40 | 97.35 | 99.72 | 100.00 | 100.00 | 97.49 | 92.90 | 98.38 | 98.53 |
| | 14 | 96.24 | 97.97 | 90.61 | 96.36 | 98.33 | 97.75 | 96.38 | 90.07 | 96.19 | 96.80 | 100.00 | 99.27 | 98.96 | **99.53** | 99.51 |
| | 15 | 93.72 | 93.22 | 79.16 | 90.20 | 94.10 | 88.63 | 82.35 | 91.01 | 96.99 | **98.62** | 61.19 | 84.62 | 96.16 | 98.30 | 98.47 |
| | 16 | 97.08 | 87.50 | 92.31 | 96.39 | 98.25 | 99.41 | 89.53 | 97.11 | 94.95 | **100.00** | 97.56 | 90.71 | 96.08 | 87.74 | 96.43 |
| | OA | 82.53 | 90.87 | 76.57 | 94.93 | 94.41 | 89.84 | 86.54 | 91.39 | 96.63 | 85.67 | 92.15 | 87.29 | 94.37 | 97.39 | **98.11** |
| | AA | 82.75 | 88.91 | 73.57 | 90.23 | 91.89 | 90.01 | 86.12 | 89.43 | 96.82 | 93.05 | 90.46 | 77.55 | 78.24 | 95.02 | **95.19** |
| | κ | 80.63 | 89.61 | 73.46 | 90.59 | 93.62 | 88.37 | 84.62 | 90.18 | 96.15 | 83.66 | 91.12 | 85.46 | 93.58 | 97.02 | **97.85** |
| | TRAIN (S) | 9.39 | 37.50 | 77.14 | 141.74 | 83.59 | 514.22 | 110.61 | 59.35 | 14.69 | 27.53 | 20.19 | 29.87 | 8.63 | 110.70 | 31.11 |
| | TEST (S) | 1.17 | 34.33 | 31.78 | 53.29 | 29.29 | 26.68 | 41.67 | 6.72 | 8.03 | 5.98 | 1.05 | 0.94 | 22.68 | 4.23 | 1.65 |

TABLE IV

CLASSIFICATION RESULTS ON THE IP DATASET (PART 2: 15% AND 20%)

| TRAIN SAMPLE | CLASS | MODEL | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D | DBDA | DWT | FDSSC | HAMIDA | HE | LEE | RVT | CS2DT | DSFORMER | HYPERKAN | SWIN-HSI | MAMBAHSI | DGPF-RENET | OURS |
| 15% | 1 | 98.70 | 96.00 | 93.33 | 91.67 | 83.58 | 67.80 | 47.62 | 79.41 | 100.00 | 98.39 | 94.74 | 60.24 | 27.35 | 100.00 | **97.50** |
| | 2 | 97.46 | 96.27 | 96.61 | 99.42 | 95.35 | 96.47 | 86.25 | 95.72 | 99.47 | 83.41 | 98.55 | 85.71 | 98.08 | **99.56** | 99.18 |
| | 3 | 99.29 | **99.86** | 96.97 | 98.47 | 96.11 | 96.07 | 86.60 | 98.31 | 98.88 | 93.51 | 77.97 | 91.86 | 94.46 | 96.99 | **98.73** |
| | 4 | 98.04 | 99.01 | 98.75 | 99.50 | 93.81 | 93.26 | 69.32 | 94.30 | 99.11 | 100.00 | 85.64 | 86.13 | 93.67 | 100.00 | **100.00** |
| | 5 | 99.63 | 99.16 | 98.15 | 97.27 | 97.54 | 97.01 | 89.88 | 98.52 | 99.56 | 95.93 | 79.60 | 93.88 | 90.75 | 93.78 | **99.27** |
| | 6 | 99.52 | **99.68** | 99.36 | 98.96 | 98.89 | 99.44 | 97.63 | 98.94 | 98.51 | 99.53 | 99.33 | 97.82 | 98.43 | 99.32 | **99.36** |
| | 7 | 100.00 | 82.93 | 97.87 | 40.00 | 97.87 | 97.96 | 75.00 | 97.87 | 100.00 | 100.00 | 100.00 | 83.87 | 2.90 | 100.00 | **100.00** |
| | 8 | 100.00 | 100.00 | 99.88 | 100.00 | 98.90 | 97.36 | 94.78 | 98.90 | 99.84 | 100.00 | 95.17 | 96.88 | 100.00 | 100.00 | **100.00** |
| | 9 | 100.00 | 96.97 | 93.75 | 0.00 | 78.57 | 90.32 | 64.00 | 97.14 | 100.00 | 100.00 | 100.00 | 66.67 | 0.00 | 100.00 | **100.00** |
| | 10 | 98.66 | 91.36 | 97.28 | 99.40 | 93.78 | 93.04 | 87.87 | 96.14 | 97.02 | 87.64 | 91.73 | 87.94 | 90.91 | **99.87** | 99.52 |
| | 11 | 98.67 | 95.16 | 98.60 | **99.64** | 96.76 | 95.22 | 89.59 | 98.50 | 98.89 | 85.31 | 94.54 | 93.12 | 98.04 | **99.49** | 98.71 |
| | 12 | 98.51 | 98.18 | 93.07 | 99.70 | 94.75 | 96.82 | 79.42 | 95.30 | 96.82 | 93.64 | 92.20 | 80.79 | 96.93 | **100.00** | 99.60 |
| | 13 | 100.00 | 100.00 | 98.58 | 99.13 | 100.00 | 98.58 | 98.56 | 98.86 | 100.00 | 100.00 | 100.00 | 100.00 | 94.22 | 99.39 | **100.00** |
| | 14 | 99.86 | 99.68 | 98.89 | 99.81 | 98.54 | 98.00 | 93.68 | 98.35 | 99.44 | 97.83 | 99.90 | 96.84 | 99.31 | 99.90 | **100.00** |
| | 15 | 99.85 | 100.00 | 96.67 | 99.85 | 94.50 | 93.60 | 68.16 | 93.72 | 99.27 | 98.82 | 60.88 | 83.92 | 95.60 | 99.03 | **99.70** |
| | 16 | 96.93 | 96.89 | 98.06 | 98.11 | 98.75 | 98.75 | 98.14 | 94.04 | 98.36 | 100.00 | 96.10 | 93.81 | 94.09 | 90.54 | **98.77** |
| | OA | 98.90 | 97.14 | 97.75 | 99.12 | 96.48 | 96.09 | 88.50 | 97.34 | 98.81 | 91.07 | 92.36 | 91.20 | 95.89 | 99.07 | **99.29** |
| | AA | 99.07 | 96.95 | 97.24 | 88.81 | 94.86 | 94.36 | 82.91 | 95.88 | 99.07 | 95.88 | 91.65 | 87.47 | 79.67 | 98.62 | **99.40** |
| | κ | 98.74 | 96.75 | 97.44 | 98.99 | 95.98 | 95.53 | 86.80 | 96.96 | 98.64 | 89.70 | 91.35 | 89.92 | 95.31 | 98.94 | **99.19** |
| | TRAIN (S) | 73.43 | 132.67 | 1117.64 | 342.73 | 128.92 | 266.71 | 211.83 | 123.38 | 98.46 | 41.75 | 27.08 | 44.81 | 8.07 | 301.05 | 38.14 |
| | TEST (S) | 0.17 | 1.20 | 129.62 | 0.43 | 0.23 | 0.48 | 0.57 | 2.37 | 6.54 | 6.51 | 1.08 | 0.85 | 20.94 | 5.64 | 1.43 |
| 20% | 1 | 96.10 | 100.00 | 95.77 | 100.00 | 95.77 | 91.18 | 81.69 | 98.63 | 100.00 | 98.39 | 97.30 | 69.81 | 46.30 | 100.00 | **100.00** |
| | 2 | 98.99 | 98.95 | 99.43 | 99.52 | 97.67 | 95.40 | 90.48 | 97.91 | 98.31 | 91.40 | 98.55 | 90.82 | 96.60 | 98.51 | **99.48** |
| | 3 | 98.65 | 99.47 | **99.85** | 98.72 | 96.30 | 95.40 | 88.99 | 99.40 | 99.59 | 94.86 | 77.80 | 92.75 | 96.95 | 98.80 | **99.55** |
| | 4 | 98.17 | 99.22 | 100.00 | 99.22 | 97.93 | 94.59 | 84.62 | 99.21 | 98.61 | 99.57 | 85.33 | 94.84 | 93.44 | 100.00 | **100.00** |
| | 5 | 98.69 | 100.00 | 99.74 | 99.36 | 98.18 | 99.09 | 88.78 | 98.46 | 98.83 | 98.21 | 83.47 | 95.89 | 93.47 | 98.45 | **98.45** |
| | 6 | 100.00 | 99.83 | 99.74 | 98.97 | 99.83 | 99.74 | 98.36 | 98.81 | 99.00 | 99.75 | 99.65 | 97.85 | 99.42 | 99.32 | **99.49** |
| | 7 | 97.78 | 100.00 | 100.00 | 97.67 | 92.68 | 100.00 | 97.67 | 87.18 | 100.00 | 100.00 | 100.00 | 85.71 | 0.00 | 81.82 | **100.00** |
| | 8 | 100.00 | 100.00 | 100.00 | 100.00 | 99.74 | 99.48 | 98.18 | 100.00 | 100.00 | 100.00 | 95.15 | 97.92 | 100.00 | 100.00 | **100.00** |
| | 9 | 100.00 | 96.77 | 100.00 | 0.00 | 86.67 | 100.00 | 90.32 | 64.00 | 100.00 | 100.00 | 93.75 | 60.87 | 0.00 | 100.00 | **100.00** |
| | 10 | 99.81 | 98.98 | 98.72 | **99.87** | 95.87 | 96.15 | 90.13 | 98.52 | 98.09 | 92.02 | 92.44 | 90.76 | 91.68 | 98.20 | **99.74** |
| | 11 | 99.52 | 99.49 | 99.49 | 99.62 | 98.25 | 96.93 | 91.42 | 98.57 | **99.66** | 83.00 | 94.64 | 95.83 | 98.90 | 99.69 | 98.57 |
| | 12 | 99.15 | 99.79 | 99.58 | 99.26 | 96.36 | 97.90 | 86.49 | 95.91 | 98.62 | 96.62 | 92.39 | 89.74 | 95.74 | **100.00** | 99.37 |
| | 13 | 100.00 | 99.39 | 99.70 | 100.00 | 99.69 | 100.00 | 99.09 | 100.00 | 99.60 | 100.00 | 100.00 | 100.00 | 94.65 | 100.00 | **100.00** |
| | 14 | 99.80 | 99.75 | 99.70 | 99.95 | 98.87 | 98.92 | 95.54 | 98.62 | 99.74 | 98.21 | 99.90 | 97.89 | 99.30 | 100.00 | **100.00** |
| | 15 | 99.36 | 99.02 | 98.20 | **100.00** | 96.26 | 95.81 | 84.46 | 95.24 | 99.79 | 99.81 | 61.67 | 88.75 | 95.97 | **100.00** | 99.68 |
| | 16 | 97.33 | 98.63 | 99.33 | 97.33 | 99.33 | 98.01 | 96.10 | 97.30 | 99.17 | 92.95 | **100.00** | 91.89 | 94.14 | 94.59 | **100.00** |
| | OA | 99.37 | 99.43 | 99.37 | 99.43 | 97.84 | 97.21 | 91.63 | 98.30 | 99.16 | 92.39 | 92.71 | 94.37 | 96.39 | 99.23 | **99.48** |
| | AA | 98.96 | 99.33 | 99.33 | 93.09 | 96.84 | 97.41 | 91.40 | 95.49 | 99.31 | 96.55 | 92.00 | 90.08 | 81.04 | 98.09 | **99.64** |
| | κ | 99.28 | 99.35 | 99.28 | 99.35 | 97.54 | 96.82 | 90.46 | 98.07 | 99.04 | 91.22 | 91.75 | 93.57 | 95.87 | 99.12 | **99.40** |
| | TRAIN (S) | 98.33 | 175.61 | 1475.39 | 415.43 | 159.35 | 318.36 | 248.47 | 147.48 | 146.20 | 50.65 | 34.57 | 56.55 | 8.14 | 415.36 | 47.95 |
| | TEST (S) | 0.17 | 1.25 | 124.66 | 0.44 | 0.23 | 0.48 | 0.65 | 2.28 | 5.63 | 6.51 | 1.26 | 0.83 | 20.85 | 5.30 | 1.52 |

At 5% training, LizardNet requires 11.42 s for training and 1.78 s for testing. Its training time is 47.95 s at the 20% setting. Under the corresponding protocols, DwT, FDSSC, and He require 1475.39, 415.43, and 318.36 s for training,

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TABLE V
CLASSIFICATION RESULTS ON THE DFC2013 HOUSTON DATASET (PART 1: 1%, 5%, AND 10%)

| TRAIN SAMPLE | CLASS | MODEL | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D | DBDA | DWT | FDSSC | HAMIDA | HE | LEE | RVT | CS2DT | DSFORMER | HYPERKAN | SWIN-HSI | MAMBAHSI | DGPF-RENET | OURS |
| 1% | 1 | 90.62 | 87.61 | 89.81 | 87.73 | 88.64 | 91.63 | 87.77 | 92.36 | 85.25 | 42.63 | **95.00** | 94.12 | 92.86 | 81.91 | 91.03 |
| | 2 | 87.30 | 88.82 | 91.56 | 91.22 | 87.40 | 80.03 | 83.87 | 88.00 | 91.46 | 50.52 | 85.05 | **94.38** | 93.33 | 89.20 | 93.31 |
| | 3 | **100.00** | 99.93 | 99.49 | 99.35 | 78.57 | 98.16 | 75.40 | 91.73 | 98.68 | 96.58 | 97.98 | 96.15 | 99.32 | 99.13 | 98.84 |
| | 4 | 93.27 | **96.55** | 95.24 | 95.64 | 88.76 | 84.62 | 92.23 | 93.32 | 90.81 | 91.09 | 95.95 | 88.64 | 94.81 | 81.74 | 96.59 |
| | 5 | 96.81 | 96.20 | 98.06 | 98.15 | 95.75 | 91.05 | 96.14 | 96.84 | 96.19 | **100.00** | 98.13 | 98.37 | 99.43 | 98.13 | 97.32 |
| | 6 | 89.18 | 94.43 | 85.82 | 86.22 | 71.51 | 74.34 | 85.22 | 64.33 | **100.00** | 81.59 | 87.62 | 78.19 | 83.70 | 76.40 | 84.78 |
| | 7 | **88.29** | 85.69 | 87.51 | 83.85 | 74.68 | 78.27 | 77.66 | 76.94 | 79.99 | 70.95 | 61.21 | 74.56 | 84.62 | 84.22 | 73.47 |
| | 8 | 66.04 | 70.55 | 73.47 | 79.56 | 70.05 | 58.49 | 56.47 | 75.09 | **88.32** | 10.09 | 58.43 | 76.48 | 54.13 | 42.78 | 72.24 |
| | 9 | 77.91 | 81.11 | **85.31** | 83.84 | 72.45 | 74.61 | 62.34 | 81.10 | 71.26 | 70.97 | 43.32 | 72.91 | 82.87 | 56.50 | 74.74 |
| | 10 | 73.83 | 57.55 | 83.53 | 72.18 | 67.52 | 68.49 | 44.63 | 67.37 | 70.38 | 12.04 | 73.21 | 74.63 | **84.82** | 47.90 | 75.31 |
| | 11 | 78.25 | 76.43 | 69.08 | 79.48 | 65.58 | 59.30 | 60.44 | 67.75 | 66.98 | 60.33 | **98.61** | 82.47 | 84.23 | 47.75 | 77.02 |
| | 12 | 78.58 | 77.58 | 77.91 | 67.56 | 68.74 | 62.57 | 50.38 | 75.64 | 63.56 | 42.07 | 74.00 | 85.92 | 70.61 | **97.13** | 88.94 |
| | 13 | 75.84 | 79.19 | 79.53 | 21.97 | 83.90 | 51.56 | 34.55 | 61.43 | **96.07** | 91.29 | 76.39 | 23.64 | 77.66 | 0.00 | 92.46 |
| | 14 | 94.81 | **100.00** | 99.18 | 97.58 | 78.49 | 76.15 | 96.45 | 97.34 | 96.97 | 97.61 | 98.12 | 96.72 | 99.52 | 93.63 | 98.58 |
| | 15 | 99.09 | 97.76 | 94.24 | 99.15 | 95.80 | 98.56 | 95.37 | 97.61 | 95.22 | 100.00 | 100.00 | 94.95 | 95.06 | 100.00 | **100.00** |
| | OA | 85.13 | 84.07 | 86.08 | 84.44 | 78.85 | 76.88 | 72.80 | 82.27 | 81.38 | 61.90 | 80.83 | 84.52 | 85.59 | 73.60 | **86.34** |
| | AA | 85.99 | 85.96 | 87.32 | 82.90 | 79.19 | 76.52 | 73.26 | 81.79 | 86.08 | 67.85 | 82.87 | 82.14 | 86.46 | 73.09 | **87.64** |
| | K | 83.92 | 82.76 | 84.97 | 83.14 | 77.11 | 74.98 | 70.60 | 80.80 | 79.85 | 58.95 | 79.28 | 83.21 | 84.43 | 71.41 | **85.24** |
| | TRAIN (S) | 40.77 | 55.49 | 122.10 | 134.10 | 71.86 | 127.84 | 145.95 | 66.55 | 11.55 | 11.16 | 2.58 | 8.84 | 138.72 | 23.29 | 13.22 |
| | TEST (S) | 5.09 | 44.50 | 4074.61 | 10.53 | 7.65 | 14.99 | 21.88 | 71.79 | 10.14 | 208.05 | 39.72 | 1.48 | 723.39 | 6.45 | 3.84 |
| 5% | 1 | 91.17 | 89.75 | 91.56 | 86.34 | 97.58 | **99.66** | 91.10 | 88.17 | 91.99 | 86.59 | 99.27 | 96.48 | 92.43 | 95.93 | 97.85 |
| | 2 | 91.76 | 89.82 | 92.74 | 90.85 | 98.12 | **99.12** | 90.18 | 91.76 | 96.14 | 99.04 | 87.09 | 98.24 | 97.72 | 92.33 | 98.36 |
| | 3 | 98.78 | 100.00 | 88.09 | 99.85 | 99.62 | 100.00 | 99.02 | 96.99 | 99.92 | 96.68 | 98.98 | 99.18 | 99.09 | 99.64 | 99.59 |
| | 4 | 96.41 | 96.19 | 94.51 | 88.27 | 97.74 | **99.70** | 96.43 | 90.92 | 93.95 | 99.12 | 97.79 | 96.72 | 95.68 | 95.65 | 96.50 |
| | 5 | 99.16 | 99.92 | 94.93 | 99.08 | 97.94 | 97.86 | 96.91 | 98.23 | 95.56 | 99.71 | 98.20 | **100.00** | 99.77 | 91.20 | 99.99 |
| | 6 | 52.31 | 97.27 | 40.29 | 99.02 | 94.25 | 92.70 | 92.33 | 92.88 | **99.63** | 98.36 | 86.25 | 99.14 | 98.53 | 91.20 | 95.91 |
| | 7 | 86.40 | 91.73 | 80.14 | 89.32 | 89.56 | 92.56 | 68.62 | 91.16 | 88.78 | 92.86 | 84.91 | 93.41 | **93.55** | 88.40 | 92.91 |
| | 8 | 82.69 | 89.33 | 62.60 | 84.98 | **92.47** | 89.30 | 76.37 | 84.80 | 87.72 | 73.32 | 71.14 | 85.57 | 72.33 | 80.47 | 89.92 |
| | 9 | 88.02 | 89.66 | 76.40 | 90.83 | 88.79 | 89.20 | 67.07 | **91.62** | 85.51 | 84.57 | 86.75 | 89.69 | 89.34 | 83.23 | 83.65 |
| | 10 | 81.83 | 82.36 | 39.90 | 69.14 | 90.85 | 92.37 | 80.31 | 87.33 | 86.85 | 91.16 | 96.19 | 96.72 | 93.33 | **97.24** | 94.59 |
| | 11 | 86.88 | 91.65 | 61.19 | 92.03 | 86.57 | 92.23 | 74.87 | 85.55 | 96.03 | 87.97 | 94.48 | 96.54 | 88.90 | **96.79** | 93.21 |
| | 12 | 78.58 | 90.21 | 65.31 | 81.52 | 93.36 | 91.18 | 64.78 | 87.73 | 84.64 | 89.52 | 87.87 | 91.68 | 92.05 | 89.22 | **97.19** |
| | 13 | 92.69 | 89.83 | 62.72 | 72.14 | 92.69 | 94.27 | 75.23 | 89.94 | 91.60 | 95.70 | 93.28 | 94.45 | **95.91** | 78.54 | 89.10 |
| | 14 | 99.88 | 100.00 | 95.97 | 99.88 | 97.93 | 98.76 | 96.98 | 97.82 | 95.93 | 99.60 | 100.00 | 99.12 | 98.00 | 100.00 | 99.98 |
| | 15 | 97.06 | 99.29 | 95.51 | 97.74 | 99.52 | 99.21 | 98.27 | 96.68 | 94.59 | 100.00 | 100.00 | 99.18 | 98.00 | 100.00 | 99.95 |
| | OA | 87.98 | 92.06 | 76.64 | 88.28 | 94.00 | 94.91 | 83.11 | 90.66 | 91.64 | 91.64 | 91.45 | 95.02 | 92.64 | 92.42 | **94.92** |
| | AA | 88.24 | 93.13 | 76.12 | 89.40 | 94.47 | 95.21 | 84.56 | 91.44 | 92.59 | 92.95 | 92.15 | 95.74 | 93.64 | 91.99 | **95.25** |
| | K | 87.04 | 91.42 | 74.85 | 87.35 | 93.52 | 94.45 | 81.74 | 89.91 | 90.97 | 90.96 | 90.76 | 94.57 | 92.04 | 91.80 | **94.51** |
| | TRAIN (S) | 5.38 | 22.21 | 54.26 | 58.61 | 330.98 | 355.41 | 113.83 | 39.28 | 6.99 | 28.71 | 4.87 | 16.32 | 141.95 | 16.77 | 25.53 |
| | TEST (S) | 36.99 | 1003.03 | 1345.14 | 1661.38 | 909.32 | 856.80 | 1344.07 | 217.09 | 9.55 | 189.31 | 42.66 | 1.41 | 801.26 | 5.16 | 2.33 |
| 10% | 1 | 85.93 | 90.58 | 85.78 | 96.24 | 97.72 | 98.18 | 96.11 | 97.45 | 95.33 | 97.40 | 98.91 | 99.16 | **99.28** | 98.21 | 98.92 |
| | 2 | 89.51 | 93.11 | 92.96 | 98.79 | 98.14 | 98.62 | 98.80 | 97.15 | 98.34 | 98.55 | 98.24 | 99.23 | 98.56 | 97.40 | **99.67** |
| | 3 | 97.89 | 99.84 | 98.54 | 99.44 | 99.52 | 100.00 | 100.00 | 98.35 | 99.92 | 98.54 | 99.70 | 100.00 | 99.92 | 99.87 | 99.74 |
| | 4 | 89.60 | 96.65 | 85.22 | 97.34 | 97.49 | 99.16 | 96.49 | 97.04 | 96.21 | **99.23** | 98.82 | 99.14 | 98.76 | 96.95 | 98.82 |
| | 5 | 97.81 | 97.39 | 92.23 | 100.00 | 99.15 | 99.55 | 98.60 | 99.06 | 99.55 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 6 | 89.30 | 98.45 | 88.70 | **99.83** | 93.84 | 93.95 | 97.81 | 95.15 | 99.31 | 99.59 | 98.66 | 100.00 | 98.69 | 97.60 | 98.66 |
| | 7 | 92.00 | 95.07 | 77.84 | **97.29** | 92.82 | 94.85 | 92.63 | 93.44 | 96.52 | 93.73 | 96.39 | 97.46 | 96.94 | 95.76 | 96.39 |
| | 8 | 90.69 | 90.43 | 81.91 | 96.96 | 94.16 | 96.84 | 89.30 | 95.33 | 97.93 | 91.84 | 94.70 | 95.34 | 88.55 | 93.25 | **98.02** |
| | 9 | 92.10 | 90.85 | 79.38 | 96.79 | 93.52 | 94.02 | 87.97 | 92.97 | **97.98** | 86.48 | 94.70 | 96.31 | 91.89 | 90.19 | 94.49 |
| | 10 | 91.28 | 91.43 | 71.13 | 95.68 | 97.29 | 96.82 | 87.78 | 96.49 | 93.73 | 98.30 | 95.07 | 97.27 | 99.35 | **99.51** | 98.89 |
| | 11 | 97.68 | 94.86 | 52.81 | 98.40 | 93.90 | 96.34 | 86.38 | 93.13 | 96.80 | **99.61** | 90.49 | 99.28 | 95.22 | 99.53 | 96.53 |
| | 12 | 96.19 | 94.05 | 71.44 | 96.88 | 95.14 | 96.96 | 90.55 | 93.00 | 93.15 | 94.15 | 96.71 | 98.32 | 95.65 | 96.43 | **98.68** |
| | 13 | 92.32 | 93.60 | 81.79 | 93.51 | 95.84 | 97.01 | 91.32 | 92.23 | 92.91 | 96.02 | 93.55 | 97.16 | 91.52 | 91.66 | **97.23** |
| | 14 | 100.00 | 99.74 | 99.48 | 100.00 | 98.68 | 99.48 | 98.59 | 99.08 | 100.00 | 100.00 | 99.48 | 100.00 | 100.00 | 100.00 | 99.95 |
| | 15 | 98.59 | 97.70 | 94.21 | 97.46 | 99.66 | 98.91 | 98.90 | 95.65 | 99.67 | 100.00 | 100.00 | 100.00 | 98.39 | 100.00 | **100.00** |
| | OA | 92.97 | 94.20 | 82.27 | 97.52 | 96.29 | 97.35 | 93.35 | 95.64 | 96.85 | 96.37 | 96.74 | 97.82 | 96.67 | 96.96 | **98.24** |
| | AA | 93.39 | 94.92 | 83.56 | 97.64 | 96.46 | 97.38 | 94.08 | 95.70 | 97.16 | 96.89 | 97.03 | 98.58 | 96.85 | 97.09 | **98.40** |
| | K | 92.40 | 93.73 | 80.84 | 97.32 | 95.60 | 97.13 | 92.81 | 95.28 | 96.59 | 96.07 | 96.47 | 97.62 | 96.40 | 96.71 | **98.10** |
| | TRAIN (S) | 10.36 | 45.04 | 108.77 | 122.68 | 754.60 | 686.74 | 238.90 | 75.95 | 11.85 | 42.68 | 11.15 | 36.17 | 141.55 | 32.42 | 44.15 |
| | TEST (S) | 36.34 | 1004.55 | 1345.44 | 1774.80 | 1060.53 | 873.94 | 1363.40 | 212.66 | 9.13 | 189.30 | 41.46 | 1.44 | 787.86 | 4.78 | 2.21 |

respectively. These measurements report the computational cost associated with the accuracy values above.

*2) Results on the DFC2013 Houston:* The DFC2013 Houston dataset contains 15 urban land-cover categories with complex spatial structure, spectral heterogeneity, and high within-class variability. We evaluate LizardNet at 1%, 5%, 10%, 15%, and 20% training proportions to measure its performance across different amounts of supervision. Tables V and VI report the results.

At the 1% and 5% settings, classes 6, 8, and 10 exhibit substantial spectral overlap. Cnn2D, DWT, and DSFormer show large accuracy decreases for some of these classes. LizardNet's results across the five training proportions are summarized as follows.

1) *Low-Data Regimes (1%, 5%, and 10%):* At 1% training, DSFormer obtains 61.90% OA, and LizardNet obtains 86.34% OA, 87.64% AA, and 85.24% $\kappa$. LizardNet's OA increases to 94.92% at 5% training and 98.24% at 10%. These are the highest OA values among the compared methods at the three sampling rates.
2) *Higher Data Regimes (15% and 20%):* LizardNet obtains 99.66% OA at 15% training. At 20%, it obtains 99.75% OA, with AA and $\kappa$ above 99.70%. Ten of the 15 classes reach 100.00% classwise accuracy under the 20% setting (classes 2, 3, 5, 7, 8, 10, 11, 13, 14, and 15).

At the 10% setting, LizardNet requires 44.15 s for training; Hamida and He require 754.60 and 686.74 s, respectively. LizardNet's testing times are 3.84, 2.33, and 2.21 s at the 1%, 5%, and 10% settings. These values provide the time measurements corresponding to the reported Houston accuracies.

*3) Results on the WHU-Hi-HongHu:* The WHU-Hi-HongHu dataset is a large agricultural HSI benchmark with many crop types, strong between-class spectral similarity, and within-class variability. We evaluate 25, 50, 75, and 100

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TABLE VI
CLASSIFICATION RESULTS ON THE DFC2013 HOUSTON DATASET (PART 2: 15% AND 20%)

| TRAIN SAMPLE | CLASS | MODEL | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D | DBDA | DWT | FDSSC | HAMIDA | HE | LEE | RVT | CS2DT | DSFORMER | HYPERKAN | SWIN-HSI | MAMBAHSI | DGPF-RENET | OURS |
| 15% | 1 | 94.39 | 99.30 | 94.55 | 99.11 | 98.58 | 98.71 | 92.95 | 97.85 | 99.78 | 97.23 | 99.45 | 98.35 | 99.52 | 99.72 | **99.91** |
| | 2 | 95.61 | 99.24 | 96.72 | 99.15 | 99.58 | 99.44 | 91.33 | 97.67 | 99.89 | 97.92 | 100.00 | 98.42 | 99.27 | 100.00 | 99.91 |
| | 3 | 100.00 | 99.92 | 99.92 | 100.00 | 100.00 | 100.00 | 99.49 | 98.75 | 100.00 | 99.91 | 99.83 | 100.00 | 100.00 | 99.83 | **100.00** |
| | 4 | 99.34 | 99.81 | 96.31 | 99.86 | 98.18 | 99.01 | 96.84 | 98.76 | 99.66 | 98.77 | 99.26 | **100.00** | 99.62 | 99.43 | 99.91 |
| | 5 | 99.58 | 98.12 | 100.00 | 100.00 | 99.76 | 100.00 | 97.63 | 99.76 | 100.00 | 99.95 | 98.05 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 6 | 100.00 | 98.90 | 99.64 | 98.90 | 98.90 | 95.17 | 99.09 | 97.30 | 99.78 | 99.43 | 100.00 | 99.14 | 98.66 | 97.83 | 97.83 |
| | 7 | 98.30 | 98.25 | 99.16 | 98.88 | 94.91 | 95.04 | 90.46 | 97.75 | 99.66 | 99.15 | 97.81 | 98.48 | 99.09 | 100.00 | **100.00** |
| | 8 | 89.83 | 98.48 | 99.15 | 99.34 | 96.33 | 97.60 | 89.77 | 96.94 | 99.94 | 96.98 | 98.24 | 98.37 | 97.86 | 99.91 | **100.00** |
| | 9 | 97.86 | 97.36 | **99.06** | 98.65 | 96.73 | 94.69 | 84.31 | 96.68 | 98.10 | 97.01 | 98.80 | 97.26 | 97.56 | 97.74 | 97.74 |
| | 10 | 90.56 | 99.33 | 98.89 | 97.61 | 98.71 | 98.90 | 92.14 | 98.08 | 98.16 | 93.55 | 100.00 | 98.48 | 99.42 | 99.81 | **100.00** |
| | 11 | 99.00 | 98.87 | 99.06 | 98.22 | 95.02 | 95.33 | 89.28 | 97.75 | 99.77 | 99.59 | 98.50 | 100.00 | 97.72 | 98.86 | **100.00** |
| | 12 | 98.49 | 95.84 | 98.57 | 99.24 | 98.07 | 97.53 | 80.77 | 98.02 | 98.31 | 98.38 | **99.53** | 98.41 | 98.52 | 99.05 | 98.95 |
| | 13 | 96.91 | 97.87 | 97.37 | 95.72 | 96.40 | 96.71 | 92.67 | 95.78 | 100.00 | 97.02 | 100.00 | 99.25 | 96.46 | 99.50 | **100.00** |
| | 14 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 99.86 | 99.45 | 99.59 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 15 | 100.00 | 99.12 | 99.91 | 100.00 | 99.20 | 99.20 | 99.20 | 97.80 | 100.00 | 100.00 | 100.00 | 100.00 | 97.84 | 100.00 | **100.00** |
| | OA | 96.85 | 98.59 | 98.36 | 99.02 | 97.83 | 97.78 | 91.82 | 97.93 | 99.44 | 98.08 | 99.13 | 98.98 | 98.82 | 99.48 | **99.66** |
| | AA | 97.32 | 98.69 | 98.55 | 98.98 | 98.02 | 97.81 | 93.03 | 97.90 | 99.54 | 98.33 | 99.30 | 99.08 | 98.77 | 99.44 | **99.62** |
| | K | 96.59 | 98.48 | 98.22 | 98.94 | 97.66 | 97.61 | 91.16 | 97.76 | 99.39 | 97.92 | 99.06 | 98.89 | 98.73 | 99.43 | **99.63** |
| | TRAIN (S) | 56.66 | 111.03 | 1631.63 | 309.68 | 136.14 | 268.96 | 265.38 | 153.10 | 172.78 | 81.06 | 32.51 | 62.15 | 138.84 | 257.86 | 58.25 |
| | TEST (S) | 5.06 | 39.66 | 4551.05 | 11.00 | 6.01 | 14.84 | 17.06 | 71.49 | 7.38 | 215.08 | 39.91 | 1.35 | 721.99 | 5.30 | 1.99 |
| 20% | 1 | 93.53 | 99.30 | 96.44 | 98.64 | 98.95 | 97.60 | 96.40 | 98.95 | 99.42 | 96.29 | 99.12 | 99.21 | 99.73 | **99.90** | 99.80 |
| | 2 | 92.36 | 99.20 | 99.05 | 99.65 | 99.35 | 99.21 | 99.50 | 99.25 | 100.00 | 99.29 | 100.00 | 99.28 | 98.69 | 100.00 | **100.00** |
| | 3 | 100.00 | 100.00 | 99.91 | 100.00 | 100.00 | 100.00 | 98.84 | 100.00 | 100.00 | 99.80 | 99.83 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 4 | 97.32 | 99.85 | 96.89 | 98.85 | 99.50 | 98.37 | 96.54 | 99.29 | 99.94 | 99.04 | **100.00** | 99.47 | 99.66 | 99.60 | 99.90 |
| | 5 | 100.00 | 100.00 | 100.00 | 100.00 | 99.70 | 100.00 | 98.83 | 99.95 | 100.00 | 100.00 | 98.43 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 6 | 99.81 | 100.00 | 99.81 | 100.00 | 96.00 | 97.20 | 99.61 | 93.07 | 100.00 | 100.00 | 100.00 | 99.18 | 98.05 | 97.69 | 97.69 |
| | 7 | 100.00 | 99.26 | 99.31 | 98.76 | 96.82 | 97.47 | 94.55 | 97.33 | 99.61 | 99.63 | 97.79 | 99.42 | 98.57 | 100.00 | **100.00** |
| | 8 | 99.85 | 99.40 | 98.41 | 98.95 | 96.95 | 97.85 | 94.14 | 99.60 | 100.00 | 96.74 | 98.33 | 99.24 | 97.49 | 100.00 | **100.00** |
| | 9 | 99.25 | 98.70 | 98.90 | **100.00** | 96.08 | 97.81 | 94.76 | 98.80 | 98.77 | 98.81 | 97.96 | 98.54 | 98.75 | 97.11 | 98.30 |
| | 10 | 99.54 | 98.54 | 98.58 | 99.54 | 99.75 | 98.69 | 95.25 | 99.39 | 99.93 | 99.07 | 100.00 | 99.43 | 99.38 | 99.90 | **100.00** |
| | 11 | 99.29 | 99.34 | 99.35 | 99.50 | 96.98 | 98.24 | 93.35 | 99.35 | 99.93 | 99.76 | 98.52 | 100.00 | 97.95 | 98.48 | **100.00** |
| | 12 | 99.80 | 99.39 | 98.73 | 99.80 | 98.33 | 99.29 | 94.83 | 98.10 | 98.85 | 99.18 | **100.00** | 99.16 | 99.01 | 99.90 | 99.59 |
| | 13 | 99.20 | 97.01 | 96.69 | 97.08 | 97.04 | 98.26 | 89.18 | 96.90 | 99.25 | 99.81 | 100.00 | 99.31 | 95.88 | 100.00 | **100.00** |
| | 14 | 100.00 | 100.00 | 100.00 | 100.00 | 99.71 | 100.00 | 99.71 | 99.71 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 15 | 100.00 | 100.00 | 99.91 | 100.00 | 99.52 | 100.00 | 98.96 | 100.00 | 99.87 | 100.00 | 100.00 | 100.00 | 97.64 | 100.00 | **100.00** |
| | OA | 98.37 | 99.33 | 98.69 | 99.38 | 98.33 | 98.60 | 96.10 | 98.91 | 99.67 | 98.93 | 99.17 | 99.28 | 98.83 | 99.53 | **99.75** |
| | AA | 98.65 | 99.33 | 98.80 | 99.38 | 98.31 | 98.67 | 96.30 | 98.65 | 99.69 | 99.16 | 99.33 | 99.48 | 98.72 | 99.50 | **99.69** |
| | K | 98.24 | 99.27 | 98.59 | 99.33 | 98.19 | 98.49 | 95.78 | 98.82 | 99.65 | 98.83 | 99.11 | 99.23 | 98.74 | 99.49 | **99.73** |
| | TRAIN (S) | 69.35 | 130.19 | 2127.72 | 399.22 | 167.57 | 338.84 | 299.10 | 188.97 | 227.23 | 106.68 | 45.90 | 79.52 | 138.59 | 349.10 | 75.47 |
| | TEST (S) | 5.10 | 41.23 | 4619.66 | 12.74 | 7.51 | 15.37 | 20.18 | 73.00 | 6.39 | 217.03 | 39.45 | 1.30 | 715.40 | 5.21 | 1.88 |

labeled samples per class. Tables VII and VIII report the results.

With 25 or 50 samples per class, several baselines show large classwise variation for classes 5, 8, 11, and 21. The LizardNet results at each sample size are reported as follows.

1) *Few-Shot Regimes (25 and 50 Samples per Class):* With 25 samples per class, LizardNet obtains the highest AA of 90.62%, together with 87.73% OA and 84.92% $\kappa$. Swin-HSI obtains the highest OA of 89.19% and $\kappa$ of 86.24% in this setting. LizardNet's classwise accuracies for classes 8 and 21 are 92.91% and 97.97%. With 50 samples per class, LizardNet obtains 92.04% OA, 94.07% AA, and 90.12% $\kappa$; the corresponding Swin-HSI OA is 92.19%. The AA values indicate more uniform performance across the 22 classes.
2) *Higher Data Regimes (75 and 100 Samples per Class):* With 75 samples per class, LizardNet obtains 94.17% OA, 95.32% AA, and 92.70% $\kappa$; classes 8, 17, and 21 reach 100.00% accuracy. With 100 samples per class, it obtains 96.57% OA, 97.11% AA, and 95.67% $\kappa$. DGPF-RENet and DBDA obtain 95.19% and 95.91% OA, respectively. LizardNet's classwise accuracies are 98.92% for class 2, 99.90% for class 8, and 100.00% for class 21.

With 100 samples per class, LizardNet requires 163.64 s for training. DWT, CS2DT, DGPF-RENet, and Swin-HSI require 1565.67, 871.22, 803.50, and 550.52 s, respectively. This comparison reports training cost for the same 22-class setting.

*4) Results on the Trento:* The Trento dataset is a small urban HSI benchmark with six land-cover categories and relatively high between-class separability. We evaluate 1%, fixed-sample, 15%, and 20% training settings to measure performance across different sample sizes. Table IX reports the results.

Under scenarios with limited training data (1% and fixed), most methods achieve relatively strong performance due to the limited number of classes and clear spectral separability. Nevertheless, differences in robustness and classwise consistency remain evident across all variations.

1) *Few-Shot Regimes (1% and Fixed):* At 1% training, DGPF-RENet and FDSSC obtain 0.00% accuracy on class 3. LizardNet obtains 97.59% OA, 95.43% AA, and 96.78% $\kappa$, the highest aggregate values in this setting. Under the fixed-sample setting, its accuracies on classes 2 and 6 are 95.89% and 95.03%, and its OA is 98.75%.
2) *Higher Data Regimes (15% & 20%):* LizardNet obtains 99.82% OA at 15% training. At 20%, it obtains 99.90% OA, 99.85% AA, and 99.87% $\kappa$. Classes 1, 3, 4, and 5 reach 100.00% classwise accuracy.

At the 20% Trento setting, LizardNet requires 163.15 s for training. DWT, FDSSC, and Lee require 4155.20, 534.10, and 477.72 s, respectively. LizardNet's testing time ranges from 6.27 to 7.94 s across the evaluated settings.

In addition, since the Trento dataset has an associated LiDAR modality, it provides a natural extension toward hyperspectral–LiDAR multimodal fusion. Incorporating elevation and structural information can further enhance class

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TABLE VII
CLASSIFICATION RESULTS ON THE WHU-HI-HONGHU DATASET (PART 1: 25 AND 50)

| TRAIN SAMPLE | CLASS | MODEL | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D | DBDA | DWT | FDSSC | HAMIDA | HE | LEE | RVT | CS2DT | DSFORMER | HYPERKAN | SWIN-HSI | MAMBAHSI | DGPF-RENET | OURS |
| 25 | 1 | 94.44 | 88.36 | 86.63 | 93.73 | 92.88 | 88.08 | 79.63 | 96.51 | 93.81 | 79.17 | 84.53 | 97.12 | 84.31 | 79.10 | 89.39 |
| | 2 | 78.96 | 66.92 | 63.34 | 67.04 | 76.96 | 67.10 | 59.12 | 82.66 | 61.14 | 84.47 | 86.39 | 76.45 | 88.94 | 90.65 | **91.46** |
| | 3 | 89.90 | 87.07 | 79.37 | 86.82 | 87.31 | 81.53 | 13.33 | 75.58 | 74.88 | 53.26 | 82.30 | 93.21 | 81.32 | **92.14** | 81.85 |
| | 4 | 94.72 | 88.29 | 89.70 | 95.25 | 94.95 | 87.88 | 81.30 | 93.24 | 95.21 | 93.34 | 65.99 | 99.85 | 95.21 | 83.59 | **88.06** |
| | 5 | 63.34 | 32.33 | 8.84 | 46.58 | 52.51 | 22.99 | 16.50 | 41.11 | 47.99 | 79.55 | 87.34 | 46.51 | 81.98 | **96.33** | 96.00 |
| | 6 | 91.96 | 90.83 | 84.97 | 87.55 | 92.00 | 92.45 | 79.01 | 94.60 | 83.00 | 91.49 | 93.17 | 96.34 | 88.51 | 92.44 | **93.31** |
| | 7 | 51.32 | 72.00 | 43.64 | 61.27 | 76.65 | 70.22 | 36.10 | 81.60 | 61.94 | 71.61 | 84.39 | 80.17 | 65.40 | 60.39 | **78.25** |
| | 8 | 40.37 | 34.78 | 21.57 | 46.75 | 34.78 | 24.53 | 18.03 | 42.57 | 100.00 | 64.22 | 59.01 | 48.62 | 76.14 | 80.64 | **92.91** |
| | 9 | 84.23 | 97.49 | 94.19 | 96.27 | 95.37 | 94.18 | 90.55 | 97.18 | 93.87 | 94.92 | 92.07 | 94.19 | 85.13 | 94.45 | **93.98** |
| | 10 | 74.05 | 74.77 | 44.28 | 81.71 | 71.65 | 68.48 | 31.22 | 76.59 | 71.42 | 57.26 | 71.28 | 80.84 | 40.82 | 78.29 | **86.24** |
| | 11 | 49.18 | 66.66 | 48.48 | 61.82 | 57.90 | 48.16 | 33.92 | 75.49 | 61.12 | 49.93 | 75.71 | 62.37 | 38.67 | 83.17 | **85.75** |
| | 12 | 51.79 | 60.03 | 44.04 | 57.65 | 65.96 | 58.98 | 48.77 | 65.75 | 42.59 | 41.71 | 61.93 | 77.42 | 67.08 | 51.95 | **78.57** |
| | 13 | 70.98 | 68.16 | 60.32 | 63.19 | 74.12 | 65.03 | 48.23 | 72.70 | 58.16 | 63.27 | 71.54 | 67.81 | 75.65 | 68.30 | **76.89** |
| | 14 | 83.31 | 88.80 | 68.83 | 75.84 | 77.35 | 72.36 | 44.14 | 86.72 | 85.00 | 69.73 | 97.23 | 78.29 | 74.72 | 77.42 | **91.55** |
| | 15 | 81.93 | 16.86 | 61.44 | 53.06 | 58.36 | 14.53 | 15.23 | 87.32 | 91.78 | 97.80 | 97.83 | 35.63 | 97.78 | **99.69** | 98.85 |
| | 16 | 95.38 | 89.16 | 27.44 | 95.96 | 91.87 | 83.61 | 60.54 | 93.48 | 61.70 | 88.46 | 95.47 | 78.91 | 96.91 | 92.68 | **95.90** |
| | 17 | 83.16 | 74.24 | 39.82 | 84.00 | 82.13 | 73.20 | 52.49 | 84.53 | 16.28 | 44.81 | 99.64 | 84.15 | 83.73 | **96.98** | 98.98 |
| | 18 | 23.50 | 57.60 | 30.63 | 38.48 | 65.91 | 58.37 | 39.01 | 74.34 | 72.13 | 98.03 | 98.76 | 79.52 | 79.89 | **99.72** | 96.38 |
| | 19 | 83.07 | 79.73 | 81.21 | 87.37 | 78.23 | 72.11 | 72.98 | 89.06 | 89.49 | 91.70 | 90.73 | 89.26 | 80.69 | **93.79** | 88.08 |
| | 20 | 69.46 | 65.98 | 56.24 | 52.37 | 66.31 | 57.92 | 65.29 | 80.78 | 76.72 | 92.89 | 93.33 | 77.48 | 89.97 | **95.78** | 96.34 |
| | 21 | 43.64 | 9.56 | 10.40 | 12.87 | 41.81 | 32.15 | 9.25 | 24.73 | 33.52 | 98.35 | 94.12 | 12.76 | 52.94 | **99.62** | 97.97 |
| | 22 | 34.90 | 53.55 | 24.96 | 44.77 | 49.27 | 51.86 | 34.83 | 57.26 | 84.64 | 97.91 | 98.83 | 61.35 | 80.75 | **95.92** | 96.87 |
| | OA | 81.57 | 78.74 | 72.87 | 81.54 | 84.47 | 76.97 | 59.37 | 84.53 | 82.89 | 82.26 | 76.73 | 89.19 | 84.28 | 83.07 | **87.73** |
| | AA | 69.71 | 66.51 | 53.20 | 67.74 | 72.01 | 62.99 | 46.79 | 76.08 | 70.75 | 77.45 | 85.53 | 73.56 | 77.57 | 86.50 | **90.62** |
| | K | 77.24 | 74.33 | 66.70 | 77.18 | 80.70 | 72.19 | 52.29 | 80.99 | 78.06 | 77.96 | 72.68 | 86.24 | 80.31 | 79.37 | **84.92** |
| | TRAIN (S) | 5.85 | 21.37 | 39.27 | 49.80 | 290.34 | 303.95 | 98.82 | 77.10 | 2.54 | 23.99 | 411.41 | 50.05 | 64.40 | 47.65 | 73.19 |
| | TEST (S) | 24.14 | 709.95 | 866.54 | 1102.70 | 609.31 | 571.35 | 886.48 | 62.17 | 99.50 | 125.48 | 23.83 | 37.43 | 470.29 | 317.26 | 235.99 |
| 50 | 1 | 96.05 | 93.52 | 94.26 | 93.19 | 93.82 | 90.99 | 93.93 | 97.33 | 95.21 | 89.93 | 84.07 | 98.14 | 97.22 | 91.19 | 92.25 |
| | 2 | 83.95 | 73.90 | 75.61 | 74.14 | 76.92 | 73.15 | 69.23 | 89.39 | 94.00 | 94.58 | 90.79 | 61.56 | 94.20 | 89.34 | **96.40** |
| | 3 | 90.81 | 84.72 | 89.65 | 71.52 | 88.87 | 88.86 | 18.81 | 91.60 | 93.89 | 85.16 | 82.26 | 95.32 | 75.69 | **94.67** | 84.91 |
| | 4 | 94.60 | 98.54 | 96.41 | 93.39 | 90.37 | 93.24 | 93.55 | 97.20 | 98.21 | 94.55 | 84.07 | 99.81 | 98.79 | **94.49** | 92.84 |
| | 5 | 54.52 | 75.12 | 57.63 | 33.85 | 34.88 | 40.93 | 37.00 | 66.20 | 47.91 | 91.76 | 88.48 | 51.47 | 73.89 | **96.01** | 98.51 |
| | 6 | 94.20 | 92.03 | 92.15 | 94.18 | 93.87 | 93.91 | 89.21 | 95.72 | 83.00 | 91.51 | 96.77 | 98.25 | 90.88 | 92.82 | **96.72** |
| | 7 | 75.91 | 74.77 | 74.47 | 81.46 | 75.85 | 76.69 | 73.31 | 86.51 | 78.50 | 71.91 | 85.27 | 86.19 | 77.78 | 77.99 | **83.65** |
| | 8 | 41.16 | 40.41 | 23.74 | 52.15 | 49.73 | 44.96 | 24.26 | 52.57 | 62.43 | 86.00 | 76.14 | 49.63 | 89.10 | 88.09 | **98.76** |
| | 9 | 96.77 | 97.09 | 97.14 | 99.27 | 97.30 | 97.38 | 94.54 | 99.04 | 83.27 | 94.86 | 95.56 | 94.52 | 94.79 | 94.20 | **94.31** |
| | 10 | 85.53 | 80.19 | 61.23 | 86.67 | 81.93 | 80.00 | 47.65 | 86.15 | 85.69 | 88.88 | 80.74 | 84.28 | 70.99 | 88.33 | **90.00** |
| | 11 | 70.93 | 70.83 | 68.41 | 83.02 | 75.74 | 65.66 | 54.71 | 83.31 | 67.12 | 82.07 | 83.99 | 80.71 | 58.98 | 86.27 | **91.02** |
| | 12 | 69.52 | 69.19 | 45.87 | 70.42 | 68.92 | 61.88 | 46.67 | 79.38 | 82.84 | 64.94 | 87.39 | 66.45 | 83.24 | 63.06 | **88.76** |
| | 13 | 79.66 | 73.55 | 65.44 | 65.60 | 78.99 | 75.65 | 61.81 | 81.29 | 71.43 | 68.76 | 61.28 | 81.18 | 73.97 | 69.46 | **82.98** |
| | 14 | 89.91 | 92.00 | 78.87 | 85.56 | 89.33 | 86.49 | 55.55 | 91.71 | 88.45 | 90.41 | 98.86 | 90.36 | 84.52 | 89.24 | **95.55** |
| | 15 | 91.10 | 72.54 | 82.69 | 92.59 | 51.36 | 59.32 | 43.41 | 82.33 | 62.41 | 99.37 | 98.19 | 78.59 | 99.14 | **98.42** | 99.01 |
| | 16 | 86.58 | 93.90 | 80.18 | 97.14 | 94.57 | 93.61 | 71.17 | 96.26 | 94.14 | 86.16 | 93.92 | 97.41 | 91.46 | 91.85 | **97.65** |
| | 17 | 73.23 | 79.32 | 14.60 | 84.10 | 87.79 | 85.08 | 56.57 | 92.35 | 33.51 | 76.60 | 100.00 | 75.82 | 97.31 | 99.49 | **99.67** |
| | 18 | 77.84 | 46.14 | 73.22 | 88.12 | 75.27 | 70.63 | 63.78 | 81.71 | 33.52 | 99.07 | 99.49 | 89.15 | 91.43 | **99.97** | 97.70 |
| | 19 | 90.71 | 88.28 | 86.72 | 86.93 | 85.09 | 83.70 | 79.65 | 92.39 | 84.62 | 95.51 | 87.85 | 94.67 | 89.35 | **94.94** | 93.47 |
| | 20 | 59.74 | 62.49 | 89.72 | 71.96 | 72.19 | 77.88 | 69.49 | 90.38 | 76.70 | 97.96 | 95.57 | 82.34 | 90.96 | **97.41** | 97.49 |
| | 21 | 55.93 | 32.95 | 27.20 | 16.46 | 45.49 | 55.21 | 9.56 | 55.54 | 33.35 | 98.00 | 98.36 | 16.82 | 83.29 | **100.00** | 99.80 |
| | 22 | 60.90 | 81.84 | 61.96 | 74.98 | 57.39 | 54.10 | 49.09 | 65.98 | 84.64 | 95.16 | 97.57 | 86.29 | 88.77 | 96.19 | **98.17** |
| | OA | 86.93 | 87.85 | 84.26 | 83.60 | 83.19 | 84.61 | 75.38 | 91.40 | 91.41 | 88.99 | 85.65 | 92.19 | 89.81 | 90.47 | **92.04** |
| | AA | 78.16 | 76.06 | 69.87 | 77.12 | 75.71 | 74.97 | 59.23 | 84.29 | 74.31 | 88.33 | 89.39 | 79.95 | 86.17 | 90.61 | **94.07** |
| | K | 83.82 | 84.73 | 80.40 | 79.85 | 79.54 | 81.07 | 69.70 | 89.26 | 87.67 | 86.25 | 82.52 | 90.08 | 87.13 | 88.08 | **90.12** |
| | TRAIN (S) | 11.56 | 41.50 | 78.91 | 100.38 | 583.74 | 635.65 | 195.06 | 147.07 | 12.70 | 40.31 | 410.25 | 98.13 | 71.89 | 80.87 | 95.85 |
| | TEST (S) | 25.45 | 707.25 | 867.07 | 1099.90 | 607.70 | 602.42 | 870.67 | 60.81 | 102.77 | 125.36 | 23.78 | 34.93 | 501.87 | 314.72 | 230.41 |

separability, particularly in urban environments with similar spectral signatures. Therefore, multimodal fusion represents a meaningful direction for future improvement and extension of the current framework.

The WHU-Hi-HongHu and Trento results provide two additional evaluations. With 25 WHU-Hi-HongHu samples per class, LizardNet obtains 90.62% AA; Lee and DwT show larger decreases on several classes. On Trento, LizardNet obtains 98.75% OA in the fixed-sample setting and 99.90% OA at 20% training. These results cover agricultural and urban scenes with different spatial resolutions and class structures.

### *E. Computational Complexity*

Before evaluating computational complexity, the overall performance of the ablation variants is presented in Table X. Table XI compares model complexity using a batch size of one and a 15 × 15 spatial patch. The dataset-specific input shapes are $1 \times 15 \times 15 \times C$, where $C$ is 200 for IP after noisy-band removal, 144 for DFC2013 Houston, 270 for WHU-Hi-HongHu, and 63 for Trento. FLOPs are obtained with TensorFlow's profiler using the `float_operation` option; a multiply and an addition are counted as two floating-point operations. All models are profiled under the same input convention for a given dataset. LizardNet uses 0.75M–0.94M parameters. DwT exceeds 60M parameters, and RvT and DSFormer use approximately 9M–18M.

LizardNet requires 2.5M–2.9M FLOPs across the four datasets. FDSSC, DBDA, and CS2DT exceed 100M FLOPs on some datasets, while MambaHSI requires 40.17M–52.15M FLOPs across the four datasets. DGPF-RENet and RvT also require more FLOPs than LizardNet in the reported settings. The accuracy tables provide the corresponding predictive results.

Swin-HSI has a small parameter count, and its FLOPs exceed those of LizardNet on IP and Trento. HyperKAN shows larger accuracy variation across the small-sample settings. LizardNet combines the parameter counts, FLOPs, and accuracies listed in the same tables through a shallow spectral–spatial architecture.

At the system level, parameter count determines checkpoint storage and FLOPs estimate arithmetic work. Table XII supplies direct measurements of latency, throughput, power, and energy on the Jetson Orin Nano.

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TABLE VIII
CLASSIFICATION RESULTS ON THE WHU-HI-HONGHU DATASET (PART 2: 75 AND 100)

| TRAIN SAMPLE | CLASS | MODEL | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D | DBDA | DWT | FDSSC | HAMIDA | HE | LEE | RVT | CS2DT | DSFORMER | HYPERKAN | SWIN-HSI | MAMBAHSI | DGPF-RENET | OURS |
| 75 | 1 | 97.29 | 97.97 | 98.06 | 97.49 | 93.56 | 95.98 | 92.94 | 96.44 | 93.82 | 94.91 | 87.87 | 98.21 | 96.78 | 96.71 | 96.78 |
| | 2 | 90.43 | 83.69 | 86.88 | 89.72 | 79.88 | 85.80 | 81.79 | 86.17 | 87.91 | 95.81 | 92.73 | 85.43 | 94.05 | **98.05** | 97.29 |
| | 3 | 95.11 | 95.11 | 90.11 | 91.94 | 90.37 | 92.51 | 83.52 | 91.49 | **96.26** | 86.16 | 86.88 | 89.15 | 84.99 | 92.08 | 93.52 |
| | 4 | 93.03 | 92.45 | 97.77 | 91.52 | 93.70 | 95.07 | 86.04 | 97.07 | 92.63 | 95.38 | 95.05 | 91.38 | **99.11** | 95.75 | 94.59 |
| | 5 | 37.38 | 76.84 | 72.17 | 79.95 | 42.31 | 52.02 | 22.33 | 63.75 | 89.10 | 95.09 | **98.45** | 80.52 | 81.16 | 98.34 | 97.72 |
| | 6 | 96.57 | 96.03 | 97.37 | 98.11 | 93.71 | 95.90 | 88.97 | 96.53 | 91.85 | 92.74 | **98.85** | 96.47 | 94.08 | 98.16 | 97.83 |
| | 7 | 89.07 | 91.62 | 89.67 | 86.26 | 80.11 | 77.95 | 71.07 | 87.14 | 88.02 | 81.53 | 81.65 | **95.23** | 81.40 | 87.37 | 93.38 |
| | 8 | 64.78 | 77.91 | 72.29 | 88.01 | 52.26 | 63.75 | 27.67 | 61.79 | 91.37 | 90.21 | 95.73 | 50.68 | 85.53 | 96.73 | **100.00** |
| | 9 | **99.13** | 91.53 | 98.65 | 98.59 | 97.70 | 98.27 | 96.93 | **99.13** | 98.62 | 97.67 | 95.90 | 97.35 | 94.94 | 98.86 | 91.49 |
| | 10 | 89.37 | 89.18 | 91.25 | **95.93** | 80.85 | 86.28 | 69.92 | 84.86 | 94.83 | 93.44 | 88.21 | 87.41 | 82.04 | 91.01 | 93.96 |
| | 11 | 88.33 | 92.87 | 88.23 | 91.94 | 69.50 | 77.05 | 61.35 | 89.60 | 93.25 | 84.79 | 90.47 | 89.62 | 67.04 | 95.56 | **97.39** |
| | 12 | 81.51 | 85.73 | 79.64 | 88.45 | 72.98 | 74.63 | 58.94 | 78.49 | **92.77** | 82.18 | 86.79 | 86.25 | 87.79 | 82.05 | 90.01 |
| | 13 | 86.51 | 84.96 | 84.77 | **92.76** | 79.41 | 80.04 | 62.11 | 84.08 | 85.67 | 76.60 | 83.49 | 85.39 | 74.56 | 78.35 | 78.18 |
| | 14 | 85.29 | 95.91 | 95.00 | 96.76 | 88.69 | 92.66 | 77.88 | 90.21 | 97.33 | 96.79 | **97.79** | 96.18 | 91.11 | 94.97 | 97.61 |
| | 15 | 95.96 | 95.66 | 88.41 | 96.11 | 50.07 | 87.41 | 61.04 | 88.96 | 94.66 | 99.19 | **100.00** | 78.54 | 98.75 | 99.89 | 99.24 |
| | 16 | 97.59 | 91.21 | 83.71 | 90.17 | 95.61 | 96.63 | 85.26 | 97.08 | 98.42 | 94.82 | **98.61** | 98.23 | 92.22 | 96.55 | 98.12 |
| | 17 | 94.49 | 90.47 | 72.56 | 85.66 | 85.40 | 94.37 | 74.29 | 93.48 | 94.11 | 95.62 | 99.76 | 95.71 | 98.29 | 100.00 | **100.00** |
| | 18 | 94.15 | 95.83 | 92.39 | 88.39 | 75.56 | 79.22 | 70.62 | 87.00 | 95.39 | 97.84 | 98.60 | 93.46 | 92.67 | **100.00** | 99.62 |
| | 19 | 92.29 | 95.40 | 93.79 | 87.81 | 85.89 | 89.20 | 83.59 | 92.10 | 88.47 | 95.86 | 90.08 | 93.68 | 89.67 | **96.19** | 95.43 |
| | 20 | 93.76 | 94.73 | 87.54 | 95.18 | 77.11 | 91.37 | 76.26 | 90.85 | 96.63 | 97.79 | **99.00** | 96.14 | 94.43 | 97.68 | 92.70 |
| | 21 | 84.18 | 79.86 | 50.58 | 67.28 | 50.25 | 57.77 | 30.42 | 61.55 | 85.13 | 98.92 | 100.00 | 44.59 | 77.82 | 100.00 | **100.00** |
| | 22 | 77.74 | 86.25 | 77.37 | 88.45 | 59.50 | 62.45 | 56.45 | 76.03 | 92.55 | 93.98 | 99.27 | 89.37 | 90.05 | **100.00** | 92.08 |
| | OA | 89.67 | 93.42 | 92.85 | 93.81 | 86.27 | 89.12 | 75.80 | 92.02 | 93.54 | 92.00 | 92.96 | 93.61 | 92.05 | 94.16 | **94.17** |
| | AA | 87.45 | 90.06 | 85.83 | 90.29 | 77.02 | 83.02 | 69.06 | 86.08 | 92.67 | 92.61 | 93.87 | 87.23 | 88.57 | 95.20 | **95.32** |
| | K | 87.30 | 91.50 | 91.03 | 92.12 | 83.07 | 86.51 | 71.03 | 90.03 | 91.85 | 89.98 | 91.19 | 92.15 | 89.94 | 92.69 | **92.70** |
| | TRAIN (S) | 62.75 | 376.75 | 1152.14 | 392.62 | 170.35 | 341.73 | 231.27 | 134.93 | 464.07 | 60.58 | 29.30 | 301.30 | 90.02 | 596.14 | 135.60 |
| | TEST (S) | 3.79 | 25.61 | 3198.09 | 7.53 | 4.14 | 9.98 | 11.09 | 48.25 | 198.34 | 139.29 | 24.46 | 37.54 | 410.95 | 446.27 | 247.65 |
| 100 | 1 | 94.21 | 97.94 | 97.22 | 98.60 | 95.44 | 95.07 | 95.55 | 97.04 | 93.56 | 96.78 | 86.29 | **99.23** | 96.95 | 97.82 | 96.34 |
| | 2 | 91.30 | 94.87 | 88.05 | 83.47 | 79.44 | 85.15 | 81.50 | 89.14 | 90.69 | 94.74 | 95.60 | 77.58 | 92.72 | 97.71 | **98.92** |
| | 3 | 88.48 | 97.41 | 96.12 | 95.92 | 91.50 | 91.09 | 81.92 | 93.99 | **98.05** | 88.55 | 85.89 | 94.16 | 87.29 | 92.56 | 96.49 |
| | 4 | 96.05 | 93.44 | 98.42 | 93.33 | 93.45 | 96.53 | 89.26 | 96.17 | **99.67** | 95.11 | 94.48 | 94.62 | 99.28 | 95.67 | 97.71 |
| | 5 | 55.75 | 92.08 | 76.97 | 92.12 | 55.20 | 59.44 | 30.93 | 54.60 | 95.39 | 96.85 | 99.00 | 88.35 | 82.35 | **99.79** | 97.53 |
| | 6 | 97.38 | 98.18 | 97.32 | 98.02 | 93.76 | 94.69 | 92.75 | 95.13 | 94.41 | 95.48 | 98.65 | 97.12 | 93.72 | **98.96** | 98.31 |
| | 7 | 92.96 | **95.20** | 85.85 | 95.40 | 82.31 | 80.73 | 77.15 | 91.90 | 91.39 | 82.42 | 85.23 | 92.41 | 83.47 | 89.44 | 90.78 |
| | 8 | 79.84 | 90.94 | 77.59 | 88.50 | 51.42 | 58.37 | 33.13 | 63.66 | 94.06 | 90.38 | 98.28 | 75.64 | 88.78 | 98.28 | **99.90** |
| | 9 | 99.44 | **99.57** | 99.05 | 94.67 | 98.20 | 95.93 | 97.13 | 98.88 | 99.19 | 95.22 | 97.00 | 98.25 | 93.19 | 98.96 | 99.02 |
| | 10 | 94.94 | 96.91 | 89.64 | **96.93** | 85.17 | 84.13 | 70.03 | 86.58 | 92.60 | 92.08 | 89.22 | 93.57 | 82.08 | 93.77 | 94.74 |
| | 11 | 92.84 | 91.82 | 91.81 | 96.42 | 66.44 | 71.83 | 62.84 | 88.83 | **96.58** | 87.90 | 92.83 | 91.43 | 73.32 | 95.24 | 94.02 |
| | 12 | 87.08 | 87.99 | 77.97 | 88.23 | 71.01 | 72.89 | 60.65 | 85.49 | 94.04 | 86.23 | 90.33 | 91.29 | 90.03 | 93.69 | **95.54** |
| | 13 | 89.80 | 92.10 | 88.30 | 89.91 | 80.12 | 78.35 | 69.20 | 85.75 | 91.24 | 83.34 | 89.90 | **94.27** | 73.73 | 83.89 | 88.83 |
| | 14 | 96.85 | 92.59 | 96.10 | 98.26 | 91.33 | 88.97 | 79.97 | 93.66 | 98.58 | 97.40 | **98.59** | 95.48 | 92.88 | 96.83 | 96.73 |
| | 15 | 95.00 | 94.27 | 94.93 | 94.85 | 48.80 | 85.23 | 49.57 | 91.48 | 89.51 | 99.50 | 100.00 | 78.69 | 98.51 | 100.00 | 99.11 |
| | 16 | **99.74** | 99.24 | 83.11 | 94.31 | 96.22 | 96.21 | 88.16 | 97.19 | 99.34 | 97.49 | 98.56 | 99.16 | 94.90 | 96.76 | 98.90 |
| | 17 | 99.38 | 98.49 | 73.11 | 94.04 | 90.39 | 88.16 | 74.83 | 94.97 | 97.98 | 95.81 | 99.04 | 95.84 | 98.23 | **100.00** | 99.97 |
| | 18 | 92.84 | 92.73 | 82.65 | 93.32 | 83.71 | 76.98 | 70.31 | 85.36 | 98.92 | 99.05 | 98.88 | 93.52 | 93.43 | **100.00** | 99.71 |
| | 19 | 94.90 | **96.78** | 94.68 | **96.78** | 88.16 | 84.61 | 78.67 | 90.39 | 93.99 | 96.68 | 91.04 | 93.73 | 89.86 | 96.69 | 96.27 |
| | 20 | 89.00 | 92.96 | 92.77 | 92.97 | 83.24 | 81.56 | 76.75 | 94.78 | 91.97 | 98.23 | **99.20** | 84.61 | 90.75 | 99.08 | 99.14 |
| | 21 | 68.20 | 94.50 | 87.77 | 92.40 | 63.73 | 57.53 | 27.01 | 65.32 | 93.30 | 99.80 | 100.00 | 5.48 | 76.26 | 100.00 | **100.00** |
| | 22 | 67.19 | 91.85 | 84.06 | 90.12 | 53.53 | 62.70 | 55.41 | 81.53 | 96.79 | 99.38 | 100.00 | 95.27 | 94.80 | 100.00 | 98.50 |
| | OA | 92.92 | 95.91 | 93.96 | 95.85 | 87.24 | 89.19 | 80.06 | 92.03 | 95.82 | 93.10 | 93.49 | 95.41 | 92.63 | 95.19 | **96.57** |
| | AA | 89.24 | 94.63 | 88.80 | 93.57 | 79.21 | 81.19 | 70.12 | 87.36 | 95.06 | 94.02 | 94.91 | 87.71 | 89.39 | 96.60 | **97.11** |
| | K | 91.19 | 94.60 | 92.40 | 94.55 | 84.28 | 86.51 | 75.82 | 90.08 | 94.88 | 91.36 | 91.87 | 94.13 | 90.67 | 93.97 | **95.67** |
| | TRAIN (S) | 77.67 | 496.47 | 1565.67 | 498.15 | 207.60 | 418.99 | 269.20 | 155.17 | 871.22 | 78.94 | 38.75 | 550.52 | 90.08 | 803.50 | 163.64 |
| | TEST (S) | 5.39 | 29.81 | 3199.25 | 7.23 | 4.03 | 10.78 | 11.53 | 46.50 | 183.35 | 141.07 | 23.83 | 41.08 | 413.71 | 477.21 | 242.93 |

In summary, LizardNet uses 0.75M–0.94M parameters and 2.52M–2.90M FLOPs across the four datasets while retaining the accuracies reported above. These values characterize its accuracy and complexity under the evaluated protocols.

### F. On-Board Deployment Evaluation

*1) Platform and Measurement Protocol:* We conducted on-board measurements on a Jetson Orin Nano 8 GB using CPU-only FP32 inference with batch size 1. Four of the six CPU cores (CPU 0–3) were fixed for both models. The board operated in `MAXN_SUPER` mode with `jetson_clocks` enabled, and the CPU upper frequency was locked at approximately 1.728 GHz. Each model was warmed up for 100 iterations and then timed for 1000 single-sample inferences. Board input power (`VDD_IN`) was sampled every 50 ms during a 20-s continuous-inference run. The same protocol was used for LizardNet and DGPF-RENet.

The checkpoint-loaded weight counts in Table XII are deployment statistics for the loaded weight files; they are reported separately from the architecture-level parameter counts in Table XI. Per-sample incremental energy is obtained from the measured incremental mean power multiplied by the measured mean latency.

*2) Measured Deployment Results:* LizardNet completes one inference in 39.78 ms on average, compared with 314.22 ms for DGPF-RENet. This gives a 7.90× latency ratio and an 87.34% reduction for LizardNet. Its P95 and P99 latencies are 42.57 and 46.31 ms; the corresponding DGPF-RENet values are 336.29 and 365.81 ms. LizardNet sustains 25.14 samples/s, which is 7.91× the 3.18 samples/s of DGPF-RENet.

The measured continuous-inference board power is 10.60 W for LizardNet and 11.70 W for DGPF-RENet. Relative to the common 5.80-W idle level, the incremental mean power is 4.80 versus 5.90 W and the incremental peak power is 8.90 versus 10.20 W. Combining measured latency and incremental power gives 0.191 J per sample for LizardNet and 1.854 J for DGPF-RENet; LizardNet therefore reduces incremental inference energy by 89.70% (9.71×). Its posttest board temperature is 50 °C, compared with 54 °C for DGPF-RENet. Under the same CPU-only protocol,

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TABLE IX
CLASSIFICATION RESULTS ON THE TRENTO DATASET

| TRAIN SAMPLE | CLASS | MODEL | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | CNN2D | DBDA | DWT | FDSSC | HAMIDA | HE | LEE | RVT | CS2DT | DSFORMER | HYPERKAN | SWIN-HSI | MAMBAHSI | DGPF-RENET | OURS |
| 1% | 1 | 96.44 | 97.24 | 98.65 | 96.88 | 98.90 | 99.27 | 91.37 | 98.60 | 95.04 | 97.00 | 99.52 | 99.00 | 99.85 | **99.97** | 99.84 |
| | 2 | 80.38 | 89.67 | 81.13 | 90.79 | 89.40 | 89.39 | 65.34 | 86.11 | 85.37 | 82.32 | 89.57 | 88.42 | **91.03** | 88.38 | 88.36 |
| | 3 | 91.98 | 72.65 | 97.52 | 0.00 | 86.91 | 94.07 | 89.14 | 89.43 | **98.60** | 98.11 | 66.60 | 27.51 | 71.27 | 0.00 | 95.41 |
| | 4 | 99.99 | 99.98 | 99.69 | 99.96 | 99.93 | 99.88 | 98.86 | 99.68 | 96.67 | 99.33 | 97.88 | 100.00 | 100.00 | 100.00 | 99.94 |
| | 5 | 99.99 | 99.76 | 99.26 | 99.41 | 99.72 | 99.06 | 96.34 | 99.40 | 95.57 | 99.69 | 99.63 | 100.00 | 99.81 | 100.00 | 99.92 |
| | 6 | **97.20** | 90.71 | 82.99 | 91.57 | 88.55 | 87.44 | 76.83 | 87.95 | 87.45 | 82.60 | 92.79 | 77.46 | 88.08 | 81.25 | 89.09 |
| | OA | 97.21 | 97.23 | 95.83 | 96.76 | 97.31 | 97.14 | 91.54 | 96.76 | 94.10 | 95.75 | 96.88 | 95.62 | 97.34 | 95.33 | **97.59** |
| | AA | 94.33 | 91.67 | 93.21 | 79.77 | 93.90 | 94.85 | 86.31 | 93.53 | 93.12 | 93.18 | 91.00 | 82.07 | 91.67 | 78.27 | **95.43** |
| | K | 96.28 | 96.29 | 94.44 | 95.66 | 96.41 | 96.17 | 88.69 | 95.67 | 92.05 | 94.32 | 95.85 | 94.18 | 96.45 | 93.74 | **96.78** |
| | TRAIN (S) | 33.97 | 49.15 | 240.29 | 146.07 | 74.70 | 117.25 | 140.74 | 77.35 | 6.47 | 7.80 | 4.72 | 14.18 | 22.82 | 12.39 | 14.00 |
| | TEST (S) | 0.93 | 5.51 | 618.21 | 1.95 | 0.90 | 2.17 | 3.06 | 10.33 | 16.22 | 30.58 | 5.14 | 3.22 | 90.13 | 5.16 | 7.94 |
| FIXED | 1 | **99.96** | 98.00 | 97.02 | 99.29 | 99.53 | 99.54 | 78.60 | 97.64 | 91.09 | 97.93 | 98.06 | 99.15 | 98.57 | 99.56 | 97.88 |
| | 2 | 92.49 | 88.42 | 87.98 | 70.00 | 88.80 | 92.14 | 72.05 | 88.49 | 88.30 | 84.49 | 92.62 | 83.21 | 94.47 | 95.78 | **95.89** |
| | 3 | 99.15 | 96.02 | 97.49 | 97.02 | 97.46 | 98.04 | 87.24 | 97.52 | 96.15 | 98.83 | 97.73 | 13.56 | 98.27 | 98.34 | **99.81** |
| | 4 | 99.98 | 99.97 | 99.80 | 99.88 | 99.83 | 99.55 | 98.03 | 99.39 | 99.84 | 99.19 | 97.79 | 100.00 | 99.88 | 99.98 | 99.91 |
| | 5 | 99.96 | 99.34 | 98.86 | 99.66 | 99.93 | 99.63 | 89.68 | 98.62 | 97.95 | 99.77 | 99.72 | 100.00 | 99.88 | 99.65 | 99.91 |
| | 6 | 93.07 | 90.60 | 87.25 | 82.28 | 89.21 | 93.59 | 63.06 | 88.57 | 86.28 | 85.95 | 94.10 | 74.52 | 90.69 | 90.95 | **95.03** |
| | OA | 98.53 | 97.39 | 96.66 | 95.31 | 97.66 | 98.25 | 86.14 | 96.71 | 95.42 | 96.43 | 97.61 | 94.67 | 98.21 | 98.45 | **98.75** |
| | AA | 97.44 | 95.39 | 94.73 | 91.36 | 95.79 | 97.08 | 81.44 | 95.04 | 93.27 | 94.36 | 96.67 | 78.41 | 96.96 | 97.38 | **98.07** |
| | K | 98.03 | 96.50 | 95.53 | 93.70 | 96.86 | 97.65 | 81.76 | 95.54 | 93.87 | 95.23 | 96.83 | 92.51 | 97.61 | 97.92 | **98.33** |
| | TRAIN (S) | 3.72 | 20.02 | 59.35 | 28.59 | 15.86 | 313.85 | 21.69 | 0.96 | 2.44 | 12.42 | 33.81 | 27.61 | 18.56 | 357.31 | 10.55 |
| | TEST (S) | 5.60 | 161.39 | 158.92 | 272.25 | 143.93 | 127.95 | 212.81 | 31.91 | 8.26 | 28.11 | 4.84 | 2.74 | 95.86 | 34.88 | 3.94 |
| 15% | 1 | 99.97 | 99.97 | 99.97 | 100.00 | 99.94 | 99.97 | 99.32 | 99.97 | 99.97 | 99.54 | 100.00 | 100.00 | 99.54 | 99.48 | 99.91 |
| | 2 | 97.95 | 98.53 | 97.62 | 92.46 | 95.97 | 96.77 | 87.49 | 95.61 | 97.48 | 90.69 | 93.20 | 96.42 | 97.17 | **99.76** | 98.91 |
| | 3 | 98.42 | 98.51 | 99.74 | 100.00 | 99.61 | 100.00 | 96.90 | 99.88 | 100.00 | 100.00 | 95.94 | 97.18 | 93.56 | 98.03 | 96.81 |
| | 4 | 100.00 | 100.00 | 99.99 | 100.00 | 100.00 | 99.98 | 99.83 | 99.99 | 100.00 | 99.79 | 97.90 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 5 | 99.93 | 99.94 | 99.99 | 100.00 | 99.99 | 99.99 | 99.66 | 99.99 | 99.86 | 99.17 | 99.67 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 6 | 98.04 | 98.64 | 97.82 | 91.98 | 96.19 | 96.89 | 89.66 | 95.90 | 98.29 | 90.22 | 97.39 | 96.25 | 94.48 | 98.96 | **99.93** |
| | OA | 99.54 | 99.67 | 99.52 | 98.43 | 99.19 | 99.35 | 97.41 | 99.13 | 99.52 | 97.69 | 98.26 | 99.18 | 98.98 | 99.77 | **99.82** |
| | AA | 99.05 | 99.27 | 99.20 | 97.41 | 98.62 | 98.93 | 95.48 | 98.56 | 99.27 | 96.57 | 97.35 | 98.31 | 97.46 | **99.37** | 99.26 |
| | K | 99.39 | 99.55 | 99.37 | 97.91 | 98.92 | 99.13 | 96.55 | 98.84 | 99.37 | 96.90 | 97.69 | 98.89 | 98.64 | 99.69 | **99.77** |
| | TRAIN (S) | 56.38 | 132.81 | 3102.87 | 432.79 | 127.05 | 264.82 | 379.41 | 234.07 | 181.82 | 21.52 | 62.16 | 120.66 | 22.83 | 166.87 | 131.69 |
| | TEST (S) | 0.85 | 5.59 | 611.66 | 1.56 | 0.87 | 2.14 | 2.45 | 10.32 | 14.39 | 30.57 | 5.01 | 2.51 | 90.71 | 4.54 | 6.84 |
| 20% | 1 | 100.00 | 100.00 | 99.97 | 99.19 | 99.88 | 100.00 | 98.99 | 99.86 | 99.95 | 99.41 | 100.00 | 100.00 | 99.61 | 99.81 | **100.00** |
| | 2 | 98.35 | 98.75 | 97.94 | 99.78 | 97.24 | 97.23 | 89.23 | 96.73 | 99.10 | 96.08 | 93.55 | 95.84 | 96.41 | **99.91** | 99.53 |
| | 3 | 98.76 | 100.00 | 99.75 | 100.00 | 99.88 | 100.00 | 98.38 | 99.22 | 99.39 | 98.75 | 96.15 | 98.27 | 92.89 | 98.69 | **100.00** |
| | 4 | 100.00 | 100.00 | 99.98 | 100.00 | 99.95 | 100.00 | 99.73 | 100.00 | 100.00 | 99.41 | 97.85 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 5 | 100.00 | 100.00 | 99.99 | 100.00 | 99.99 | 100.00 | 99.52 | 99.98 | 99.88 | 99.39 | 99.66 | 100.00 | 100.00 | 100.00 | **100.00** |
| | 6 | 98.45 | 98.84 | 98.03 | **99.72** | 97.46 | 97.43 | 90.19 | 96.93 | 99.21 | 89.78 | 97.67 | 96.48 | 95.58 | 99.25 | 99.57 |
| | OA | 99.63 | 99.75 | 99.58 | 99.84 | 99.43 | 99.46 | 97.53 | 99.32 | 99.77 | 98.08 | 98.31 | 99.21 | 99.02 | 99.87 | **99.90** |
| | AA | 99.26 | 99.60 | 99.28 | 99.78 | 99.07 | 99.11 | 96.01 | 98.79 | 99.59 | 97.14 | 97.48 | 98.43 | 97.42 | 99.61 | **99.85** |
| | K | 99.51 | 99.67 | 99.43 | 99.79 | 99.24 | 99.28 | 96.71 | 99.10 | 99.69 | 97.43 | 97.75 | 98.91 | 98.70 | 99.82 | **99.87** |
| | TRAIN (S) | 64.26 | 160.71 | 4155.20 | 534.10 | 146.48 | 306.98 | 477.72 | 324.89 | 262.33 | 25.29 | 82.75 | 160.30 | 23.02 | 224.65 | 163.15 |
| | TEST (S) | 0.82 | 5.38 | 556.15 | 1.55 | 0.94 | 2.20 | 2.54 | 11.89 | 11.41 | 30.47 | 4.89 | 2.58 | 90.50 | 4.38 | 6.27 |

TABLE X
ABLATION STUDY OF NINE CONFIGURATIONS ON IP WITH 5% TRAINING SAMPLES. THE BASE NETWORK KEEPS ALL SPECTRAL CHANNELS (S1 DISABLED), A PLAIN 3 × 3 CONVOLUTIONAL BLOCK (S2 DISABLED), AND A SINGLE-PATH FEATURE FLOW WITH THE NUCLEAR-NORM TERM DISABLED (S3 DISABLED). THE PCA-BASED INPUT PROJECTION, STANDARD 3 × 3 CONVOLUTIONAL BACKBONE, AND LINEAR CLASSIFICATION HEAD REMAIN FIXED. WE EVALUATE THE BASE NETWORK AND ALL COMBINATIONS OF **S1** (CONNECTOMICS-INSPIRED SSG), **S2** (LIAD), AND **S3** (TMI). DGPF-RENET IS INCLUDED AS AN EXTERNAL REFERENCE. RESULTS ARE AVERAGES OVER TEN INDEPENDENT RUNS

| MODEL CONFIGURATION | INNOVATIONS | | | METRICS (%) | | |
|---|---|---|---|---|---|---|
| | S1 | S2 | S3 | OA | AA | $\kappa$ |
| **DGPF-RENET (EXTERNAL SOTA)** | - | - | - | 94.44 | 87.78 | 93.66 |
| 1. BASE NETWORK | × | × | × | 86.72 | 79.84 | 84.91 |
| 2. BASE + S1 | ✓ | × | × | 90.18 | 85.07 | 88.74 |
| 3. BASE + S2 | × | ✓ | × | 88.46 | 82.59 | 86.83 |
| 4. BASE + S3 | × | × | ✓ | 87.91 | 81.36 | 86.20 |
| 5. BASE + S1 + S2 | ✓ | ✓ | × | 92.74 | 87.42 | 91.68 |
| 6. BASE + S1 + S3 | ✓ | × | ✓ | 91.83 | 86.51 | 90.69 |
| 7. BASE + S2 + S3 | × | ✓ | ✓ | 90.05 | 84.27 | 88.71 |
| **8. LIZARDNET (OURS)** | ✓ | ✓ | ✓ | **94.63** | **89.26** | **93.87** |

LizardNet records lower latency, power, energy, and posttest temperature.

### G. Ablation Study

We evaluate the contribution of each module on IP with a fixed 5% training ratio. DGPF-RENet [66] serves as the external baseline, and all $2^3$ combinations of SSG, LIAD, and TMI are included.

The proposed framework is constructed upon three core innovations: S1: connectomics-inspired SSG, S2: LIAD module, and S3: TMI module.

*1) Overall Contribution and Interaction of Innovations:* Table X presents the performance of the nine experimental configurations (one external baseline + eight internal variants). The "Base Network" is the standard CNN backbone with SSG, LIAD, and TMI disabled.

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TABLE XI
COMPARISON OF MODEL COMPLEXITY IN TERMS OF PARAMS (M) AND FLOPS (M)

| Model | Indian Pines | | DFC2013 Houston | | WHU-Hi-HongHu | | Trento | |
|---|---|---|---|---|---|---|---|---|
| | Params | FLOPs | Params | FLOPs | Params | FLOPs | Params | FLOPs |
| Cnn2D | 1.45 | 3.76 | 1.45 | 3.45 | 1.46 | 1.29 | 1.43 | 2.64 |
| DBDA | 0.38 | 107.97 | 0.28 | 59.51 | 0.51 | 19.00 | 0.13 | 26.02 |
| DwT | 61.92 | 110.70 | 61.86 | 93.94 | 61.99 | 27.29 | 61.78 | 40.04 |
| FDSSC | 1.19 | 174.29 | 0.85 | 106.68 | 1.61 | 30.30 | 0.37 | 41.64 |
| Hamida | 0.42 | 30.64 | 0.30 | 107.07 | 0.71 | 30.00 | 0.09 | 23.09 |
| He | 0.62 | 53.58 | 0.41 | 87.10 | 1.14 | 31.21 | 0.08 | 60.32 |
| Lee | 0.39 | 134.66 | 0.32 | 94.33 | 0.48 | 28.96 | 0.21 | 31.47 |
| RvT | 8.94 | 38.20 | 8.87 | 16.91 | 9.02 | 7.57 | 8.78 | 11.03 |
| CS2DT | 1.98 | 245.41 | 1.80 | 184.97 | 1.60 | 73.23 | 1.63 | 110.33 |
| DSFormer | 17.96 | 32.10 | 9.90 | 15.03 | 9.92 | 4.37 | 11.12 | 7.21 |
| HyperKAN | 0.15 | 85.47 | 0.15 | 3.86 | 0.15 | 18.67 | 0.15 | 10.80 |
| Swin-HSI | 0.09 | 116.77 | 0.09 | 1.41 | 0.09 | 3.74 | 0.09 | 8.53 |
| MambaHSI | 0.43 | 48.05 | 0.42 | 44.86 | 0.44 | 52.15 | 0.41 | 40.17 |
| DGPF-RENet | 10.88 | 21.87 | 10.14 | 20.39 | 11.81 | 23.72 | 9.07 | 18.25 |
| **Ours** | **0.87** | **2.77** | **0.82** | **2.67** | **0.94** | **2.90** | **0.75** | **2.52** |

TABLE XII
ON-BOARD CPU-ONLY DEPLOYMENT MEASUREMENTS ON THE JETSON ORIN NANO 8 GB (FP32 AND BATCH SIZE 1)

| Metric | LizardNet | DGPF-RENet |
|---|---|---|
| Checkpoint-loaded weights (M) | 0.918 | 10.88 |
| FLOPs (M) | 2.77 | 21.87 |
| Mean latency (ms) | **39.78** | 314.22 |
| Median latency (ms) | **39.53** | 312.32 |
| P95 latency (ms) | **42.57** | 336.29 |
| P99 latency (ms) | **46.31** | 365.81 |
| Maximum latency (ms) | **52.26** | 412.87 |
| Throughput (samples/s) | **25.14** | 3.18 |
| Idle board input power, VDD_IN (W) | 5.80 | 5.80 |
| Continuous-inference board power (W) | **10.60** | 11.70 |
| Incremental mean power (W) | **4.80** | 5.90 |
| Incremental peak power (W) | **8.90** | 10.20 |
| Per-sample incremental energy (J) | **0.191** | 1.854 |
| Post-test board temperature (°C) | **50** | 54 |

*a) Analysis of single modules:* Among rows 2–4, S1 gives the largest individual gain, increasing OA by 3.46 percentage points over the Base Network. S2 and S3 increase OA by 1.74 and 1.19 points, respectively. Thus, all three modules give positive individual gains, with the largest contribution from spectral gating in this experiment.

*b) Analysis of combined modules:* Rows 5–7 evaluate pairwise module interactions. The S1 + S2 configuration reaches 92.74% OA. This result is consistent with S1 supplying a spectrally filtered feature map to the boundary-aware S2 operator.

*c) Comparison with sota:* Each individual module improves the Base Network. The complete LizardNet in Row 8 obtains 94.63% OA, compared with 94.44% for DGPF-RENet. The full configuration therefore gives the highest OA in this ablation table.

*2) S1 (Connectomics-Inspired SSG):* To verify the bio-computational efficiency of the proposed spectral selection mechanism, we benchmark our ODE-inspired steady-state scoring against standard dimensionality reduction and functional attention methods.

Table XIII reports a controlled three-way comparison that keeps the backbone, global spatial aggregation, data split, and training protocol fixed. Replacing continuous soft weights with Top-$K$ binary gating increases OA from 91.34% to 94.42%, a gain of 3.08 percentage points, while retaining 0.88M parameters and reducing FLOPs from 2.90M to 2.77M. This result isolates the benefit of hard spectral sparsification. With the same Top-$K$ gating and the same 2.77M FLOPs, the full SSG further increases OA to 94.63%, exceeding the non-steady-state Top-$K$ variant by 0.21 percentage points while using 0.87M parameters. Overall, full SSG improves OA by 3.29 percentage points over GAP + soft gate with slightly fewer parameters and FLOPs. These results demonstrate that Top-$K$ hard gating provides the main sparsification gain and that the ODE-inspired steady-state score contributes additional discriminative accuracy beyond hard gating alone.

*3) S2 (LIAD):* We compare LIAD with fixed convolution, isotropic diffusion, and spatial attention under the same backbone.

Table XIV compares the spatial operators under the same backbone. The Base setting, 3 × 3 convolution, isotropic diffusion, spatial attention, and LIAD obtain 91.83%, 92.41%, 93.18%, 93.86%, and 94.63% OA, respectively. LIAD uses gradient-adaptive conductance to limit cross-boundary propagation and gives the highest OA in this comparison.

Table XV evaluates the number of LIAD updates. Increasing $T$ from 1 to 2 raises OA from 94.63% ± 0.21% to 94.81% ± 0.19% and increases FLOPs from 2.77M to 2.95M (1.07× cost). At $T = 3$, OA is 94.74% ± 0.23% with 3.13M FLOPs. At $T = 5$, OA decreases to 94.28% ± 0.27% and FLOPs increase to 3.49M (1.26 × ). The parameter count remains 0.87M because the diffusion operator shares parameters across updates. Table XIV shows that one LIAD update already provides a clear boundary-refinement gain, obtaining 94.63% OA compared with 92.41% for 3 × 3 convolution, 93.18% for isotropic diffusion, and 93.86% for spatial attention. A second update repeats the same parameter-shared diffusion operator. It adds no new boundary cue, learnable parameter, or processing mechanism, and obtains the additional 0.18

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TABLE XIII

CONTROLLED ABLATION OF SSG SCORING AND GATING ON IP (5% TRAINING). ALL VARIANTS USE THE SAME BACKBONE, GLOBAL SPATIAL AGGREGATION, DATA SPLIT, AND TRAINING PROTOCOL, AND DIFFER ONLY IN THEIR SCORING/GATING MECHANISM

| VARIANT | SCORING / GATING | OA (%) | PARAMS (M) | FLOPS (M) |
|---|---|---|---|---|
| GAP + SOFT GATE | GAP, SAME SCORE, CONTINUOUS SOFT WEIGHTS | 91.34 | 0.88 | 2.90 |
| GAP + TOP-$K$ HARD GATE | SAME SCORE, TOP-$K$ BINARY GATING | 94.42 | 0.88 | **2.77** |
| **FULL SSG** | **ODE-INSPIRED STEADY-STATE SCORE + TOP-$K$** | **94.63** | **0.87** | **2.77** |

TABLE XIV

INTERNAL ABLATION OF S2: COMPARISON OF SPATIAL ENHANCEMENT STRATEGIES ON IP (5% TRAINING). ALL VARIANTS SHARE THE SAME BACKBONE EQUIPPED WITH S1 AND S3, AND DIFFER ONLY IN THE SPATIAL ENHANCEMENT MODULE PLACED BETWEEN THEM

| METHOD | MECHANISM TYPE | OA (%) |
|---|---|---|
| NO SPATIAL | POINT-WISE ONLY | 91.83 |
| STANDARD CONV | FIXED RECEPTIVE FIELD | 92.41 |
| ISOTROPIC DIFFUSION | UNIFORM SMOOTHING | 93.18 |
| SPATIAL ATTENTION | GLOBAL CORRELATION | 93.86 |
| **OURS (LIAD)** | **EDGE-AWARE PDE** | **94.63** |

TABLE XV

EFFECT OF THE NUMBER OF LIAD DIFFUSION STEPS ON IP WITH 5% TRAINING SAMPLES. THE PARAMETER COUNT REMAINS UNCHANGED BECAUSE THE DIFFUSION OPERATOR SHARES PARAMETERS ACROSS ITERATIONS

| **Steps $T$** | **OA (%)** | **Params (M)** | **FLOPs (M)** | **Relative Cost** |
|---|---|---|---|---|
| 1 | $94.63 \pm 0.21$ | 0.87 | 2.77 | $1.00\times$ |
| 2 | $\mathbf{94.81 \pm 0.19}$ | 0.87 | 2.95 | $1.07\times$ |
| 3 | $94.74 \pm 0.23$ | 0.87 | 3.13 | $1.13\times$ |
| 5 | $94.28 \pm 0.27$ | 0.87 | 3.49 | $1.26\times$ |

TABLE XVI

SENSITIVITY TO THE ADDITIVE EDGE-RESIDUAL COEFFICIENT $\lambda$ ON IP WITH 5% TRAINING SAMPLES. BECAUSE $\lambda$ CHANGES ONLY THE RESIDUAL AMPLITUDE, PARAMETERS AND FLOPS REMAIN CONSTANT

| $\lambda$ | **OA (%)** | **AA (%)** | $\kappa$ **(%)** | **Params (M)** | **FLOPs (M)** |
|---|---|---|---|---|---|
| 0.0 | $94.01 \pm 0.26$ | 88.57 | 93.18 | 0.87 | 2.77 |
| 0.1 | $94.16 \pm 0.25$ | 88.73 | 93.35 | 0.87 | 2.77 |
| 0.2 | $94.31 \pm 0.24$ | 88.89 | 93.52 | 0.87 | 2.77 |
| 0.3 | $94.44 \pm 0.23$ | 89.04 | 93.67 | 0.87 | 2.77 |
| 0.4 | $94.56 \pm 0.22$ | 89.17 | 93.80 | 0.87 | 2.77 |
| **0.5** | $\mathbf{94.63 \pm 0.21}$ | **89.26** | **93.87** | 0.87 | 2.77 |
| 0.6 | $94.58 \pm 0.22$ | 89.20 | 93.81 | 0.87 | 2.77 |
| 0.7 | $94.45 \pm 0.23$ | 89.05 | 93.65 | 0.87 | 2.77 |
| 0.8 | $94.29 \pm 0.24$ | 88.88 | 93.47 | 0.87 | 2.77 |
| 0.9 | $94.10 \pm 0.26$ | 88.66 | 93.26 | 0.87 | 2.77 |
| 1.0 | $93.89 \pm 0.28$ | 88.41 | 93.02 | 0.87 | 2.77 |

percentage points through 7% more sequential computation. Additional iterations do not produce a sustained gain, and the decrease at $T = 5$ indicates that repeated propagation can oversmooth discriminative boundaries and small structures. We retain $T = 1$ as the default deployment setting because it completes the intended edge-conditioned residual correction at the lowest computational cost. Using repeated updates to pursue a marginal additional gain would weaken the lightweight design objective of the proposed biologically inspired framework. The $T = 2$ setting remains an optional high-accuracy configuration when the additional computation is acceptable.

Table XVI shows a unimodal trend. Increasing $\lambda$ from 0 to 0.5 raises OA from 94.01% $\pm$ 0.26% to 94.63% $\pm$ 0.21%, AA from 88.57% to 89.26%, and $\kappa$ from 93.18% to 93.87%. Relative to $\lambda = 0$, the selected value improves OA by 0.62 percentage points and both AA and $\kappa$ by 0.69 points. Values above 0.5 reduce all three metrics; at $\lambda = 1.0$, OA, AA, and $\kappa$ are 93.89% $\pm$ 0.28%, 88.41%, and 93.02%. The results indicate that a moderate edge residual complements the learned multiscale branch and that a large residual suppresses task-adaptive features. Parameters and FLOPs remain 0.87M and 2.77M because $\lambda$ changes the residual amplitude and leaves the architecture fixed. We therefore select $\lambda = 0.5$.

*4) S3 (TMI):* Finally, we analyze the impact of the Low-Rank constraint imposed on the integration weights.

TABLE XVII

INTERNAL ABLATION OF S3 ON IP (5% TRAINING): INTEGRATION STRATEGIES FOR THE TECTAL MULTIPATH ROUTING. ALL VARIANTS SHARE THE SAME BACKBONE EQUIPPED WITH S1 AND S2

| INTEGRATION STRATEGY | RANK CONSTRAINT | OA (%) |
|---|---|---|
| SINGLE PATH | – | 92.74 |
| MULTI-PATH | FULL RANK | 93.71 |
| **MULTI-PATH** | **LOW-RANK** ($\|\cdot\|_*$) | **94.63** |

Table XVII compares TMI with its unregularized counterpart under the same backbone. Nuclear-norm regularization encourages a lower effective rank and gives a higher OA in this comparison. It does not reduce the stored parameter count or inference FLOPs. Table XVIII separately evaluates the regularization coefficient $\alpha$ across all four datasets.

### H. Visualization

To provide an intuitive comparison of classification performance, this section visualizes the classification maps generated by LizardNet and 14 representative models on four benchmark hyperspectral datasets. Compared with quantitative metrics, classification maps offer a direct assessment of spatial consistency and misclassification patterns. As shown in the results, LizardNet produces more coherent classification maps with clearer spatial structures across different datasets, which is consistent with the quantitative performance reported earlier.

*1) Results on the IP:* Fig. 3 shows the IP classification maps. Several competing models contain salt-and-pepper errors in the fragmented "corn" and "soybean" regions. The LizardNet map contains more homogeneous regions and follows the ground-truth boundaries more closely under the 5% training setting.

*2) Results on the DFC2013 Houston:* Fig. 4 shows the DFC2013 Houston classification maps. Several baselines contain fragmented predictions around the spectrally similar "residential" and "commercial" regions and discontinuities along the narrow "road," "highway," and "railway" structures.

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TABLE XVIII

CROSS-DATASET SENSITIVITY TO THE NUCLEAR-NORM COEFFICIENT $\alpha$. (A) OA AND THE TRAINING–VALIDATION OA GAP UNDER THE STANDARD SETTING OF EACH DATASET. (B) MODEL COMPLEXITY AT THE SELECTED $\alpha = 1 \times 10^{-3}$. BECAUSE $\alpha$ CHANGES ONLY THE TRAINING OBJECTIVE, PARAMETER COUNTS AND FLOPS ARE CONSTANT WITH RESPECT TO $\alpha$ FOR EACH DATASET

**(a)**

| $\alpha$ | Indian Pines OA / Gap (%) | DFC2013 Houston OA / Gap (%) | WHU-Hi-HongHu OA / Gap (%) | Trento OA / Gap (%) |
|---|---|---|---|---|
| 0 | 93.71/6.84 | 94.08/7.12 | 86.91/9.46 | 98.31/2.74 |
| $1 \times 10^{-5}$ | 93.96/6.18 | 94.26/6.45 | 87.16/8.71 | 98.42/2.38 |
| $1 \times 10^{-4}$ | 94.27/5.21 | 94.55/5.38 | 87.49/7.35 | 98.57/1.91 |
| $5 \times 10^{-4}$ | 94.49/4.35 | 94.78/4.46 | 87.68/5.92 | 98.69/1.47 |
| $\mathbf{1 \times 10^{-3}}$ | **94.63/3.76** | **94.92/3.81** | **87.73/4.86** | **98.75/1.22** |
| $5 \times 10^{-3}$ | 93.91/2.63 | 94.31/2.74 | 87.05/3.41 | 98.39/0.94 |
| $1 \times 10^{-2}$ | 92.84/1.97 | 93.46/2.06 | 85.92/2.53 | 97.81/0.72 |

**(b)**

| Dataset | Selected $\alpha$ | Params (M) | FLOPs (M) |
|---|---|---|---|
| Indian Pines | $1 \times 10^{-3}$ | 0.87 | 2.77 |
| DFC2013 Houston | $1 \times 10^{-3}$ | 0.82 | 2.67 |
| WHU-Hi-HongHu | $1 \times 10^{-3}$ | 0.94 | 2.90 |
| Trento | $1 \times 10^{-3}$ | 0.75 | 2.52 |

*Discussion:* Table XVIII shows a consistent cross-dataset optimum at $\alpha = 1 \times 10^{-3}$. Relative to $\alpha = 0$, the selected coefficient improves OA by 0.92, 0.84, 0.82, and 0.44 percentage points on Indian Pines, DFC2013 Houston, WHU-Hi-HongHu, and Trento, respectively. The training–validation gap decreases by 3.08, 3.31, 4.60, and 1.52 percentage points. These two changes indicate reduced overfitting at moderate regularization. Values above $1 \times 10^{-3}$ continue to shrink the gap and reduce OA on every dataset. At $\alpha = 1 \times 10^{-2}$, OA falls to 92.84%, 93.46%, 85.92%, and 97.81%, indicating underfitting from excessive rank suppression. We therefore use $\alpha = 1 \times 10^{-3}$ for all datasets. The coefficient appears only in the loss, so the architecture, parameter count, and FLOPs remain fixed. At the selected value, the four datasets require 0.87M/2.77M, 0.82M/2.67M, 0.94M/2.90M, and 0.75M/2.52M parameters/FLOPs, respectively.

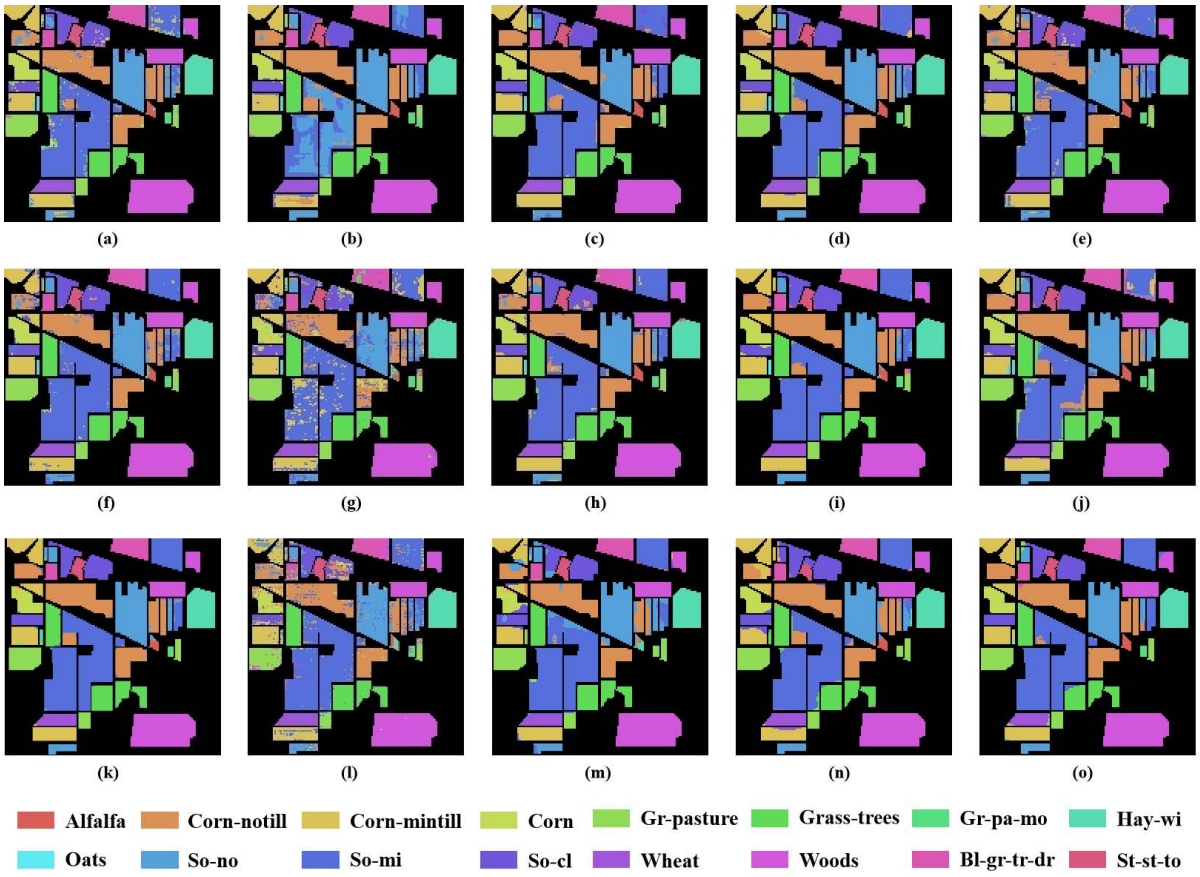



Fig. 3. Classification maps on IP. (a) Cnn2D. (b) DBDA. (c) DwT. (d) FDSSC. (e) Hamida. (f) He. (g) Lee. (h) RvT. (i) CS2DT. (j) DSFormer. (k) HyperKAN. (l) Swin-HSI. (m) MambaHSI. (n) DGPF-RENet. (o) LizardNet.

The LizardNet map preserves more continuous linear structures and produces more coherent predictions within the urban blocks. These spatial patterns agree with the classwise and aggregate results reported in Tables V and VI.

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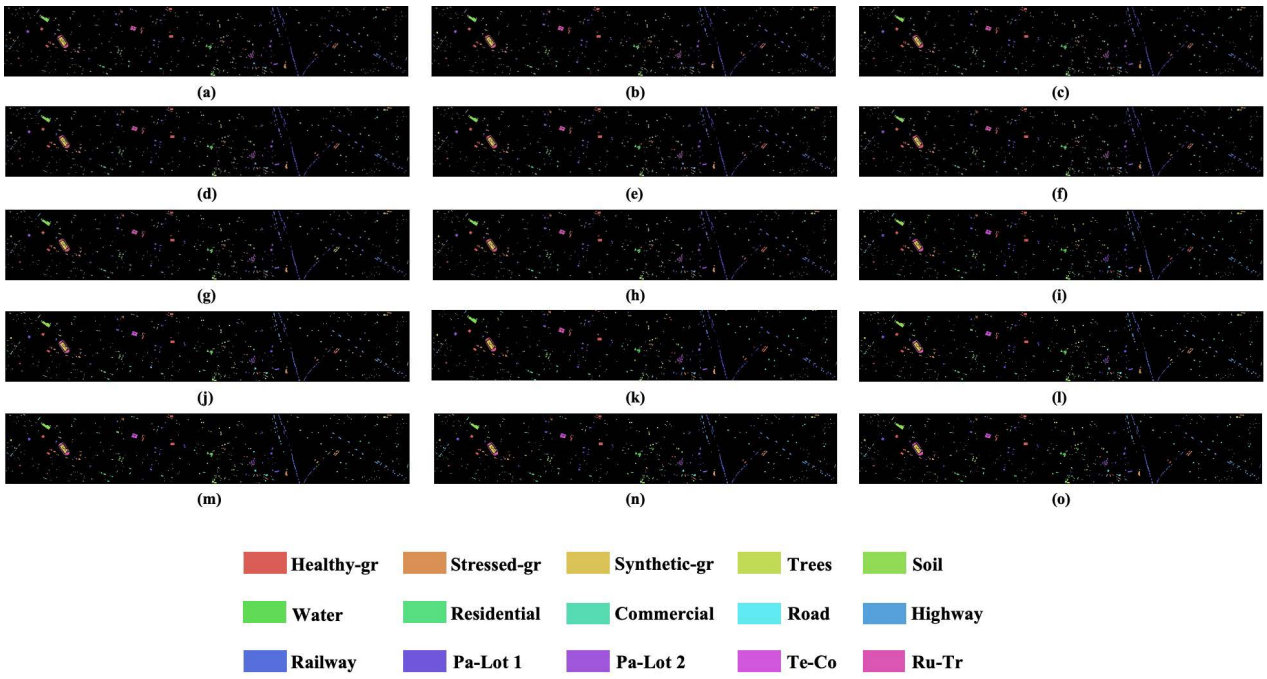



Fig. 4. Classification maps on DFC2013 Houston. (a) Cnn2D. (b) DBDA. (c) DwT. (d) FDSSC. (e) Hamida. (f) He. (g) Lee. (h) RvT. (i) CS2DT. (j) DSFormer. (k) HyperKAN. (l) Swin-HSI. (m) MambaHSI. (n) DGPF-RENet. (o) LizardNet.

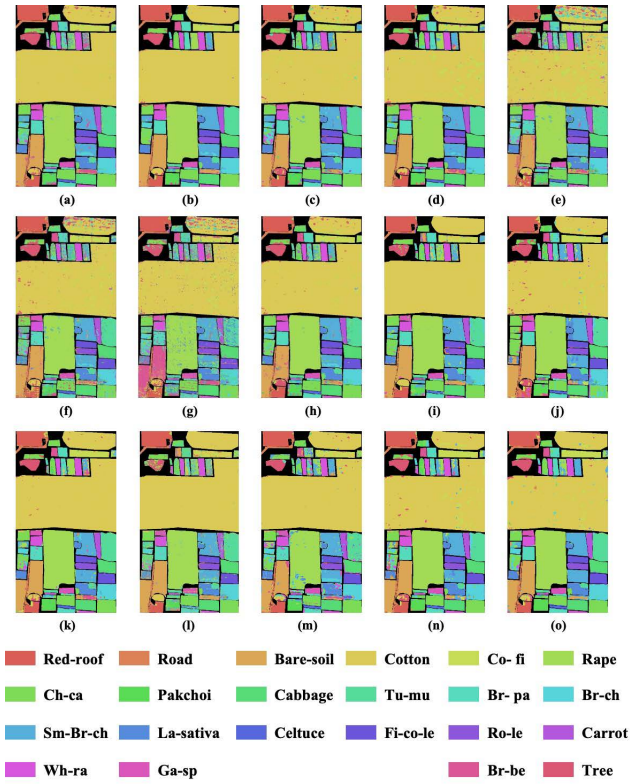



Fig. 5. Classification maps on WHU-Hi-HongHu. (a) Cnn2D. (b) DBDA. (c) DwT. (d) FDSSC. (e) Hamida. (f) He. (g) Lee. (h) RvT. (i) CS2DT. (j) DSFormer. (k) HyperKAN. (l) Swin-HSI. (m) MambaHSI. (n) DGPF-RENet. (o) LizardNet.

*3) Results on the WHU-Hi-HongHu:* Fig. 5 shows the WHU-Hi-HongHu classification maps. Several baselines produce irregular transitions along narrow plot boundaries, including the "cabbage" and "pakchoi" regions. The LizardNet map shows smoother within-plot predictions and clearer plot boundaries.

*4) Results on the Trento:* Fig. 6 shows the Trento classification maps. Several models produce fragmented predictions within the "vineyards" and "woods" regions. LizardNet produces more contiguous regions that follow the ground-truth land-use patterns.

*5) Dataset-Specific t-SNE and Silhouette Analysis:* Figs. 7–10 present high-resolution, class-colored t-SNE visualizations for IP, DFC2013 Houston, WHU-Hi-HongHu, and Trento, respectively. All panels are generated from the corresponding real model outputs using the same latent-feature evaluation pipeline. Within each dataset, all fifteen methods use the same samples and ground-truth class colors. Each panel is identified by a method letter, includes its SC value, and is accompanied by a complete class-name legend

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{22}$$

where $a(i)$ is the mean distance from sample $i$ to the other samples in its own class and $b(i)$ is the minimum mean distance from sample $i$ to any other class. The overall SC is the mean of $s(i)$ across samples; a larger value indicates greater intraclass compactness and interclass separation. The calculated Silhouette Coefficients across all methods on the four datasets are summarized in Table XIX.

*6) Qualitative Comparison With Lightweight Models:* The qualitative classification maps of different lightweight HSI classifiers are presented in Figs. 11–14. Compared with the ground-truth maps, these visual results provide intuitive evaluations of spatial consistency, boundary preservation, and robustness to spectral ambiguity.

Fig. 11 shows more compact prediction regions for LizardNet on IP. Several lightweight baselines contain fragmented regions and isolated errors among spectrally similar vegetation

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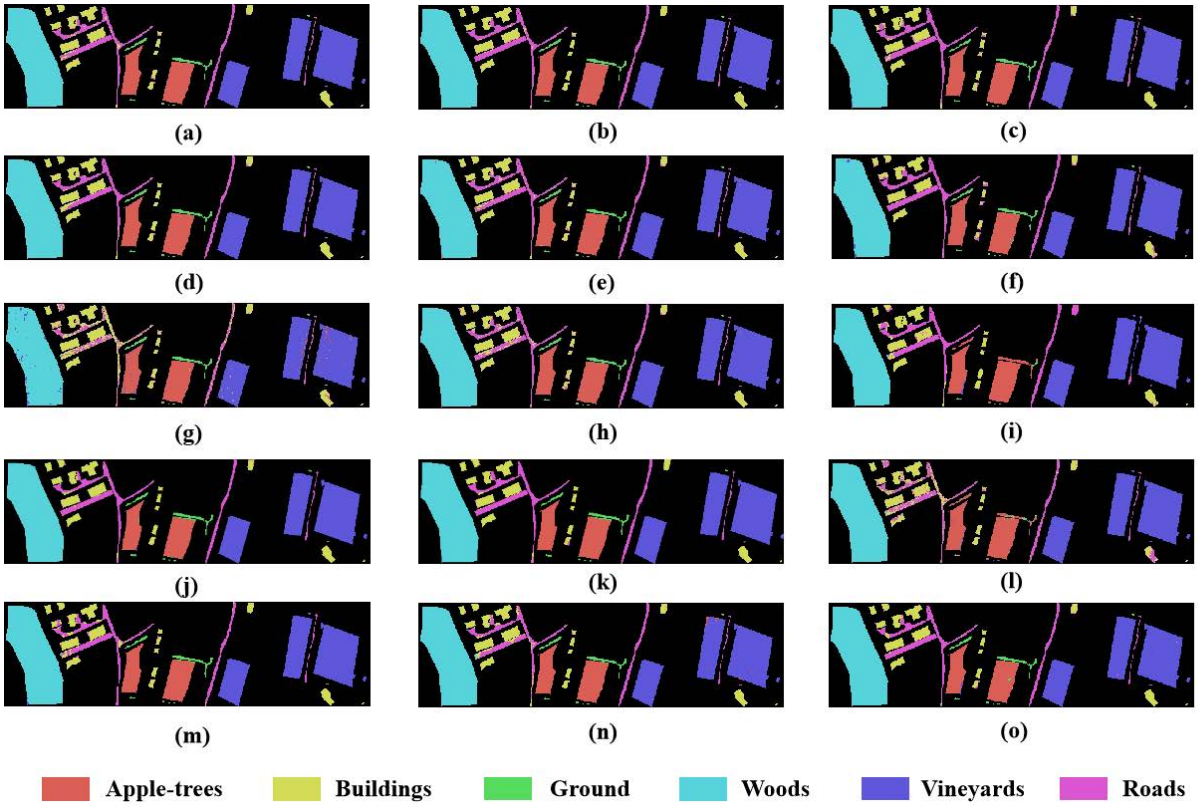



Fig. 6. Classification visible results for the Trento HSI by (a) Cnn2D, (b) DBDA, (c) DwT, (d) FDSSC, (e) Hamida, (f) He, (g) Lee, (h) RvT, (i) CS2DT, (j) DSFormer, (k) HyperKAN, (l) Swin-HSI, (m) MambaHSI, (n) DGPF-RENet, and (o) Ours.

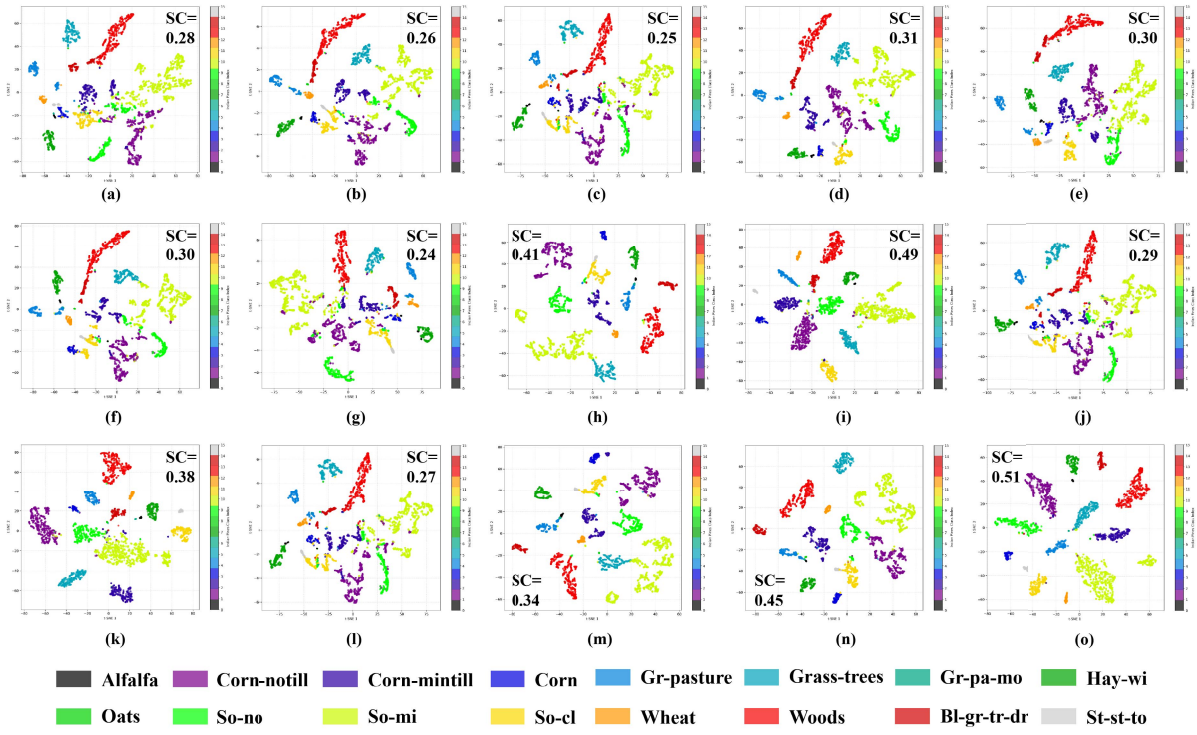



Fig. 7. Dataset-specific t-SNE feature visualization on IP. (a) Cnn2D. (b) DBDA. (c) DwT. (d) FDSSC. (e) Hamida. (f) He. (g) Lee. (h) RvT. (i) CS2DT. (j) DSFormer. (k) HyperKAN. (l) Swin-HSI. (m) MambaHSI. (n) DGPF-RENet. (o) LizardNet. Colors denote ground-truth classes, and SC denotes the Silhouette Coefficient computed from the corresponding latent features.

classes. This visual pattern is consistent with the OA and classwise results in the preceding tables.

Fig. 13 shows the WHU-Hi-HongHu classification maps. LizardNet produces more continuous agricultural regions and

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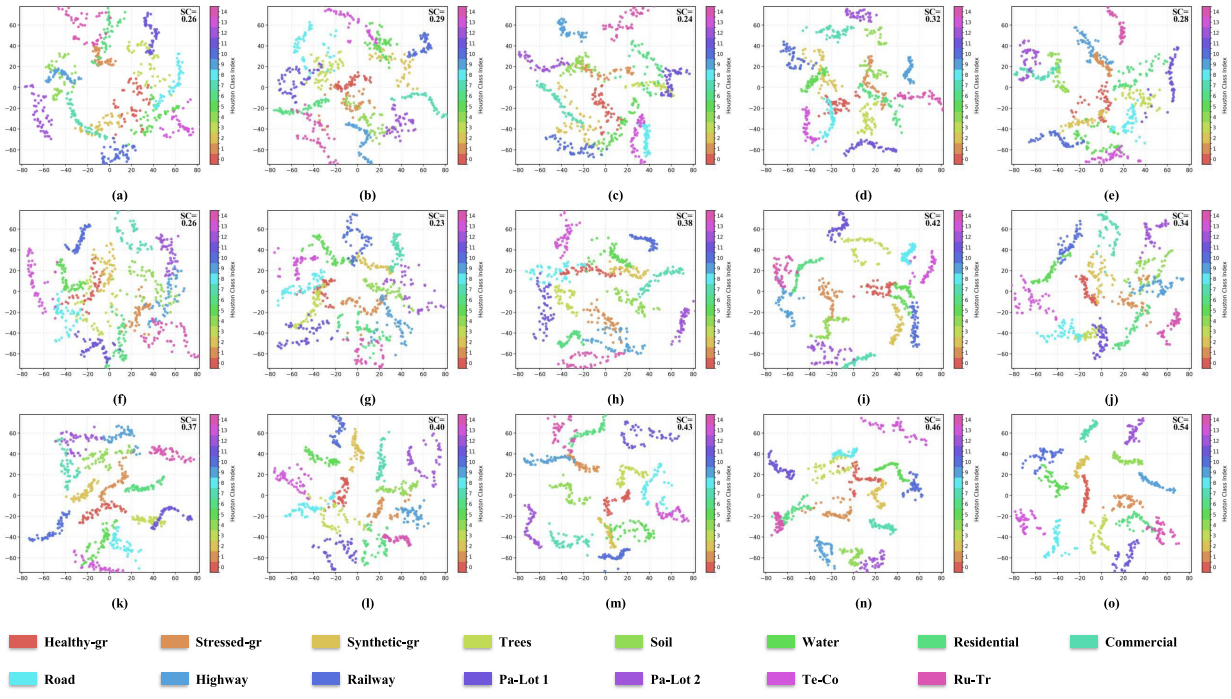



Fig. 8. Dataset-specific t-SNE feature visualization on DFC2013 Houston. (a) Cnn2D. (b) DBDA. (c) DwT. (d) FDSSC. (e) Hamida. (f) He. (g) Lee. (h) RvT. (i) CS2DT. (j) DSFormer. (k) HyperKAN. (l) Swin-HSI. (m) MambaHSI. (n) DGPF-RENet. (o) LizardNet. Colors denote the 15 ground-truth classes; the class-name legend and panelwise SC values are provided in the figure.

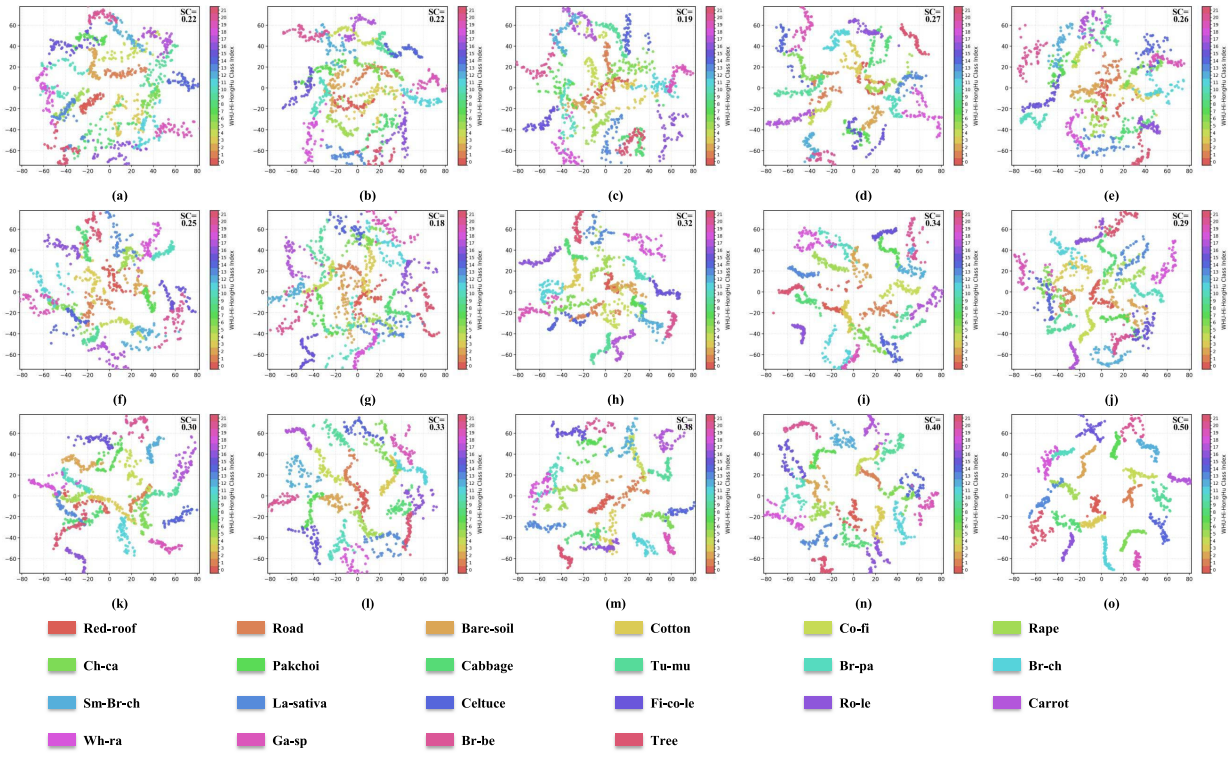



Fig. 9. Dataset-specific t-SNE feature visualization on WHU-Hi-HongHu. (a) Cnn2D. (b) DBDA. (c) DwT. (d) FDSSC. (e) Hamida. (f) He. (g) Lee. (h) RvT. (i) CS2DT. (j) DSFormer. (k) HyperKAN. (l) Swin-HSI. (m) MambaHSI. (n) DGPF-RENet. (o) LizardNet. Colors denote the 22 ground-truth classes; the class-name legend and panelwise SC values are provided in the figure.

fewer irregular class transitions than the compared lightweight models. This visual result agrees with the quantitative table under the same training condition.

For DFC2013 Houston and Trento (see Figs. 12 and 14), the LizardNet maps retain finer structures and clearer class boundaries. Several lightweight baselines show

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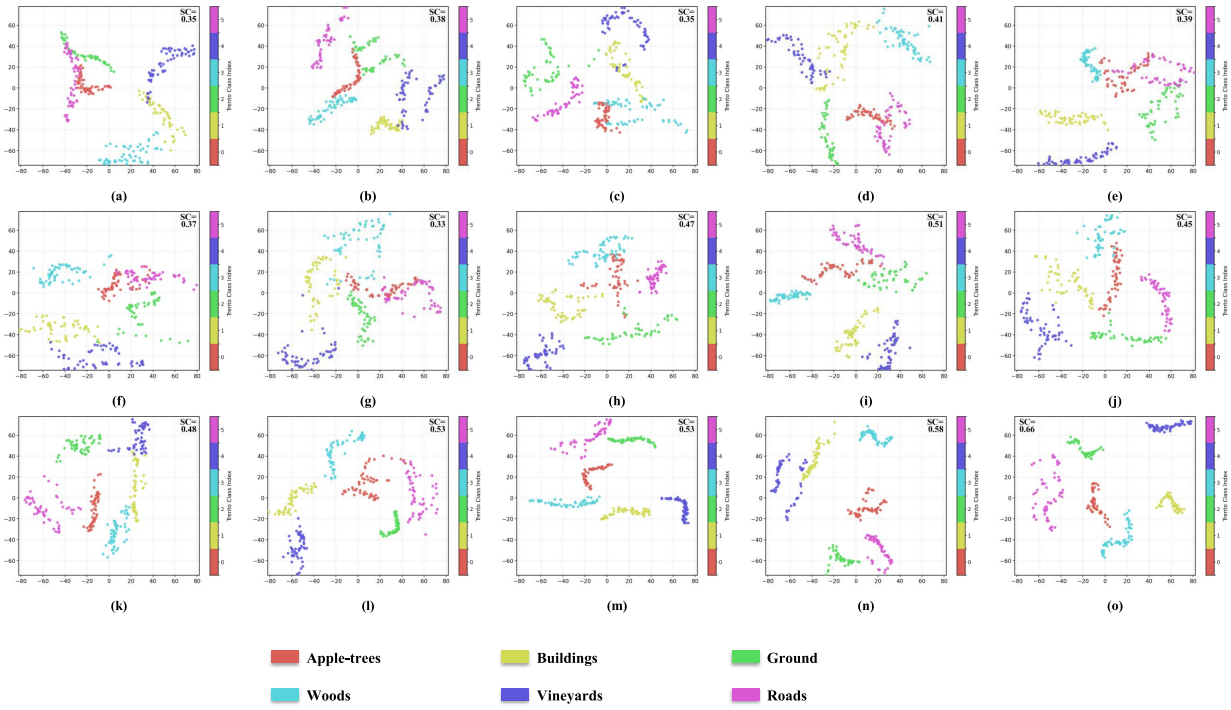



Fig. 10. Dataset-specific t-SNE feature visualization on Trento. (a) Cnn2D. (b) DBDA. (c) DwT. (d) FDSSC. (e) Hamida. (f) He. (g) Lee. (h) RvT. (i) CS2DT. (j) DSFormer. (k) HyperKAN. (l) Swin-HSI. (m) MambaHSI. (n) DGPF-RENet. (o) LizardNet. Colors denote the six ground-truth classes; the class-name legend and panelwise SC values are provided in the figure.

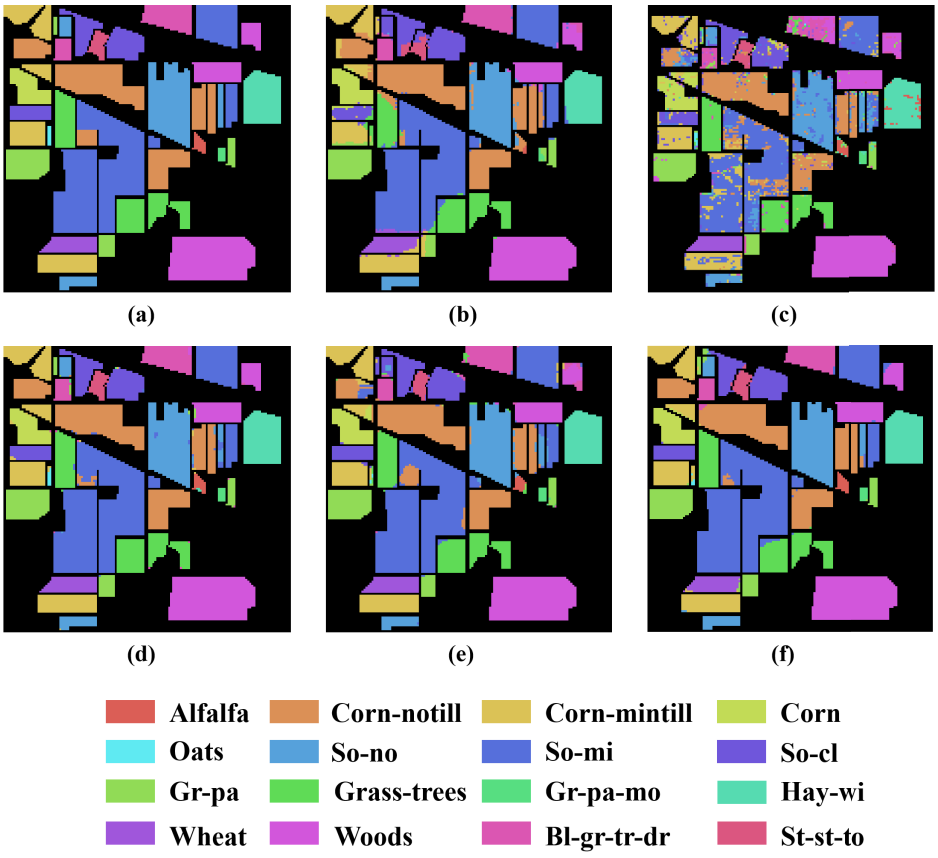



Fig. 11. Qualitative comparison on IP using lightweight HSI classification networks. (a) Ground truth. (b) MADANet. (c) SpectralFormer-Light. (d) SSFTT. (e) SSTN. (f) LizardNet.

boundary smoothing or local fluctuations in urban and mixed-landscape regions. LIAD supplies boundary-conditioned spatial refinement, and TMI supplies compact feature aggregation.

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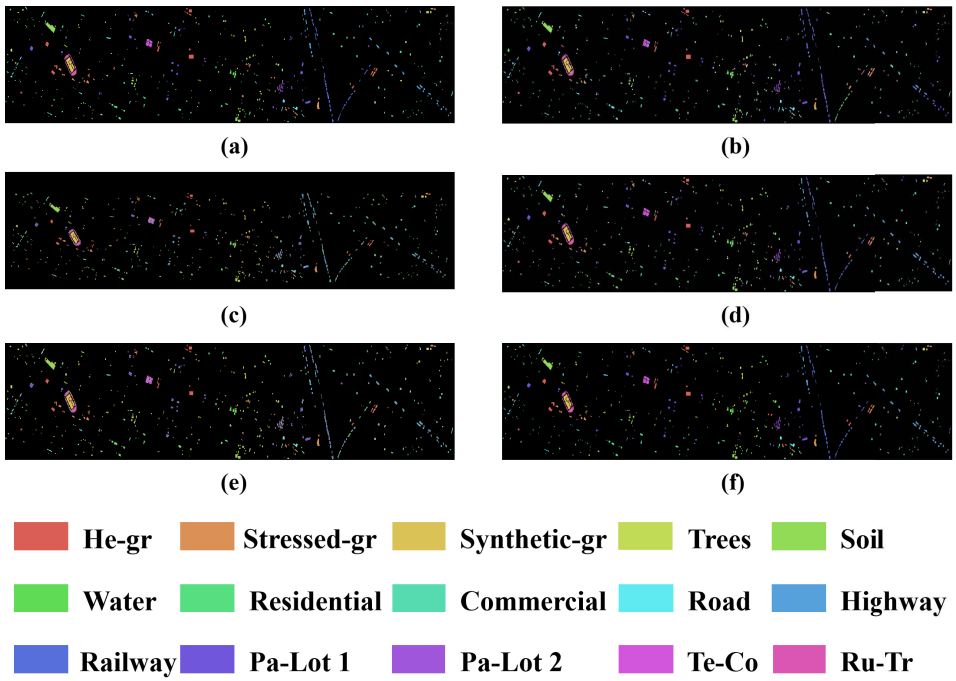



Fig. 12. Qualitative comparison on DFC2013 Houston using lightweight HSI classification networks. (a) Ground truth. (b) MADANet. (c) SpectralFormer-Light. (d) SSFTT. (e) SSTN. (f) LizardNet.

Across the four visualization sets, the LizardNet maps contain fewer isolated errors and more continuous class regions. These observations agree with the corresponding quantitative accuracy results.

LizardNet combines the reported classification accuracies with 0.75M–0.94M parameters and 2.52M–2.90M FLOPs across the four datasets. The results support explicit spectral, spatial, and low-rank constraints as a design option for lightweight HSI classifiers.

*7) Discussion:* The experiments evaluate representation capability and computational cost jointly. Very small parameter counts reduce memory consumption, and the accuracy results show the associated representation limits for some compressed models.

LizardNet achieves an effective accuracy–efficiency tradeoff by introducing biologically inspired structural constraints into feature learning. The SSG module reduces redundant spectral interactions through adaptive feature selection, the LIAD module enhances spatial discrimination by considering local structural variations, and the TMI module promotes low-effective-rank feature integration and small-sample generalization.

## *I. Comprehensive Accuracy-Efficiency Analysis*

Computational efficiency affects deployment on resource-constrained platforms such as UAVs and satellites. We compare LizardNet with four lightweight HSI classifiers that cover different design strategies. SSFTT [8] uses spectral–spatial tokenization, SSTN [79] uses compact spectral–spatial interaction, MADANet [80] combines convolution and attention, and SpectralFormer-Light reduces Transformer depth and embedding dimension. The comparison reports accuracy, parameters, and FLOPs for each model.

*1) Accuracy Evaluation Under Limited Training Samples:* Tables XX–XXIII summarize the classwise accuracies, OA, AA, and $\kappa$ coefficients obtained by different lightweight classifiers under various training settings.

On DFC2013 Houston (see Table XXI), LizardNet obtains 86.34% OA with 1% training samples, 4.26 percentage points above SSFTT. Its OA reaches 99.75% at the 20% training ratio.

On WHU-Hi-HongHu (see Table XXII), LizardNet obtains 87.73% OA with 25 samples per class, jointly the highest value with MADANet among the compared lightweight networks. For class 8, LizardNet, SSFTT, and SSTN obtain 92.91%, 71.68%, and 50.68% accuracy, respectively.

Similar improvements are observed on the Trento dataset (see Table XXIII). LizardNet achieves 98.75% OA under the fixed training setting and further increases to 99.90% with 20% training samples, demonstrating its robustness under both limited and sufficient supervision conditions.

Overall, these results show that LizardNet extracts separable spectral–spatial representations with the limited training sets used in these experiments.

*2) Computational Complexity and Accuracy–Cost Tradeoff:* To further investigate the computational characteristics of lightweight HSI classifiers, the model complexity and accuracy–cost tradeoff are analyzed in Table XXIV. The number of parameters, FLOPs, and an empirical FLOPs efficiency indicator are jointly reported to provide a comprehensive comparison between classification performance and computational burden

$$\eta_{\text{FLOPs}} = \frac{OA\,(\%)}{\text{FLOPs}\,(M)} \tag{23}$$

where $OA(\%)$ denotes the OA, and FLOPs($M$) represents the computational cost measured in millions of floating-point

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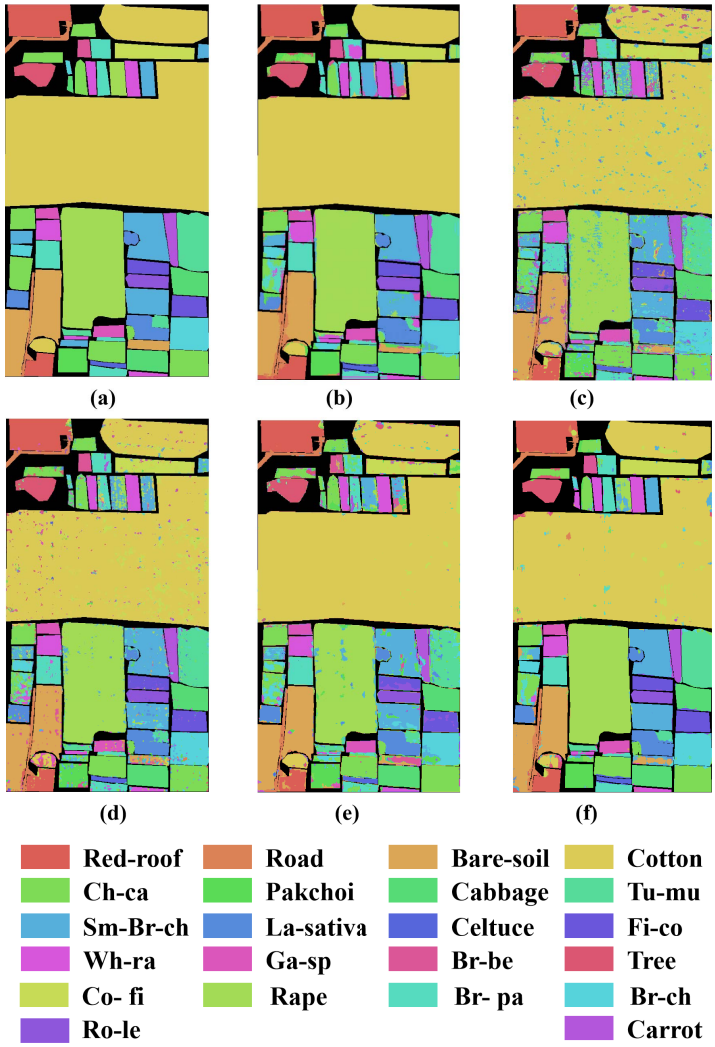



Fig. 13. Qualitative comparison on WHU-Hi-HongHu using lightweight HSI classification networks. (a) Ground truth. (b) MADANet. (c) SpectralFormer-Light. (d) SSFTT. (e) SSTN. (f) LizardNet.

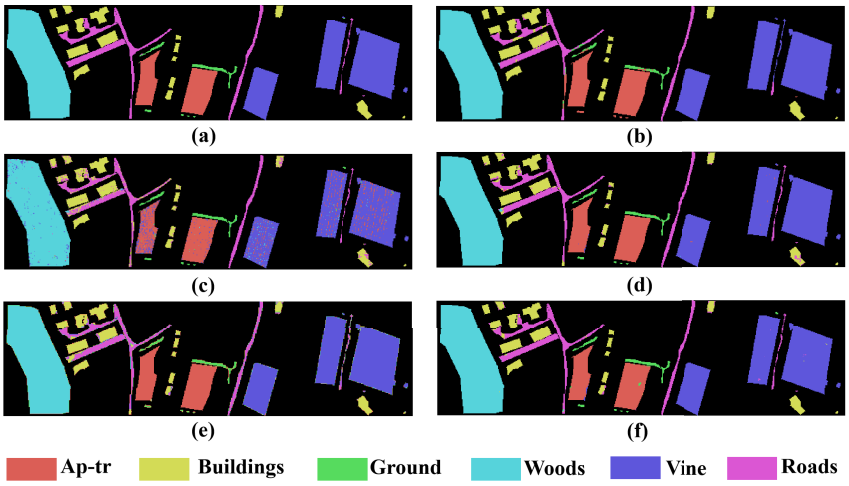



Fig. 14. Qualitative comparison on Trento using lightweight HSI classification networks. (a) Ground truth. (b) MADANet. (c) SpectralFormer-Light. (d) SSFTT. (e) SSTN. (f) LizardNet.

operations. A higher $\eta_{\text{FLOPs}}$ indicates that more classification accuracy is obtained per unit of computational cost. This metric is only used as an empirical indicator for relative efficiency comparison under identical experimental settings.

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TABLE XIX
SILHOUETTE COEFFICIENTS COMPUTED FROM THE REAL LATENT MODEL OUTPUTS USED FOR THE FOUR DATASET-SPECIFIC T-SNE VISUALIZATIONS. THE BEST VALUE FOR EACH DATASET IS SHOWN IN BOLD

| Method | Indian Pines | DFC2013 Houston | WHU-Hi-HongHu | Trento |
|---|---|---|---|---|
| Cnn2D | 0.28 | 0.26 | 0.22 | 0.35 |
| DBDA | 0.26 | 0.29 | 0.22 | 0.38 |
| DwT | 0.25 | 0.24 | 0.19 | 0.35 |
| FDSSC | 0.31 | 0.32 | 0.27 | 0.41 |
| Hamida | 0.30 | 0.28 | 0.26 | 0.39 |
| He | 0.30 | 0.26 | 0.25 | 0.37 |
| Lee | 0.24 | 0.23 | 0.18 | 0.33 |
| RvT | 0.41 | 0.38 | 0.32 | 0.47 |
| CS2DT | 0.49 | 0.42 | 0.34 | 0.51 |
| DSFormer | 0.29 | 0.34 | 0.29 | 0.45 |
| HyperKAN | 0.38 | 0.37 | 0.30 | 0.48 |
| Swin-HSI | 0.27 | 0.40 | 0.33 | 0.53 |
| MambaHSI | 0.34 | 0.43 | 0.38 | 0.53 |
| DGPF-RENet | 0.45 | 0.46 | 0.40 | 0.58 |
| **LizardNet** | **0.51** | **0.54** | **0.50** | **0.66** |

*Discussion:* Table XIX reports the results on all four datasets. LizardNet obtains the highest SC on Indian Pines (0.51), DFC2013 Houston (0.54), WHU-Hi-HongHu (0.50), and Trento (0.66). The strongest competing values are 0.49, 0.46, 0.40, and 0.58, giving margins of 0.02, 0.08, 0.10, and 0.08. WHU-Hi-HongHu contains 22 fine-grained crop classes with similar spectra, and Trento contains six classes; LizardNet gives the highest SC in both settings. The t-SNE panels show greater class mixing for several convolutional baselines and tighter within-class groups for LizardNet. The SC values are computed from the same latent outputs used in each panel and provide a numerical measure of this separation. The result is observed consistently on all four datasets.

For a consistent complexity comparison, all FLOPs in Table XXIV are computed using 15 × 15 input patches, matching the training and testing protocol. The accuracy and parameter-count results remain unchanged. LizardNet is motivated by selective information processing in reptilian visual pathways, where task-relevant responses are emphasized and redundant interactions are suppressed. FLOPs directly quantify the arithmetic workload required to realize this selective processing and therefore provide the primary measure of its computational effect. Parameter count is retained as a complementary measure of model-storage demand. Under this common protocol, LizardNet has the lowest FLOPs on every dataset: 2.77M on IP, 2.67M on DFC2013 Houston, 2.90M on WHU-Hi-HongHu, and 2.52M on Trento. The corresponding ranges are 2.66M–6.81M for SSFTT, 3.34M–5.62M for SSTN, 16.80M for MADANet, and 50.81M–219.90M for SpectralFormer-Light. LizardNet uses 0.75M–0.94M parameters, corresponding to at most 3.76 MB of raw FP32 weights, so its absolute storage demand also remains small. Compared with SSTN, LizardNet reduces FLOPs by 43.1%, 36.9%, 48.4%, and 24.6%, while increasing OA by 7.04, 11.40, 2.67, and 0.15 percentage points on the four datasets, respectively. It also achieves higher OA with lower FLOPs than SSFTT in all four settings. Its $\eta_{\text{FLOPs}}$ values are 34.16, 32.34, 30.25, and 38.73, each the highest value in the corresponding comparison. The extremely small parameter counts of SSTN and MADANet provide a storage advantage. LizardNet achieves the highest or joint-highest accuracy, the lowest arithmetic workload, and the best FLOPs efficiency under the common input protocol. These results establish the most favorable accuracy and efficiency balance among the evaluated lightweight methods.

SpectralFormer-Light has a relatively compact parameter count and a higher FLOP count because it retains dense token-wise interactions. On WHU-Hi-HongHu, it requires 219.90M FLOPs.

The corrected results also permit a direct Pareto interpretation. Relative to MADANet, LizardNet reduces FLOPs by 83.5%, 84.1%, 82.7%, and 85.0% on IP, DFC2013 Houston, WHU-Hi-HongHu, and Trento, respectively. The corresponding OA differences are +6.59, +42.55, 0.00, and +3.70 percentage points. Relative to SpectralFormer-Light, the FLOPs reductions are 98.3%, 97.7%, 98.7%, and 95.0%, with OA gains of 32.84, 25.40, 31.81, and 16.06 percentage points. These comparisons cover datasets with substantially different spectral dimensions and class structures, yet the arithmetic-cost ranking remains consistent. No evaluated baseline combines lower FLOPs with higher OA than LizardNet under the common 15 × 15 protocol. SSTN and MADANet retain an advantage in stored parameter count, which is relevant to checkpoint memory. Their higher FLOPs indicate greater arithmetic reuse of those compact weights.

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TABLE XX

CLASSIFICATION RESULTS ON THE IP DATASET (5%, 10%, 15%, AND 20%)

| 5% | | | | | | 10% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 0.00 | 0.00 | 47.73 | **82.93** | 81.82 | 1 | 14.29 | 58.54 | 76.19 | 73.17 | **96.10** |
| 2 | 68.85 | 39.01 | 89.17 | **94.40** | 92.72 | 2 | 84.61 | 71.91 | 94.00 | 93.31 | **97.57** |
| 3 | 72.70 | 43.15 | **94.42** | 70.98 | 92.18 | 3 | 89.37 | 63.05 | **98.28** | 95.72 | 97.66 |
| 4 | 61.11 | 40.44 | 84.00 | 15.04 | **91.15** | 4 | 92.22 | 77.93 | 94.91 | 99.07 | **99.30** |
| 5 | 87.16 | 36.90 | **99.56** | 93.25 | 89.61 | 5 | 90.98 | 78.11 | **99.77** | 94.71 | 95.66 |
| 6 | 79.60 | 90.91 | **99.71** | 93.52 | 97.41 | 6 | 88.27 | 93.61 | **99.70** | 98.17 | 99.16 |
| 7 | 0.00 | 3.85 | 51.85 | **100.00** | 75.56 | 7 | 0.00 | 56.00 | 100.00 | 100.00 | 85.60 |
| 8 | 97.25 | 99.12 | 100.00 | 100.00 | 99.78 | 8 | 100.00 | 72.79 | 100.00 | 100.00 | **100.00** |
| 9 | 0.00 | 0.00 | **73.68** | 73.33 | 43.16 | 9 | 0.00 | 5.56 | **77.78** | 20.00 | 66.67 |
| 10 | 84.82 | 45.61 | **97.62** | 87.12 | 93.10 | 10 | 85.50 | 73.34 | **97.85** | 90.17 | 97.21 |
| 11 | 95.71 | 80.53 | 96.40 | 95.93 | **96.50** | 11 | 97.48 | 73.43 | 97.94 | 97.60 | **98.75** |
| 12 | 70.89 | 13.14 | 79.57 | 40.60 | **91.67** | 12 | 72.00 | 53.28 | 94.44 | **98.31** | 96.35 |
| 13 | 50.32 | 89.18 | **100.00** | 93.85 | 98.26 | 13 | 83.87 | 83.70 | 99.47 | **100.00** | 98.53 |
| 14 | 97.61 | 91.92 | **99.67** | 96.92 | 97.74 | 14 | 97.81 | 89.81 | **100.00** | 96.58 | 99.51 |
| 15 | **94.88** | 21.86 | 81.47 | 94.01 | 93.95 | 15 | 96.59 | 26.80 | 96.01 | 95.98 | **98.47** |
| 16 | 59.15 | 81.82 | 82.95 | **95.45** | 93.60 | 16 | 64.79 | 85.54 | **98.82** | 89.29 | 96.43 |
| OA | 88.04 | 61.79 | 94.04 | 87.59 | **94.63** | OA | 90.34 | 73.27 | 97.53 | 95.79 | **98.11** |
| AA | 63.75 | 48.59 | 86.11 | 82.96 | **89.26** | AA | 72.36 | 66.46 | 95.32 | 90.13 | **95.19** |
| κ | 86.28 | 55.60 | 93.20 | 85.70 | **93.87** | κ | 88.98 | 69.60 | 97.19 | 95.20 | **97.85** |

| 15% | | | | | | 20% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 45.71 | 56.41 | 97.44 | 80.00 | **97.50** | 1 | 68.57 | 61.11 | 97.22 | 86.49 | **100.00** |
| 2 | 89.40 | 77.00 | 97.58 | 97.69 | **99.18** | 2 | 91.80 | 79.25 | 97.34 | 97.99 | **99.48** |
| 3 | 94.92 | 60.71 | 99.86 | 99.15 | 98.73 | 3 | 96.67 | 76.20 | 98.93 | 97.74 | **99.55** |
| 4 | 92.78 | 82.59 | 98.49 | 86.63 | **100.00** | 4 | 98.89 | 87.30 | **100.00** | 98.42 | **100.00** |
| 5 | 91.26 | 84.88 | 98.28 | 97.81 | **99.27** | 5 | 91.80 | 87.56 | 99.48 | 99.48 | 98.45 |
| 6 | 90.79 | 93.06 | **100.00** | 98.07 | 99.36 | 6 | 92.60 | 94.35 | **99.48** | 99.32 | **99.49** |
| 7 | 14.29 | 69.57 | 100.00 | 100.00 | **100.00** | 7 | 47.62 | 81.82 | 100.00 | 100.00 | **100.00** |
| 8 | 100.00 | 83.25 | 100.00 | 100.00 | **100.00** | 8 | 100.00 | 91.62 | 100.00 | 100.00 | **100.00** |
| 9 | 0.00 | 17.65 | 82.35 | **100.00** | **100.00** | 9 | 0.00 | 43.75 | 93.75 | 93.33 | **100.00** |
| 10 | 89.16 | 88.38 | 98.90 | 96.74 | **99.52** | 10 | 95.93 | 82.63 | 98.31 | 97.17 | **99.74** |
| 11 | 98.02 | 81.11 | 99.47 | 97.03 | 98.71 | 11 | 99.41 | 76.48 | 99.69 | 99.13 | 98.57 |
| 12 | 78.44 | 62.30 | 98.39 | 95.84 | **99.60** | 12 | 90.00 | 77.00 | 98.93 | 96.84 | **99.37** |
| 13 | 87.74 | 98.28 | 100.00 | 100.00 | **100.00** | 13 | 96.77 | 100.00 | 100.00 | 100.00 | **100.00** |
| 14 | 98.02 | 95.07 | 99.53 | 97.96 | **100.00** | 14 | 98.02 | 97.43 | 100.00 | 99.51 | **100.00** |
| 15 | 96.59 | 32.62 | 99.69 | 99.39 | **99.70** | 15 | 100.00 | 54.22 | 100.00 | 95.79 | **99.68** |
| 16 | 71.83 | 92.41 | 97.44 | 100.00 | 98.77 | 16 | 84.51 | 87.84 | 94.52 | 97.33 | **100.00** |
| OA | 90.76 | 79.73 | 99.07 | 97.43 | **99.29** | OA | 92.65 | 82.41 | 99.11 | 98.45 | **99.48** |
| AA | 77.43 | 73.45 | 97.96 | 96.64 | **99.40** | AA | 84.54 | 79.91 | 98.60 | 97.41 | **99.64** |
| κ | 89.44 | 76.87 | 98.94 | 97.07 | **99.19** | κ | 91.60 | 80.06 | 98.99 | 98.23 | **99.40** |

TABLE XXI

CLASSIFICATION RESULTS ON THE DFC2013 HOUSTON DATASET (1%, 5%, 15%, AND 20%)

| 1% | | | | | | 5% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 7.26 | 84.65 | **98.63** | 93.38 | 91.03 | 1 | 73.47 | 89.90 | 94.28 | **98.74** | 97.85 |
| 2 | 0.00 | 72.20 | 87.11 | 66.02 | **93.31** | 2 | 48.58 | **98.82** | 98.57 | 97.40 | 98.36 |
| 3 | 23.96 | 87.25 | **99.71** | 98.12 | 98.84 | 3 | 58.87 | 98.04 | **100.00** | 99.85 | 99.59 |
| 4 | 10.58 | 85.87 | 87.09 | 90.83 | **96.59** | 4 | 15.98 | 91.53 | **99.41** | 93.99 | 96.50 |
| 5 | 51.64 | 83.65 | **100.00** | **100.00** | 97.32 | 5 | 83.14 | 98.30 | **100.00** | **100.00** | 99.99 |
| 6 | 10.98 | 81.00 | **85.71** | 74.06 | 84.78 | 6 | 57.32 | 85.71 | 93.20 | 77.02 | **95.91** |
| 7 | 7.58 | 41.04 | 83.19 | **86.07** | 73.47 | 7 | 23.78 | 50.00 | 91.45 | 91.95 | **92.91** |
| 8 | 37.04 | 42.40 | 65.75 | 52.44 | **72.24** | 8 | 54.71 | 74.94 | **90.61** | 57.87 | 89.92 |
| 9 | 17.77 | 58.03 | 60.61 | **77.82** | 74.74 | 9 | 39.75 | 73.09 | 75.44 | 81.01 | **83.65** |
| 10 | 34.12 | 34.43 | **83.21** | 34.73 | 75.31 | 10 | 39.48 | 71.93 | 93.14 | 88.16 | **94.59** |
| 11 | 0.85 | 44.03 | **86.43** | 73.43 | 77.02 | 11 | 13.86 | 46.80 | 91.22 | 72.83 | **93.21** |
| 12 | 9.07 | 37.46 | 69.70 | 60.28 | **88.94** | 12 | 37.03 | 66.10 | 91.89 | 89.59 | **97.19** |
| 13 | 0.00 | 0.00 | 26.51 | 20.26 | **92.46** | 13 | 10.67 | 17.98 | 66.82 | 0.01 | **89.10** |
| 14 | 24.92 | 93.85 | 81.60 | 98.11 | **98.58** | 14 | 27.08 | 93.10 | 98.77 | **100.00** | 99.98 |
| 15 | 82.47 | 92.65 | 99.54 | 100.00 | **100.00** | 15 | 89.84 | 98.25 | 99.04 | **100.00** | 99.95 |
| OA | 43.79 | 60.94 | 82.08 | 74.94 | **86.34** | OA | 54.40 | 77.01 | 92.60 | 85.80 | **94.92** |
| AA | 21.22 | 62.57 | 80.99 | 75.04 | **87.64** | AA | 44.90 | 76.97 | 92.26 | 83.23 | **95.25** |
| κ | 39.36 | 57.80 | 80.60 | 72.87 | **85.24** | κ | 50.56 | 75.14 | 92.00 | 84.62 | **94.51** |

| 15% | | | | | | 20% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 94.00 | 99.15 | **100.00** | 99.53 | 99.91 | 1 | **100.00** | 95.80 | **100.00** | 92.51 | 99.80 |
| 2 | 88.35 | 94.27 | **99.91** | 90.34 | **99.91** | 2 | 92.76 | 91.72 | 99.90 | 96.02 | **100.00** |
| 3 | 79.43 | 99.83 | 100.00 | 100.00 | **100.00** | 3 | 99.43 | 99.82 | 100.00 | 99.82 | **100.00** |
| 4 | 28.04 | 93.00 | 99.43 | 94.23 | **99.91** | 4 | 44.87 | 98.69 | 99.50 | 93.47 | **99.90** |
| 5 | 98.09 | 98.29 | 99.81 | 99.62 | **100.00** | 5 | 98.20 | 98.49 | 100.00 | 98.79 | **100.00** |
| 6 | 66.67 | 96.38 | 98.19 | **99.64** | 97.83 | 6 | 72.36 | 96.92 | **100.00** | 99.62 | 97.69 |
| 7 | 56.91 | 84.03 | 96.75 | 95.92 | **100.00** | 7 | 61.47 | 86.59 | 97.64 | 93.79 | **100.00** |
| 8 | 57.99 | 91.20 | 98.49 | 84.12 | **100.00** | 8 | 68.89 | 89.05 | 99.30 | 96.08 | **100.00** |
| 9 | 70.87 | 76.79 | 95.30 | 96.43 | **97.74** | 9 | 81.18 | 72.43 | 98.10 | 98.10 | **98.30** |
| 10 | 54.51 | 90.31 | 100.00 | 99.90 | **100.00** | 10 | 63.63 | 86.34 | 100.00 | 99.49 | **100.00** |
| 11 | 37.95 | 58.72 | 97.71 | 97.43 | **100.00** | 11 | 81.45 | 69.84 | 99.80 | 94.03 | **100.00** |
| 12 | 82.28 | 74.62 | **99.71** | 95.52 | 98.95 | 12 | 94.56 | 81.44 | **99.90** | 96.25 | 99.59 |
| 13 | 42.13 | 39.70 | 99.75 | 70.68 | **100.00** | 13 | 77.81 | 37.60 | 99.20 | 94.68 | **100.00** |
| 14 | 30.77 | 92.29 | 100.00 | 99.45 | **100.00** | 14 | 85.85 | 95.32 | 100.00 | 99.71 | **100.00** |
| 15 | 94.62 | 98.22 | 100.00 | 100.00 | **100.00** | 15 | 100.00 | 97.92 | 100.00 | 100.00 | **100.00** |
| OA | 54.68 | 86.19 | 98.88 | 95.16 | **99.66** | OA | 59.54 | 87.03 | 99.48 | 96.37 | **99.75** |
| AA | 65.51 | 85.79 | 99.00 | 94.85 | **99.62** | AA | 81.50 | 86.53 | 99.56 | 96.82 | **99.69** |
| κ | 50.81 | 85.06 | 98.79 | 94.77 | **99.63** | κ | 56.14 | 85.96 | 99.44 | 96.07 | **99.73** |

LizardNet therefore occupies the strongest accuracy–FLOPs operating point in this comparison while keeping its raw FP32 weight storage below 3.76 MB. This distinction supports the lightweight claim through measured arithmetic efficiency and

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TABLE XXII

CLASSIFICATION RESULTS ON THE WHU-HI-HONGHU DATASET (25, 50, 75, AND 100)

| 25 | | | | | | 50 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 88.12 | 81.70 | **93.98** | 89.30 | 89.39 | 1 | 87.74 | 94.91 | **95.54** | 94.10 | 92.25 |
| 2 | 90.86 | 69.95 | 84.34 | 79.52 | **91.46** | 2 | **98.52** | 90.79 | 89.35 | 96.19 | 96.40 |
| 3 | **95.23** | 78.86 | 84.19 | 89.13 | 81.85 | 3 | 81.90 | 81.47 | **87.70** | 80.83 | 84.91 |
| 4 | 90.06 | 54.00 | 86.21 | **95.35** | 88.06 | 4 | 89.74 | 89.25 | 90.76 | **97.82** | 92.84 |
| 5 | **99.26** | 54.79 | 83.19 | 72.02 | 96.00 | 5 | 92.95 | 88.10 | 94.07 | 92.01 | **98.51** |
| 6 | 80.60 | 66.56 | 90.21 | 87.51 | **93.31** | 6 | 90.68 | 84.68 | 93.27 | 92.30 | **96.72** |
| 7 | 82.46 | 42.24 | 64.03 | 72.07 | 78.25 | 7 | **84.46** | 77.44 | 78.53 | 72.14 | 83.65 |
| 8 | 88.62 | 24.13 | 71.68 | 50.68 | **92.91** | 8 | 88.94 | 63.81 | 81.00 | 77.75 | **98.76** |
| 9 | **94.42** | 85.32 | 93.84 | 94.32 | 93.98 | 9 | **98.74** | 97.90 | 97.57 | 95.85 | 94.31 |
| 10 | **89.28** | 14.57 | 70.33 | 44.01 | 86.24 | 10 | 74.99 | 59.48 | 83.15 | 74.11 | **90.00** |
| 11 | 60.17 | 32.69 | 50.81 | 63.89 | **85.75** | 11 | **92.70** | 79.12 | 76.76 | 67.83 | 91.02 |
| 12 | **88.78** | 61.26 | 45.96 | 76.30 | 78.57 | 12 | **93.03** | 50.04 | 80.76 | 65.34 | 88.76 |
| 13 | **89.90** | 36.13 | 63.41 | 59.50 | 76.89 | 13 | 69.20 | 71.14 | 74.40 | 68.47 | **82.98** |
| 14 | **92.17** | 59.02 | 73.50 | 88.92 | 91.55 | 14 | **99.15** | 82.67 | 90.23 | 84.72 | 95.55 |
| 15 | **100.00** | 62.44 | 97.85 | 97.85 | 98.85 | 15 | 96.22 | 95.80 | 93.85 | 98.63 | **99.01** |
| 16 | 87.99 | 76.41 | 85.12 | 53.63 | **95.90** | 16 | 91.90 | 88.35 | 97.16 | 89.36 | **97.65** |
| 17 | **100.00** | 61.78 | 96.65 | 96.65 | 98.98 | 17 | **100.00** | 81.45 | 97.38 | 90.84 | 99.67 |
| 18 | 85.56 | 52.51 | 95.52 | 96.33 | **96.38** | 18 | 96.56 | 83.52 | 98.09 | **99.21** | 97.70 |
| 19 | **89.98** | 61.95 | 86.30 | 86.85 | 88.08 | 19 | 91.08 | 81.12 | 90.05 | 88.28 | **93.47** |
| 20 | **99.91** | 62.87 | 77.35 | 91.42 | 96.34 | 20 | 94.43 | **98.28** | 94.43 | 97.06 | 97.49 |
| 21 | **100.00** | 35.23 | 97.62 | 30.24 | 97.97 | 21 | **100.00** | 81.77 | 98.16 | 80.05 | 99.80 |
| 22 | 98.78 | 58.28 | 91.28 | **99.23** | 96.87 | 22 | 94.48 | 90.58 | 96.47 | 89.00 | **98.17** |
| OA | **87.73** | 55.92 | 81.62 | 85.06 | **87.73** | OA | 88.39 | 84.18 | 88.92 | 89.23 | **92.04** |
| AA | 90.55 | 56.03 | 81.06 | 77.94 | **90.62** | AA | 91.25 | 82.35 | 89.94 | 86.00 | **94.07** |
| K | **85.57** | 49.36 | 77.44 | 81.33 | 84.92 | K | 85.67 | 80.47 | 86.25 | 86.45 | **90.12** |

| 75 | | | | | | 100 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 81.39 | 95.62 | 94.57 | 92.23 | **96.78** | 1 | **96.50** | 94.00 | 95.85 | **96.50** | 96.34 |
| 2 | 68.97 | 91.13 | 93.00 | 89.76 | **97.29** | 2 | 96.45 | 92.94 | 95.42 | 98.83 | **98.92** |
| 3 | 82.74 | 83.52 | 86.87 | 88.21 | **93.52** | 3 | 84.91 | 82.86 | 92.73 | 80.25 | **96.49** |
| 4 | **98.47** | 88.01 | 91.25 | 98.03 | 94.59 | 4 | **99.49** | 91.18 | 91.15 | 97.76 | 97.71 |
| 5 | **99.45** | 88.80 | 90.67 | 85.19 | 97.72 | 5 | **98.80** | 93.18 | 93.35 | 94.33 | 97.53 |
| 6 | 90.37 | 87.70 | 93.02 | 88.12 | **97.83** | 6 | 91.89 | 86.79 | 95.31 | 93.95 | **98.31** |
| 7 | 78.87 | 78.82 | 81.49 | 82.28 | **93.38** | 7 | 73.21 | 71.87 | 83.99 | 76.22 | **90.78** |
| 8 | 81.86 | 73.84 | 79.99 | 84.54 | **100.00** | 8 | 94.43 | 77.64 | 87.30 | 72.58 | **99.90** |
| 9 | 94.32 | **98.51** | 97.91 | 98.32 | 91.49 | 9 | 91.90 | 97.30 | 98.45 | 97.20 | **99.02** |
| 10 | 69.17 | 65.93 | 88.01 | 73.06 | **93.96** | 10 | 78.65 | 73.63 | 88.12 | 91.97 | **94.74** |
| 11 | 57.32 | 72.21 | 84.69 | 66.82 | **97.39** | 11 | 77.57 | 81.40 | 89.53 | 82.65 | **94.02** |
| 12 | 73.79 | 77.88 | 86.51 | 70.36 | **90.01** | 12 | 88.69 | 77.56 | 82.76 | 73.23 | **95.54** |
| 13 | 68.93 | 74.81 | **81.53** | 71.38 | 78.18 | 13 | 76.12 | 77.10 | 80.09 | 83.70 | **88.83** |
| 14 | 85.76 | 87.25 | 91.77 | 95.10 | **97.61** | 14 | 90.10 | 83.05 | 93.17 | 76.23 | **96.73** |
| 15 | 93.96 | 97.20 | 97.12 | 99.14 | **99.24** | 15 | **99.89** | 97.23 | 98.86 | 99.22 | 99.11 |
| 16 | 84.26 | 93.27 | 93.90 | 79.89 | **98.12** | 16 | 97.28 | 91.55 | 96.69 | 95.03 | **98.90** |
| 17 | 100.00 | 92.88 | 96.49 | 100.00 | **100.00** | 17 | 99.32 | 88.63 | 98.16 | 99.45 | **99.97** |
| 18 | 95.24 | 92.74 | 97.85 | 90.45 | **99.62** | 18 | 98.83 | 96.70 | 97.54 | 95.00 | **99.71** |
| 19 | 75.72 | 82.11 | 93.33 | 88.51 | **95.43** | 19 | 91.41 | 91.19 | 94.15 | 93.30 | **96.27** |
| 20 | 81.02 | 98.36 | 90.93 | **99.03** | 92.70 | 20 | 95.55 | 97.49 | 97.56 | 98.70 | **99.14** |
| 21 | 99.92 | 96.33 | 99.67 | 89.62 | **100.00** | 21 | 99.92 | 97.96 | 99.83 | 97.96 | **100.00** |
| 22 | **100.00** | 96.02 | 97.61 | 96.19 | 92.08 | 22 | 96.37 | 94.82 | 96.76 | 97.54 | **98.50** |
| OA | 88.68 | 85.70 | 90.13 | 90.16 | **94.17** | OA | 92.37 | 87.21 | 91.17 | 92.07 | **96.57** |
| AA | 84.62 | 86.95 | 91.28 | 87.56 | **95.32** | AA | 91.70 | 88.00 | 93.04 | 90.53 | **97.11** |
| K | 85.77 | 82.40 | 87.74 | 87.62 | **92.70** | K | 90.37 | 84.14 | 89.03 | 90.01 | **95.67** |

TABLE XXIII

CLASSIFICATION RESULTS ON THE TRENTO DATASET (1%, FIXED, 15%, AND 20%)

| 1% | | | | | | FIXED | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 98.00 | 60.98 | 98.80 | 99.35 | **99.84** | 1 | 100.00 | 90.69 | 99.06 | 99.66 | 97.88 |
| 2 | 68.49 | 79.19 | 83.96 | **94.33** | 88.36 | 2 | **97.82** | 88.48 | 80.13 | 95.43 | 95.89 |
| 3 | 10.03 | 93.04 | 34.39 | 79.54 | **95.41** | 3 | 75.27 | 96.98 | 93.41 | 98.03 | **99.81** |
| 4 | 99.96 | 97.30 | 99.75 | **99.99** | 99.94 | 4 | 100.00 | 98.47 | 99.68 | 99.99 | 99.91 |
| 5 | 99.92 | 81.54 | 98.40 | **99.96** | 99.92 | 5 | 100.00 | 84.05 | 99.95 | 99.97 | 99.91 |
| 6 | 87.19 | 62.67 | 85.17 | 84.92 | **89.09** | 6 | 85.12 | 80.85 | **96.25** | 91.25 | 95.03 |
| OA | 93.89 | 81.53 | 95.07 | 97.44 | **97.59** | OA | 97.84 | 89.58 | 97.35 | 98.55 | **98.75** |
| AA | 77.26 | 79.12 | 83.41 | 93.01 | **95.43** | AA | 93.04 | 89.92 | 94.75 | 97.39 | **98.07** |
| K | 91.81 | 75.24 | 93.41 | 96.58 | **96.78** | K | 97.11 | 86.23 | 96.47 | 98.07 | **98.33** |

| 15% | | | | | | 20% | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS | CLASS | MADANET | SPECTRALFORMER-LIGHT | SSFTT | SSTN | OURS |
| 1 | 98.59 | 87.43 | 99.91 | **99.92** | 99.91 | 1 | 100.00 | 76.88 | 100.00 | 99.97 | **100.00** |
| 2 | 98.37 | 89.14 | **98.91** | 96.56 | **98.91** | 2 | 98.07 | 83.81 | 98.67 | 98.22 | **99.53** |
| 3 | 88.12 | **99.02** | 98.77 | 98.61 | 96.81 | 3 | 95.09 | 97.13 | 99.48 | 97.79 | **100.00** |
| 4 | 100.00 | 97.45 | 100.00 | 99.99 | **100.00** | 4 | 100.00 | 97.97 | 100.00 | 99.99 | **100.00** |
| 5 | 100.00 | 89.98 | 100.00 | 99.99 | **100.00** | 5 | 99.93 | 94.94 | 100.00 | 100.00 | **100.00** |
| 6 | 95.93 | 84.35 | 96.44 | 97.37 | **99.93** | 6 | 98.05 | 87.44 | 98.54 | 98.11 | **99.57** |
| OA | 99.04 | 91.37 | 99.49 | 99.35 | **99.82** | OA | 99.46 | 91.62 | 99.71 | 99.59 | **99.90** |
| AA | 96.83 | 91.23 | 99.01 | 98.74 | **99.26** | AA | 98.52 | 89.69 | 99.45 | 99.01 | **99.85** |
| K | 98.72 | 88.53 | 99.32 | 99.14 | **99.77** | K | 99.29 | 88.76 | 99.61 | 99.45 | **99.87** |

explicitly preserves the storage advantage of the smallest-parameter baselines.

The comparison with SSFTT gives the same result without relying on the two smallest parameter counts. LizardNet

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TABLE XXIV

COMPARISON OF LIGHTWEIGHT MODEL COMPLEXITY AND FLOPS EFFICIENCY. $\eta_{\text{FLOPs}}$ DENOTES THE ACCURACY ACHIEVED PER MILLION FLOPS. ALL FLOPS ARE RECOMPUTED USING 15 × 15 INPUT PATCHES. ACCURACY VALUES USE IP AT 5% TRAINING, DFC2013 HOUSTON AT 1% TRAINING, WHU-HI-HONGHU AT 25 SAMPLES PER CLASS, AND TRENTO AT 1% TRAINING

| Model | Indian Pines | | | DFC2013 Houston | | |
|---|---|---|---|---|---|---|
| | Params(M) | FLOPs(M) | $\eta_{\text{FLOPs}}$ | Params(M) | FLOPs(M) | $\eta_{\text{FLOPs}}$ |
| MADANet | 0.05 | 16.80 | 5.24 | 0.05 | 16.80 | 2.61 |
| SpectralFormer-Light | 0.80 | 159.40 | 0.39 | 0.80 | 115.03 | 0.53 |
| SSFTT | 0.15 | 6.81 | 13.81 | 0.15 | 5.25 | 15.63 |
| SSTN | 0.02 | 4.87 | 17.99 | 0.02 | 4.23 | 17.72 |
| **Ours** | **0.87** | **2.77** | **34.16** | **0.82** | **2.67** | **32.34** |
| Model | WHU-Hi-HongHu | | | Trento | | |
| | Params(M) | FLOPs(M) | $\eta_{\text{FLOPs}}$ | Params(M) | FLOPs(M) | $\eta_{\text{FLOPs}}$ |
| MADANet | 0.05 | 16.80 | 5.22 | 0.05 | 16.80 | 5.59 |
| SpectralFormer-Light | 0.81 | 219.90 | 0.25 | 0.79 | 50.81 | 1.60 |
| SSFTT | 0.15 | 6.81 | 11.99 | 0.15 | 2.66 | 35.74 |
| SSTN | 0.02 | 5.62 | 15.14 | 0.01 | 3.34 | 29.17 |
| **Ours** | **0.94** | **2.90** | **30.25** | **0.75** | **2.52** | **38.73** |

reduces FLOPs by 59.3%, 49.1%, 57.4%, and 5.3% while improving OA by 0.59, 4.26, 6.11, and 2.52 percentage points on the four datasets. Its $\eta_{\text{FLOPs}}$ is 1.90×, 1.83×, 2.00×, and 1.08× the strongest competing value on IP, DFC2013 Houston, WHU-Hi-HongHu, and Trento, respectively. The Trento comparison has the smallest arithmetic margin, where LizardNet still uses 0.14M fewer FLOPs and gains 2.52 OA points over SSFTT. On WHU-Hi-HongHu, which has the largest spectral input among these datasets, LizardNet uses 2.90M FLOPs and obtains 87.73% OA; SSFTT and SSTN require 6.81M and 5.62M FLOPs and obtain 81.62% and 85.06% OA. The four-dataset pattern shows that the favorable operating point persists across different input spectra and class structures. Parameter count continues to report checkpoint storage, while the corrected FLOPs and $\eta_{\text{FLOPs}}$ report the arithmetic efficiency of the implemented feature route.

Under the common 15 × 15 input protocol, LizardNet requires 2.77M, 2.67M, 2.90M, and 2.52M FLOPs on the four datasets, respectively. These are the lowest FLOPs in every comparison. Its corresponding $\eta_{\text{FLOPs}}$ values of 34.16, 32.34, 30.25, and 38.73 are also the highest. SSFTT requires 6.81M, 5.25M, 6.81M, and 2.66M FLOPs, while SSTN requires 4.87M, 4.23M, 5.62M, and 3.34M FLOPs. LizardNet therefore achieves higher OA with lower arithmetic cost than both methods across all four settings. Parameter count and FLOPs characterize different resource demands. The smaller parameter stores of SSTN and MADANet provide a memory advantage, while their weights are reused across more arithmetic operations and yield higher FLOPs on every dataset. For LizardNet, the low arithmetic workload is the computational consequence of selective gating, boundary-conditioned refinement, and constrained integration, which suppress redundant responses and interactions. Its highest or joint-highest OA, lowest FLOPs, and highest $\eta_{\text{FLOPs}}$ show that the accuracy gains are achieved without increasing the arithmetic workload.

LizardNet's complexity follows from its shallow architecture and constrained feature routing. SSG masks a subset of spectral channels, and TMI applies nuclear-norm regularization to multipath fusion during training. The regularizer promotes low effective rank and does not itself remove inference parameters. The resulting parameter counts and FLOPs are reported in Table XXIV together with the corresponding classification accuracies.

## VI. CONCLUSION

This article presented LizardNet, a lightweight neurodynamic framework for HSI classification. Its topology encodes three organizational features of the reptilian visual pathway. SSG implements ODE-inspired steady-state spectral gating, LIAD implements PDE-based anisotropic diffusion, and TMI implements nuclear-norm-regularized multipath fusion. The three modules address spectral redundancy, spatial boundary ambiguity, and small-sample generalization through low-effective-rank fusion. The overall parameter and FLOP efficiency results from the shallow constrained topology and feature routing.

Across four benchmark datasets, LizardNet obtains 98.75% OA on Trento and 94.63% OA with 5% training samples on IP. It uses 0.75M–0.94M parameters and 2.52M–2.90M FLOPs across the four datasets and records the lowest FLOPs among the evaluated lightweight models under the common 15 × 15 input protocol. On-board Jetson Orin Nano 8-GB measurements give 39.78-ms mean CPU latency, 25.14-samples/s throughput, and 0.191-J incremental energy per sample. Relative to DGPF-RENet under the same protocol, these values correspond to 87.34% lower mean latency and 89.70% lower incremental energy. The experiments quantify the accuracy, generalization, and deployment cost associated with the proposed structural constraints.

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